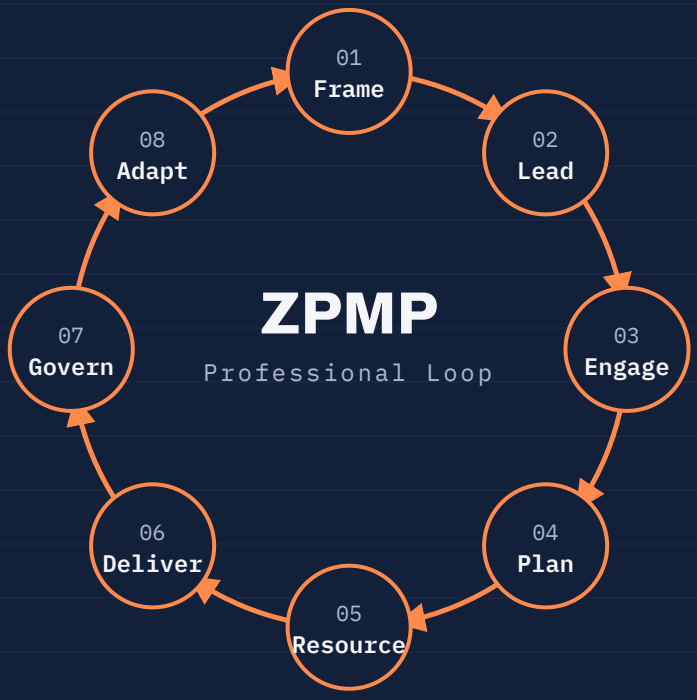
ZPMP

ZyeemZia™ Project Management Professional

Lead people, run the process, master the business environment.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 40 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

A modern project management professional programme aligned to the topics of the PMP® Examination Content Outline (July 2026): people, process and business environment across predictive, agile and hybrid delivery, with value, AI and sustainability built in and every outline task mapped to one memorable eight-stage loop.

01
Learn
Work through 40 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 40-question official exam at zpmp.zyeemzia.com, graded on the ZyeemZia server. Pass mark 80%.

04
Get certified
Download your certificate online. Anyone can verify its credential ID at zpmp.zyeemzia.com/#verify. 90%+ and the capstone earn Distinction.

Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Week 1	m1 Frame	Foundations and the outline map; draft your business case, charter and vision	☐
Week 2	m2 Lead	Team charter, RACI and leadership approach for your capstone	☐
Week 3	m3 Engage	Stakeholder register, alignment map, knowledge transfer and communication plans	☐
Week 4	m4 Plan	Integrated plan, scope breakdown and value-based priorities	☐
Week 5	m5 Resource	Estimates, schedule or release forecast, resources, budget and EVM practice	☐
Week 6	m6 Deliver	Procurement, quality, status dashboard and closure plan	☐
Week 7	m7 Govern	Governance thresholds, compliance register, change log and risk register	☐
Week 8	m8 Adapt	Impediments, improvement, change and external radar; practice exam, capstone pack and certification exam	☐

Learner name .....

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Start date ..... Target exam date .....

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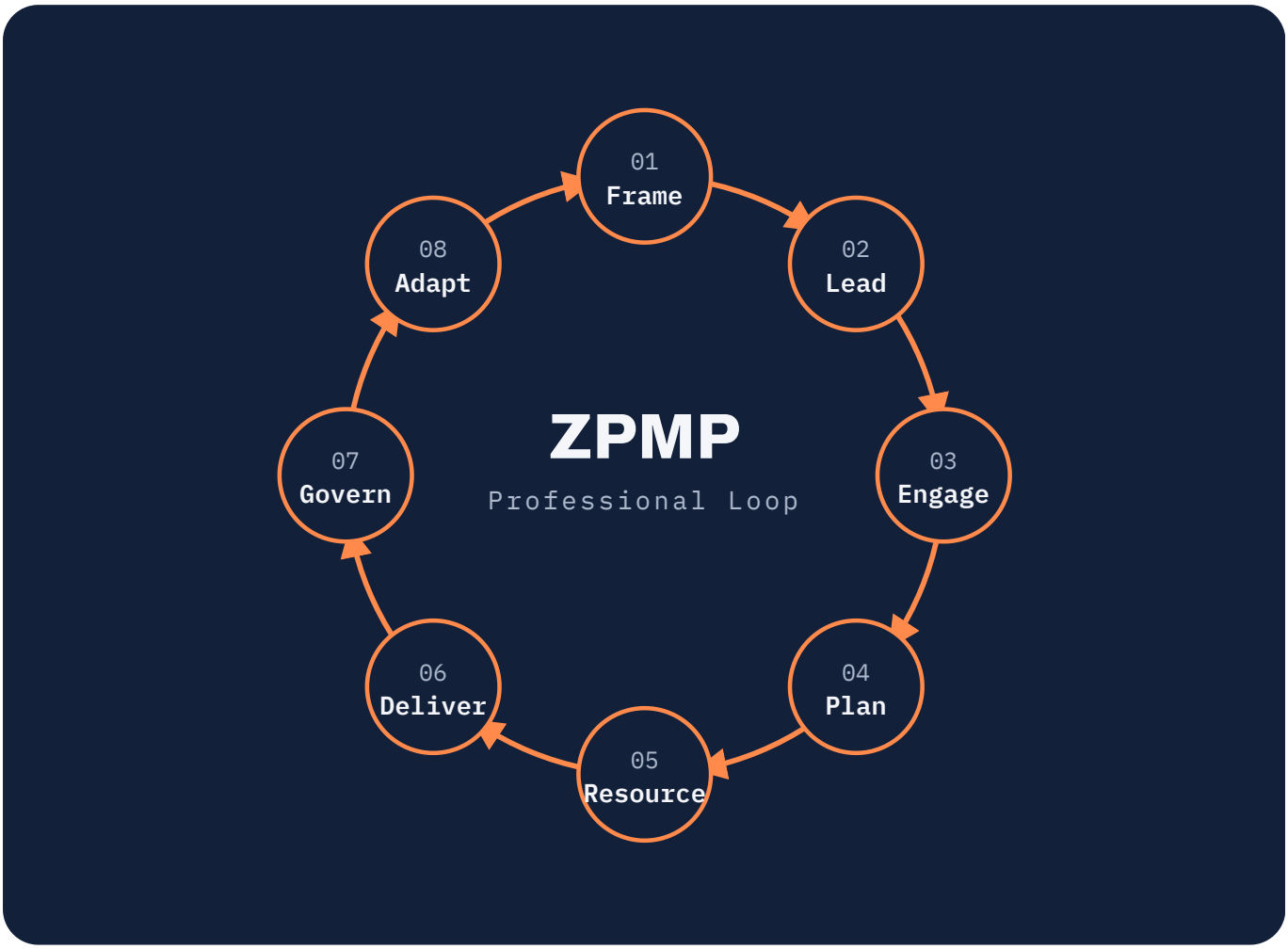
+ **Glossary · Capstone guide · Practice exam & answers · Certificate · Notes**

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THE FRAMEWORK

The ZPMP Professional Loop

Eight stages a project professional moves through again and again: frame the value, lead the team, engage stakeholders, plan the work, resource it, deliver it, govern it and adapt to change. People, process and the business environment in one continuous loop.



01 Frame
Foundations · Approaches, principles and the outline map · People Task 1 begins

02 Lead
People · Tasks 1–3 (common vision, conflict, leading the team)

03 Engage
People · Tasks 4–8 (engage stakeholders, align and manage expectations, knowledge transfer, communication)

04 Plan
Process · Tasks 1–3 (integrated plan and delivery planning, scope, value-based delivery)

05 Resource
Process · Tasks 4, 6 and 8 (resources, finance, schedule)

06 Deliver
Process · Tasks 5, 7, 9 and 10 (procurement, quality, project status, closure)

07 Govern
Business Environment · Tasks 1, 2, 3 and 5 (governance, compliance, change control, risk)

08 Adapt
Business Environment · Tasks 4, 6, 7 and 8 (impediments and issues, continuous improvement,

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organisational change, external environment), plus capstone and exam readiness

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01

MODULE 1 · FRAME

Foundations: Value, Approaches & the PMP® Outline

Foundations · Approaches, principles and the outline map · People Task 1 begins

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain what projects, programmes and portfolios are and how they create value
- Apply the principles of modern project management to everyday decisions
- Choose and tailor a predictive, agile or hybrid approach
- Build a business case, charter and shared vision for a project
- Map every task of the July 2026 outline to the ZPMP modules

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1. Projects, Value & the Project Manager's Role · 24 min
2. Principles & the Value Mindset · 25 min
3. Delivery Approaches: Predictive, Agile, Hybrid & Tailoring · 30 min
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5. How ZPMP Maps to the PMP® Exam Outline (July 2026) · 22 min

MODULE 1 · FRAME · LESSON 1 · 24 MIN

Projects, Value & the Project Manager's Role

Projects, operations, products, programmes and portfolios, and what a modern project manager is accountable for

- Define a project in modern, value-focused terms
- Distinguish projects from operations and long-lived products
- Explain how projects, programmes and portfolios connect to strategy
- Describe the responsibilities and stances of a modern project manager across approaches

Every organisation runs on two engines. One keeps today's business working: payroll, customer service, production lines. The other changes the business: a new app, a new hospital wing, a new way of working. That second engine is **project work**, and in 2026 it is where most competitive advantage is won or lost.

DEFINITION · PROJECT

A **project** is a temporary, purposeful effort that creates a unique result — a product, service, capability or change — so that an organisation can move from where it is now to a better future state. The effort ends; the result is meant to keep creating value.

Four quick tests: is it a project?

- **Temporary** — it has a planned start and a planned finish (the result may last for decades).
- **Unique** — the result, context or team has not been combined in exactly this way before.
- **Purposeful** — it exists to deliver a change that someone values, not just to keep busy.
- **Progressively elaborated** — details become clearer as you learn, so plans are refined over time.

Projects vs operations vs products

Project

- Temporary effort with an end date
- Creates something new or changes something
- Team often assembled for the effort
- Success = outcomes and benefits enabled
- Example: building a mobile wallet's savings feature

Operations

- Ongoing and repetitive
- Sustains the business as it is
- Stable, permanent teams and processes
- Success = efficiency, reliability, service levels
- Example: processing 50,000 wallet transactions a day

Product

- Long-lived offering with its own life cycle
- Evolves through many projects or continuous releases
- Often owned by a stable, cross-functional team
- Success = customer and business value over time
- Example: the wallet app itself, improved every sprint

INSIGHT · PROJECT THINKING MEETS PRODUCT THINKING

Many digital teams no longer 'finish' a project and walk away. They fund a **stable product team** and deliver change continuously. Project management skills still matter here — scope, risk, stakeholders, money and time do not disappear — they are simply applied in shorter, repeating loops.

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Where projects live: from strategy to daily work

Strategy	`Vision and mission` `Strategic objectives and OKRs` `Risk appetite`
Portfolio	`Chooses and balances investments` `Funds what best serves strategy` `Stops what no longer fits`
Program	`Coordinates related projects` `Manages shared dependencies` `Delivers combined benefits`
Projects and products	`Create the deliverables` `Enable outcomes` `Hand over to operations`
Operations	`Run the new capability` `Realise most of the benefits` `Feed back improvement ideas`

Project, program and portfolio at a glance

ASPECT	PROJECT	PROGRAM	PORTFOLIO
Main question	Are we building this right?	Are these related efforts adding up to the intended benefits?	Are we investing in the right things?
Scope	A defined result	A set of related projects and activities	All investments in a business area or company
Time frame	Defined start and finish	Usually longer, ends when benefits are secured	Ongoing
Measured by	Outcomes enabled, delivered within agreed constraints	Benefits realised across components	Strategic return and balance of the whole mix
Typical leader	Project manager or delivery lead	Program manager	Portfolio manager or investment board

The modern project manager is less a task-master and more a **value-focused integrator and leader**. You connect strategy to delivery, people to purpose and plans to reality. On agile teams some traditional PM duties are shared by a product owner, a team coach and the team itself; your contribution shifts toward coordination across teams, stakeholders, risk, budget and governance.

Six stances of the modern project manager

M1
Value navigator
Keeps every decision tied to outcomes, benefits and the business case.

M2
Integrator
Joins scope, schedule, cost, risk, quality and people into one coherent plan.

M3
Servant leader
Removes obstacles, protects focus and grows the team's capability.

M4
Adaptive planner
Plans in the right level of detail and updates plans as learning arrives.

M5
Trusted communicator
Shares honest status, early bad news and clear options with stakeholders.

M6
Ethical steward
Acts with honesty, fairness, respect and responsibility for people and planet.

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Who does what across approaches

RESPONSIBILITY	PREDICTIVE PROJECT	AGILE TEAM	HYBRID PROJECT
Setting priorities	PM with sponsor via baselines	Product owner, continuously	Product owner within PM-managed milestones
Detailed planning	PM with team, up front	Team, each iteration	Both, at different levels
Removing impediments	PM	Team coach / Scrum Master and team	PM and team coach together
Stakeholder and budget governance	PM	Often a PM, product owner or delivery lead outside the team	PM

INSIGHT · THREE ARENAS OF THE JOB

Whatever the method, a project manager works in three arenas at once: **people** (leading the team and engaging stakeholders), **process** (planning and running the technical work of delivery) and the **business environment** (governance, compliance, risk, change and the world outside the project). ZPMP is organised around exactly these three arenas, and lesson 1.5 shows how.

SCENARIO · DECIDE · PROJECT OR OPERATIONS?

You work at a Chattogram logistics company. Your manager says: "Please project-manage our monthly payroll run from now on — make a Gantt chart and run it as a project every month." Payroll has run the same way for six years without major problems. What is the best response?

(A) Agree and create a new project plan, charter and Gantt chart every month.

(B) Explain that payroll is operations best managed with a stable process, checklist and service levels — and offer to run a separate project if the goal is to improve or automate payroll.

(C) Refuse, because project managers should not touch operational work.

(D) Turn payroll into a Kanban board and call it an agile project.

Circle your choice. Answer and feedback: module answer key.

Key terms

Project
A temporary, purposeful effort that creates a unique product, service, capability or change to move an organisation toward a better future state.

Program
A group of related projects and activities managed together so their combined benefits are greater than if they were managed separately.

Product
A long-lived offering, such as an app or a service, that evolves through many releases or projects and is usually owned by a stable team.

Operations
Ongoing, repetitive work that sustains the organisation's existing products and services.

Portfolio
The full set of projects, programs and operational investments an organisation manages to achieve its strategy, selected and balanced as a whole.

Progressive elaboration
Refining plans and details step by step as information and learning increase, rather than fixing everything on day one.

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Project manager
The person assigned by the performing organisation to lead the team and integrate the work so that the project achieves its objectives and enables its intended value.

KNOWLEDGE CHECK

1. Which of these is the clearest example of a project?
   - (A) Answering customer calls at a bank's contact centre every day
   - (B) Producing 20,000 shirts per week on an existing production line
   - (C) Migrating a bank's core system to a cloud platform by next March
   - (D) Running the monthly payroll for all staff across the company's branches
2. A company groups an app launch, agent training and a marketing campaign so that together they increase rural financial inclusion. This group is best described as a:
   - (A) Portfolio
   - (B) Program
   - (C) Product
   - (D) Operation
3. What does 'progressively elaborated' mean?
   - (A) The project grows larger every month
   - (B) Stakeholders are added gradually
   - (C) The plan is fixed on day one and never changed
   - (D) Details are refined as more becomes known
4. Why do many digital organisations fund stable product teams rather than one-off projects?
   - (A) To deliver change continuously and keep product knowledge in one team
   - (B) Because product work does not need budgets or funding decisions
   - (C) Because project management is no longer needed in digital firms
   - (D) To reduce how often teams need to talk directly to customers
5. On a hybrid project, the product owner reorders the app backlog each sprint while the project manager runs the hardware contract and reports milestone progress to the steering group. What does this division of work show?
   - (A) The project manager has lost authority and should take the backlog back
   - (B) Responsibilities are shared by role, and the PM integrates them into one picture
   - (C) The project is badly organised because one person should own every decision
   - (D) The steering group should approve every backlog change before each sprint

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 2 · 25 MIN

Principles & the Value Mindset

Why modern project management starts with value, and the principles and domains that guide judgement

- Trace the chain from outputs to outcomes, benefits and value
- Explain the principles of modern project management in your own words
- Describe the performance domains and what good looks like in each

A project can finish on time, on budget and to specification — and still be a failure. If nobody uses the new system, if the clinic is built where patients cannot reach it, if the feature solves a problem customers do not have, the organisation has spent money and gained nothing. Modern project management therefore measures success by **value delivered**, not tasks completed.

From work to value

01	→	02	→	03	→	04	→	05
Activities The work the team does		**Outputs** Deliverables: the app, the building, the report		**Outcomes** What changes because people use the outputs		**Benefits** Measurable gains for the organisation or society		**Value** Worth to stakeholders after cost, risk and time

EXAMPLE · A HOSPITAL APPOINTMENT APP

Output: a booking app is live. **Outcome:** 60% of patients book online instead of queuing. **Benefit:** phone calls fall by a third and average waiting time drops by 25 minutes. **Value:** a calmer front desk, staff time redeployed to patient care and better patient experience — worth far more than the build cost.

Organisations create value through a connected **value delivery system**: portfolios choose investments, programs and projects create capabilities, products evolve them and operations run them. Information must flow both ways — strategy guides projects, and evidence from projects corrects strategy.

Twelve principles of modern project management, in ZPMP words

P1
Act as a responsible steward
Look after money, people, trust and the environment as if they were your own, and act with integrity.

P2
Build a team that works together
Create a safe, collaborative environment where people share accountability for results.

P3
Engage stakeholders genuinely
Involve the people affected early and often, and understand their interests and influence.

P4
Keep value at the centre
Constantly check that the work still leads to the intended outcomes and benefits.

P5
Think in systems
See the project as a set of interacting parts within a wider organisation; changes ripple.

P6
Lead, whatever your title
Motivate, influence and set direction — leadership is a behaviour, not a job title.

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P7
Tailor the approach
Fit methods, processes and documents to the context instead of applying one recipe everywhere.

P8
Build quality in
Prevent defects through good design and clear acceptance criteria rather than inspecting them out later.

P9
Navigate complexity
Expect surprises from human behaviour, ambiguity and technology; use feedback to find a way through.

P10
Respond to risk wisely
Manage threats and opportunities so the overall risk exposure supports the goals.

P11
Stay adaptable and resilient
Absorb setbacks, recover quickly and adjust plans as conditions change.

P12
Enable change
Help people adopt the new way of working so the future state is actually reached.

INSIGHT · PRINCIPLES GUIDE JUDGEMENT; PROCESSES PRESCRIBE STEPS
A process says *what to do*. A principle says *what matters* when the process does not fit. When you meet a situation no template covers — and you will — fall back on the principles.

Eight performance domains — where effort must go well

D1
Stakeholders
Productive relationships and agreement with the people who influence or are affected by the project.

D2
Team
Shared ownership, high performance and healthy team culture.

D3
Development approach and life cycle
A delivery approach and phase structure that suit the product and context.

D4
Planning
Enough organised, coordinated planning to deliver — no more, no less.

D5
Project work
Efficient processes, resource and supplier management, and continuous learning.

D6
Delivery
Scope and quality that achieve the intended outcomes.

D7
Measurement
Reliable data to understand performance and act early (developed in modules 5 and 6).

D8
Uncertainty
Awareness of risk, ambiguity, complexity and volatility, with responses that fit (developed in module 7).

WATCH OUT · EDITIONS CHANGE, IDEAS LAST
Professional standards are revised from time to time. Newer editions tend to group these ideas into fewer, broader principles, give more weight to a **holistic view** of the project and to **sustainability**, and organise the domains differently. Learn the ideas here, then check the exact edition and outline used by any external exam you plan to take.

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Sustainability as an everyday principle

- Ask early: what environmental, social and economic effects will this project have during delivery and across the life of what it creates?
- Treat energy use, waste, travel and supplier practices as design choices, not afterthoughts.
- Look at whole-life cost and impact, not purchase price alone.
- Make sustainability requirements testable, so the team can show they were met.

Output mindset vs value mindset

Output mindset

- "Did we deliver every feature on the list?"
- Success declared at go-live
- Change requests seen as threats
- Reports show % of tasks complete
- PM acts as task tracker

Value mindset

- "Did the change achieve the outcome we needed?"
- Success confirmed when benefits appear
- Change welcomed when it increases value
- Reports show outcomes, risks and decisions needed
- PM acts as value navigator and integrator

SCENARIO · DECIDE · DELIVERED — BUT WILL IT BE USED?

You are managing a field-sales app for a Dhaka consumer-goods company. The sponsor's main concern is go-live on 1 March. In user tests, sales officers say the order screen is too slow on low-end phones in poor-signal areas. Fixing it would take two extra weeks. What should you do FIRST?

(A) Go live on 1 March as promised; performance can be fixed later.

(B) Quietly add two weeks to the schedule and inform the sponsor afterwards.

(C) Bring the evidence to the sponsor with options — e.g. a smaller release for good-signal regions on 1 March plus the performance fix two weeks later — and let the sponsor decide based on value.

(D) Ask the developers to work weekends until it is fixed by 1 March.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a project you know that finished on time and on budget but disappointed its users or sponsor. Which link in the chain — output, outcome, benefit or value — broke, and which principle would have helped most?

Key terms

Output

A deliverable the project produces, such as an app, a building or a trained group of staff.

Outcome

The change in behaviour or performance that happens when people use the outputs.

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Value
The worth that stakeholders gain from a project, which may be financial, social, environmental or strategic.

Performance domain
A group of related activities, such as stakeholders, planning or uncertainty, that must go well for the project to deliver value.

Systems thinking
Seeing the project as a set of interacting parts so that the knock-on effects of a decision are anticipated.

Principle
A broad statement of what matters that guides judgement when no process or template fits.

Value delivery system
The connected set of portfolios, programmes, projects, products and operations through which an organisation turns strategy into value.

KNOWLEDGE CHECK

6. Patients now book appointments online instead of queuing. In the value chain this is best described as:
   - (A) An output
   - (B) An activity
   - (C) A constraint
   - (D) An outcome

7. Which principle is most directly applied when a PM adapts documents and ceremonies to a small, low-risk project?
   - (A) Build quality in
   - (B) Stay adaptable and resilient
   - (C) Tailor the approach
   - (D) Think in systems

8. A change in one team's API causes delays in two other teams. Which principle best helps the PM anticipate this?
   - (A) Act as a responsible steward
   - (B) Think in systems
   - (C) Stay adaptable
   - (D) Lead, whatever your title

9. Which performance domain focuses on choosing between predictive, adaptive and hybrid delivery?
   - (A) Development approach and life cycle
   - (B) Planning
   - (C) Delivery
   - (D) Project work

10. Why are principles especially useful to a project manager?
    - (A) They replace the need for any defined process or methodology
    - (B) They are only needed on agile projects, not on predictive ones
    - (C) They guide judgement when no process or template fits
    - (D) They guarantee success when every principle is applied in full

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 3 · 30 MIN

Delivery Approaches: Predictive, Agile, Hybrid & Tailoring

Choose a development approach with confidence and tailor it to the project, team and organisation

- Compare predictive, iterative, incremental, agile and hybrid life cycles
- Use an uncertainty matrix to select a starting approach
- Design a deliberate hybrid and tailor practices to context

A **project life cycle** is the series of phases a project passes through from start to finish. The **development approach** is how the product is created within those phases. Choosing the wrong approach is one of the most expensive early mistakes a project can make — and one of the easiest to avoid.

Five life cycles compared

APPROACH	HOW IT WORKS	REQUIREMENTS	WHEN VALUE ARRIVES	BEST FIT
Predictive	Plan the whole effort up front, then execute in sequence	Fixed early	Mostly at the end	Stable scope, high cost of change, e.g. building construction
Iterative	Repeat cycles to refine the solution through feedback	Clarified through cycles	At the end, but solution improves each cycle	Unclear solution, e.g. new service design
Incremental	Deliver usable pieces one after another	Defined per increment	Each increment adds value	Value can be split, e.g. district-by-district rollout
Agile	Short iterative and incremental cycles with constant feedback	Evolve continuously	Frequently, often every 1–4 weeks	High uncertainty, fast learning needed, e.g. digital products
Hybrid	Combine elements deliberately	Mixed: some fixed, some evolving	Depends on the design	Mixed work types, regulation plus innovation

DEFINITION · ITERATIVE VS INCREMENTAL

Iterative = making the *same thing better* through repeated cycles (sketch → prototype → refined prototype). **Incremental** = adding *more pieces* of finished value over time (module 1, then module 2). Agile combines both.

A typical hybrid life cycle

01 **Initiate** — Charter and business case approved →

02 **Design and approvals** — Architecture, compliance and vendor contracts fixed →

03 **Build in sprints** — Features delivered and demoed every two weeks →

04 **Integration and sign-off** — Regulatory and security acceptance gate →

05 **Rollout in waves** — Region by region, learning between waves →

06 **Close and transition** — Handover to operations, lessons captured

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To choose a starting approach, ask two questions. **How clear and agreed are the requirements?** And **how certain is the way we will build it** — the technology or method? Plotting your project on these two axes is a practical version of the well-known Stacey-style complexity matrix.

The ZPMP uncertainty matrix

ZONE	REQUIREMENTS	TECHNOLOGY OR METHOD	SUGGESTED STARTING APPROACH
Clear	Known and agreed	Proven	Predictive; use standard plans and checklists
Complicated	Mostly known	Needs expert analysis	Predictive or hybrid with expert reviews and prototypes for risky parts
Complex	Evolving or disputed	Partly unknown	Agile or hybrid; short cycles, frequent feedback, small experiments
Chaotic	Unclear and changing fast	Unknown	Act to stabilise first (crisis mode), then move to a complex-zone approach

Tailoring factors

FACTOR	LEANS PREDICTIVE WHEN…	LEANS ADAPTIVE WHEN…
Requirement stability	Scope is fixed by law, design or contract	Users will discover needs as they see the product
Cost of change	Late change is very expensive (concrete, hardware)	Change is cheap (software, content, process)
Delivery options	Value only exists when everything is complete	Value can be released in usable slices
Regulation and safety	Heavy documentation and formal approvals are mandatory	Lighter governance is acceptable
Stakeholder availability	Customer reviews only at fixed gates	A product owner or users are available every week
Team experience	Team is used to phase-based delivery	Team is cross-functional and experienced with agile

Tailoring in five moves

1. **Diagnose** Assess uncertainty, size, criticality, regulation, culture and team capability.
2. **Pick a base** Choose a starting approach — predictive, agile or a hybrid pattern your organisation already knows.
3. **Adjust** Add, remove or scale practices: lighter documents for small projects, extra reviews for safety-critical ones.
4. **Agree** Confirm the approach with the sponsor, governance bodies and the team so expectations match.
5. **Inspect and adapt** Review the approach itself at retrospectives and phase gates; change what is not working.

> **WATCH OUT · BEWARE THE WORST OF BOTH WORLDS**
> Some teams call themselves hybrid but simply run sprints inside a big upfront specification followed by a single big-bang release. They carry the cost of agile ceremonies **and** the risk of late feedback. Good hybrid design is deliberate: each element is there for a reason.

Most organisations in 2026 run **hybrid** delivery. Good hybrids are designed on purpose: each part of the work gets the approach that fits its uncertainty, and the joins between parts are managed explicitly.

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Five common hybrid patterns

H1

Predictive wrapper, agile core

Gates, budget and milestones are predictive; product development inside runs in sprints.

H2

Agile discovery, predictive build

Prototype and test ideas iteratively, then build the chosen solution with a fixed plan (common in construction and hardware).

H3

Parallel tracks

Agile software teams alongside predictive hardware, construction or procurement, joined by integration milestones.

H4

Predictive build, incremental rollout

The solution is completed predictively, then deployed in waves — site by site or region by region.

H5

Project plus flow

A project delivers a new capability; a Kanban team handles support and small changes in parallel.

Designing a hybrid in six steps

1. **Split the work** Divide scope into components (software, hardware, training, process, procurement).
2. **Rate uncertainty** Place each component on the uncertainty matrix earlier in this lesson.
3. **Assign approaches** Predictive for stable, high-cost-of-change parts; agile or flow for uncertain parts.
4. **Design the joins** Integration milestones, interface agreements, shared cadence for reviews.
5. **Align governance** Map phase gates and reporting to both styles — milestones and release forecasts.
6. **Review and adapt** Check at each gate and retrospective whether the design still fits.

SCENARIO · DECIDE · CHOOSING AN APPROACH FOR A BANK

A commercial bank in Dhaka must deliver a new regulatory reporting system by a fixed central-bank deadline, with requirements defined in a published circular. In the same project the bank wants a new customer self-service portal whose features are still being debated by marketing. What approach should the PM recommend?

(A) Fully predictive for both parts, with a detailed specification signed off before any work begins.

(B) Fully agile for both parts, with no fixed milestones.

(C) A hybrid: predictive planning and formal sign-off for the regulatory component, agile sprints with frequent feedback for the portal, and a shared integration and release plan.

(D) Delay the portal until the regulatory system is complete, then decide.

Circle your choice. Answer and feedback: module answer key.

Key terms

Life cycle

The series of phases a project passes through from start to finish.

Development approach

The way the product is created within the life cycle: predictive, adaptive (agile) or hybrid.

Predictive approach

An approach that defines scope, schedule and cost early and then executes the plan in sequence, controlling change through baselines.

Adaptive approach

An approach that plans in short cycles and refines scope through frequent feedback; agile methods are the most common form.

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Iterative
Improving the same result through repeated cycles of build, feedback and refinement.

Incremental
Delivering usable pieces of the result one after another, each adding value.

Tailoring
Deliberately adapting the approach, processes and artefacts to fit the project's context.

KNOWLEDGE CHECK

11. A team delivers a payroll module first, then leave management, then performance reviews — each usable on release. This is mainly:
    - (A) Iterative development
    - (B) Predictive development
    - (C) Prototype-driven development
    - (D) Incremental development

12. Requirements are disputed and the technology is new to the team. In the uncertainty matrix this project is most likely:
    - (A) Clear
    - (B) Complicated
    - (C) Complex
    - (D) Chaotic

13. Which factor most strongly favours a predictive approach?
    - (A) Users want to see early versions
    - (B) The cost of late change is very high
    - (C) The team is cross-functional
    - (D) Requirements will emerge from customer feedback

14. What is the main danger of a poorly designed hybrid?
    - (A) It may combine heavy upfront specification with late feedback
    - (B) It always costs more than a purely predictive or agile approach
    - (C) It removes the need for a sponsor and for formal governance gates
    - (D) It forbids sprints in any phase that has a fixed regulatory deadline

15. A project prototypes a new clinic layout iteratively with staff, then builds it with a fixed construction plan. This is the pattern:
    - (A) Project delivery plus continuous flow
    - (B) Parallel agile and predictive tracks
    - (C) Agile discovery, predictive build
    - (D) Pure agile from discovery to build

16. What should a PM do after choosing and tailoring an approach?
    - (A) Review it periodically and adapt it if it is not working
    - (B) Lock it in for the whole project to give stability
    - (C) Ask the team to follow it without further discussion
    - (D) Replace it every sprint to experiment with new practices

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 1 · FRAME · LESSON 4 · 30 MIN

Business Case, Charter & a Common Vision

Justify the investment, authorise the project and build a vision that everyone shares (People Task 1 begins)

- Outline a lean business case and interpret payback, ROI and NPV
- Explain the purpose and contents of a project charter
- Create, promote and refresh a shared project vision
- Find the root cause when stakeholders misread the vision

Before an organisation commits people and money, someone must answer a simple question: **why this, why now, and is it worth it?** The **business case** answers that question. It is not a one-time formality — it is the yardstick you return to whenever a major decision or change arises.

Anatomy of a lean business case

1. **Problem or opportunity** What is wrong or possible today? Use evidence: data, complaints, market signals.
2. **Options** At least three, including 'do nothing' and 'do the minimum', so the decision is a real choice.
3. **Benefits** What measurable improvements will each option bring, and when?
4. **Costs and funding** One-time and ongoing costs, and where the money comes from.
5. **Risks and assumptions** What could make the case fail, and what are we assuming to be true?
6. **Recommendation** The preferred option, the decision criteria and the next step requested.

Common investment measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Payback period	`Initial investment ÷ annual net cash inflow (for even inflows)`	How long until the money spent is recovered. Shorter is safer, but it ignores what happens after payback.
Return on investment (ROI)	`(Total benefits − Total costs) ÷ Total costs × 100%`	Gain per unit of cost. Simple, but ignores the timing of money.
Net present value (NPV)	`Σ [Cash flow in year t ÷ (1 + r)^t] − Initial investment`	Value today of future cash flows, discounted at rate r. Positive NPV means the investment beats the required return.
Benefit–cost ratio (BCR)	`Present value of benefits ÷ Present value of costs`	Greater than 1 means benefits exceed costs in today's money.

EXAMPLE · SAME PROJECT, DIFFERENT VERDICTS

A garment exporter can spend **Tk 50 lakh** on an automated cutting-room system that saves **Tk 20 lakh a year for 3 years**. Payback = 50 ÷ 20 = **2.5 years**. Simple ROI = (60 − 50) ÷ 50 = **20%**. But at a 10% discount rate the savings are worth 18.18 + 16.53 + 15.03 = **Tk 49.74 lakh** today, so NPV ≈ **−Tk 0.26 lakh**. Lesson: the timing of money can turn an apparently good case into a marginal one — present more than one measure.

OKRs (Objectives and Key Results) are a popular way to connect strategy to work. An **objective** is a short, inspiring, qualitative statement of where you want to go. **Key results** are two to five measurable outcomes that show you got there. Projects and initiatives are the bets you place to move the key results.

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The **project charter** is the birth certificate of a project. Issued by the sponsor, it formally authorises the project to exist, names the project manager and gives that person authority to use organisational resources. Without one, you are leading work nobody has officially agreed to.

TEMPLATE

One-page project charter

A lean charter the sponsor can read in five minutes and sign. Fill one row per section.

SECTION	GUIDING QUESTION	YOUR PROJECT'S ANSWER
Purpose	*Why does this project exist, and why now?*	*Field staff register flood-affected households on paper; data reaches head office 3 weeks late, delaying aid.*
Success criteria	*How will we know it worked?*	*95% of registrations synced within 48 hours; aid disbursed within 10 days of registration by Dec 2027.*

A charter authorises the work, but it does not make people **believe** in it. A **common vision** is a short, shared picture of the future the project will create: who will be better off, what will be different and how everyone will know. When the team and key stakeholders hold the same picture, hundreds of small daily decisions line up without being escalated. This is the first task of the People domain, and it starts here, before the team is even complete.

Five elements of a shared project vision

V1

Purpose

Why the project exists and why now, in one sentence that links to strategy.

V2

Beneficiaries

Who will be better off, described as real people, for example 'night-shift nurses' rather than 'users'.

V3

The change

What will be visibly different when the project succeeds.

V4

Evidence

Two or three measures that will show the vision has been reached.

V5

Boundaries

What the project will not do, so that the vision does not grow without limit.

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Keep the vision alive

1. **Co-create it** Draft the vision with the sponsor, product owner and a few team members and users, not alone at a desk.
2. **Test understanding** Ask people to explain the vision in their own words. Mismatches show where it is unclear.
3. **Promote it** Open planning sessions, sprint reviews and steering meetings with the vision and the measure it serves.
4. **Use it to decide** When choosing between options, ask which one moves the vision forward most.
5. **Refresh it** Review the vision at each phase gate or quarterly planning event, and when strategy or the market shifts.

WATCH OUT · WHEN PEOPLE MISREAD THE VISION, FIND THE ROOT CAUSE

In a Chattogram port digitisation project, the operations team kept rejecting screens the developers thought were finished. Asking 'why?' five times showed the real cause: the vision said 'faster container release', but operations measured success by **fewer manual checks**, while the developers aimed at **faster screens**. The fix was not more meetings; it was rewriting the vision's evidence measure so both groups shared it. Break a misunderstanding down into its causes before you repeat the message more loudly.

SCENARIO · DECIDE · NO CHARTER, BUT 'JUST START'

A director calls you in: "The board loves this idea. Start the vendor selection tomorrow — paperwork can come later." There is no charter, no confirmed budget and no named sponsor. What should you do FIRST?

(A) Start vendor meetings tomorrow to show enthusiasm.

(B) Refuse to do anything until a full project management plan is approved.

(C) Draft a one-page charter with the director within a day or two, confirm who the sponsor is and get it signed before engaging vendors.

(D) Email the whole company announcing the project.

Circle your choice. Answer and feedback: module answer key.

Key terms

Business case
A documented justification for an investment that compares options, benefits, costs and risks and recommends a way forward.

Project charter
A document issued by the sponsor that formally authorises a project and gives the project manager authority to apply resources.

Payback period
How long it takes for the money spent to be recovered from the project's net inflows.

Product vision
A statement of who a product is for, what need it meets and how it is different, used to guide backlog decisions.

OKR
Objectives and Key Results: a qualitative objective paired with a few measurable outcomes that show progress toward it.

Net present value (NPV)
The value today of future net cash flows, discounted at a chosen rate; a positive NPV means the investment earns more than the rate.

Common vision
A short, shared picture of the future the project will create, held by the team and key stakeholders.

KNOWLEDGE CHECK

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17. A project costs Tk 30 lakh and returns Tk 10 lakh a year in even net inflows. What is the payback period?

(A) 2 years
(B) 3 years
(C) 10 years
(D) 30 years

18. Which measure accounts for the time value of money?

(A) Simple return on investment
(B) Payback period
(C) Net present value
(D) Percent complete

19. Which is a well-written key result?

(A) Launch the redesigned app to all customers by June
(B) Increase completion rate from 42% to 70% by June
(C) Hold weekly product meetings with the sales team
(D) Hire two senior developers before the end of March

20. Who normally issues and signs the project charter?

(A) The sponsor or initiating authority
(B) The project manager, once appointed
(C) The lead vendor or delivery partner
(D) The quality manager for the programme

21. Developers and the operations team both say they support the vision 'faster container release', yet operations keeps rejecting finished screens. What should the project manager do FIRST?

(A) Repeat the vision at every meeting until both groups accept it
(B) Ask the sponsor to rule which group is right and close the matter
(C) Analyse why the groups interpret the vision differently and agree a shared measure
(D) Let operations write the acceptance criteria and accept the delay

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · FRAME · LESSON 5 · 22 MIN

How ZPMP Maps to the PMP® Exam Outline (July 2026)

Three domains, 26 tasks, one loop: how the course is organised and how to study it

- Explain the three domains of the July 2026 outline and their weights
- Locate every outline task in the ZPMP modules
- Describe the eight stages of the ZPMP Professional Loop
- Plan your own study across predictive, agile and hybrid approaches

ZPMP is organised around the **topics** of PMI's PMP® Examination Content Outline published for July 2026. The outline describes the job of a project manager in three **domains**: People, Process and Business Environment. Each domain is split into **tasks**, and each task is illustrated with a few **enablers**, examples of the work involved. ZPMP teaches every task in its own words, with its own examples, and adds the practical tools you need on real projects.

WATCH OUT · AN INDEPENDENT PROGRAMME

ZPMP (ZyeemZia™ Project Management Professional) is created and issued by ZyeemZia. It is **not** PMI's PMP® certification, does not award or count as the PMP®, and is not affiliated with or endorsed by PMI. Its curriculum follows the outline's topics to help you prepare. To earn the PMP® itself you must apply to PMI, meet its eligibility rules and pass its exam.

DEFINITION · DOMAIN, TASK AND ENABLER

A **domain** is a broad area of the job. A **task** is a responsibility inside that domain, such as managing conflicts or planning procurement. An **enabler** is an example of the work that shows what the task means in practice. Enablers are illustrations, not a complete list, so ZPMP treats them as a topic checklist rather than as a syllabus to memorise.

Domains, weights and where they live in ZPMP

DOMAIN	SHARE OF THE OUTLINE	ZPMP MODULES	ZPMP EXAM QUESTIONS (OF 40)
People	33%	m1 Frame (vision), m2 Lead, m3 Engage	13
Process	41%	m4 Plan, m5 Resource, m6 Deliver	16
Business Environment	26%	m7 Govern, m8 Adapt	11

All 26 outline tasks mapped to ZPMP (task names as published in the PMP® Examination Content Outline, July 2026, PMI)

DOMAIN	TASK	OUTLINE TASK NAME	WHERE ZPMP TEACHES IT (MODULE AND LESSON)
People	1	Develop a common vision	m1 Frame (1.4), m2 Lead (2.1)
People	2	Manage conflicts	m2 Lead (2.4)
People	3	Lead the project team	m2 Lead (2.1, 2.2, 2.3, 2.5)

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DOMAIN	TASK	OUTLINE TASK NAME	WHERE ZPMP TEACHES IT (MODULE AND LESSON)
People	4	Engage stakeholders	m3 Engage (3.1)
People	5	Align stakeholder expectations	m3 Engage (3.2)
People	6	Manage stakeholder expectations	m3 Engage (3.3)
People	7	Help ensure knowledge transfer	m3 Engage (3.4)
People	8	Plan and manage communication	m3 Engage (3.5)
Process	1	Develop an integrated project management plan and plan delivery	m4 Plan (4.1), with m1 Frame (1.3)
Process	2	Develop and manage project scope	m4 Plan (4.2, 4.3)
Process	3	Help ensure value-based delivery	m4 Plan (4.4, 4.5)
Process	4	Plan and manage resources	m5 Resource (5.3)
Process	5	Plan and manage procurement	m6 Deliver (6.1)
Process	6	Plan and manage finance	m5 Resource (5.4, 5.5)
Process	7	Plan and optimize quality of products/deliverables	m6 Deliver (6.2)
Process	8	Plan and manage schedule	m5 Resource (5.1, 5.2)
Process	9	Evaluate project status	m6 Deliver (6.3, 6.4)
Process	10	Manage project closure	m6 Deliver (6.5)
Business Environment	1	Define and establish project governance	m7 Govern (7.1)
Business Environment	2	Plan and manage project compliance	m7 Govern (7.2)
Business Environment	3	Manage and control changes	m7 Govern (7.3)
Business Environment	4	Remove impediments and manage issues	m8 Adapt (8.1)
Business Environment	5	Plan and manage risk	m7 Govern (7.4, 7.5)
Business Environment	6	Continuous improvement	m8 Adapt (8.2)
Business Environment	7	Support organizational change	m8 Adapt (8.3)
Business Environment	8	Evaluate external business environment changes	m8 Adapt (8.4)

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INSIGHT · APPROACHES RUN THROUGH EVERY DOMAIN

The outline expects predictive, agile and hybrid ways of working to appear in **every** domain, with roughly 40% of questions set in predictive contexts and 60% in agile or hybrid ones. ZPMP follows the same balance: most lessons show at least one predictive and one agile or hybrid example, and the exam mixes them.

What the 2026 outline emphasises

More weight on the business environment

- About a quarter of the outline, up from a small share in the previous (2021) outline
- Governance, compliance, change, risk and improvement
- Changes outside the project: regulation, technology, market and geopolitics

Value and outcomes

- Value-based delivery and benefits tracking
- Success judged by stakeholder value, not only time, cost and scope
- Integrated planning that keeps checking business value

New themes

- Artificial intelligence as a tool and as a source of change
- Sustainability in planning, quality, risk and compliance
- Knowledge transfer and continuous improvement

The ZPMP Professional Loop: eight stages

1 Frame
Understand value, choose the approach and create a shared vision.

2 Lead
Build and lead the team: roles, leadership, motivation, conflict and coaching.

3 Engage
Engage stakeholders, align and manage expectations, transfer knowledge and communicate.

4 Plan
Integrate the plan, define and break down scope, and steer delivery towards value.

5 Resource
Estimate and schedule the work, secure people and equipment, and manage the money.

6 Deliver
Run delivery with suppliers and built-in quality, measure status honestly and close well.

7 Govern
Set governance, meet compliance duties, control change and manage risk.

8 Adapt
Remove impediments, improve continuously, support change and respond to the outside world.

PRO TIP · WHY A LOOP, NOT A LINE

The stages overlap and repeat. A two-week sprint passes through Plan, Resource, Deliver and Govern many times; Lead and Engage never stop; what you learn in Adapt feeds the next Frame. Use the loop as a thinking checklist, not as a rigid sequence of phases.

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Reading the approach in a situational question

CUE IN THE QUESTION	USUALLY SIGNALS	WHAT GOOD ANSWERS TEND TO RESPECT
Baseline, change control board, Gantt chart, work package	Predictive	Formal change control, baselines, documented approval
Sprint, product owner, backlog, retrospective, velocity	Agile	Team self-management, the product owner's authority over priorities, short feedback loops
Phase gates plus sprints, fixed hardware with evolving software	Hybrid	Each part handled by its own approach, with the joins managed explicitly
No cue at all	Any approach	Choose the answer that analyses first and acts collaboratively

TEMPLATE

Outline coverage tracker

Track your confidence in each outline task as you work through ZPMP. One row per task.

DOMAIN AND TASK	ZPMP LESSON	CONFIDENCE (1–5)	EVIDENCE FROM MY OWN WORK	NEXT ACTION
People 2: conflicts	*2.4*	*3*	*Mediated a design dispute on a clinic app*	*Practise the collaborate-versus-compromise questions*

SCENARIO · DECIDE · STUDY THE WHOLE SPECTRUM

A study partner says: 'Since 60% of the outline is agile or hybrid, I will skip predictive topics such as critical path and earned value and focus only on Scrum.' What is the best advice?

(A) Agree, because agile questions will dominate the exam.

(B) Recommend studying all three approaches, because hybrid questions combine predictive and agile ideas and predictive still makes up a large share.

(C) Suggest studying only predictive topics, because they contain the formulas.

(D) Advise memorising the enablers word for word.

Circle your choice. Answer and feedback: module answer key.

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Key terms

Exam content outline
A published description of the domains and tasks an exam tests, with the share of questions each domain receives.

Task
A specific responsibility of the project manager within a domain.

Hybrid approach
A deliberate mix of predictive and adaptive practices, each applied where it fits best.

Value-based delivery
Ordering and releasing work so that stakeholders receive the most value as early as possible, and checking that value throughout.

Domain
A broad area of project management practice; the July 2026 outline uses People, Process and Business Environment.

Enabler
An illustrative example of the work involved in a task; not an exhaustive list.

ZPMP Professional Loop
ZyeemZia's eight-stage framework for this course: Frame, Lead, Engage, Plan, Resource, Deliver, Govern and Adapt.

KNOWLEDGE CHECK

22. Which domain carries the largest share of the July 2026 outline?

(A) People
(B) Business Environment
(C) Process
(D) All three are weighted equally

23. In the outline, what is an enabler?

(A) A mandatory step that every candidate must perform in order
(B) An illustrative example of the work involved in a task
(C) A tool licensed by PMI for project managers
(D) A domain weight used to score the exam

24. Roughly what share of the outline's questions is set in agile or hybrid contexts?

(A) About 60%
(B) About 20%
(C) About 40%
(D) About 90%

25. Which ZPMP module teaches removing impediments and managing issues?

(A) m6 Deliver
(B) m2 Lead
(C) m7 Govern
(D) m8 Adapt

26. A colleague says passing the ZPMP exam means she can write 'PMP' after her name. What is correct?

(A) She can, because ZPMP follows the same outline
(B) She cannot: ZPMP is an independent ZyeemZia programme, and the PMP® is awarded only by PMI
(C) She can after one year of project experience
(D) She can if she also completes the capstone with distinction

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MODULE 1 · FRAME

Answer key

Knowledge checks

1. C A core-system migration is temporary, unique and creates a change. The others are ongoing, repetitive operations.

2. B Related efforts coordinated for combined benefits form a program. A portfolio is the wider investment mix across many goals.

3. D Progressive elaboration means plans and requirements become more detailed as information and learning increase.

4. A Stable product teams keep knowledge and deliver value continuously. Project management skills are still applied, but in shorter loops.

5. B In agile and hybrid settings some traditional PM duties sit with other roles. The modern PM integrates scope, schedule, money, risk and people across those roles rather than owning every decision personally.

6. D An outcome is the change in behaviour or capability that results from using an output. The app itself is the output.

7. C Tailoring means fitting the approach to the context instead of applying every process by default.

8. B Systems thinking means seeing interactions and ripple effects across parts of the project and organisation.

9. A That domain covers the delivery approach, cadence and phases that suit the product and context.

10. C Principles are guidance for behaviour and decisions; they help most when the situation is new or ambiguous.

11. D Adding usable pieces of value one after another is incremental delivery.

12. C Evolving requirements with partly unknown technology place the work in the complex zone, which favours agile or hybrid approaches.

13. B When late change is very expensive — such as poured concrete or manufactured hardware — careful upfront planning pays off.

14. A A hybrid without deliberate design can carry agile overhead while still getting feedback only at the end.

15. C Iterative discovery followed by predictive execution suits work where building is expensive to change.

16. A Tailoring is continuous: inspect the approach at retrospectives and gates and adjust where needed.

17. B Payback = 30 ÷ 10 = 3 years.

18. C NPV discounts future cash flows to today's value. Simple ROI and payback ignore timing.

19. B A key result is a measurable outcome. The others are tasks or outputs.

20. A The sponsor (or equivalent authority) issues the charter, which authorises the project and the PM.

21. C When people read the same vision differently, break the misunderstanding down to its root cause, here different success measures, and agree evidence everyone shares. Repetition or escalation leaves the cause untouched.

22. C Process carries about 41%, People about 33% and Business Environment about 26%.

23. B Enablers illustrate what a task involves. They are examples, not an exhaustive or mandatory list.

24. A About 40% represents predictive approaches and the remaining 60% is split between agile and hybrid.

25. D Removing impediments and managing issues is Business Environment Task 4, taught in m8 Adapt, lesson 8.1.

26. B ZPMP follows the outline's topics but is independent. It does not award or count as the PMP®, which PMI alone issues.

Scenarios & case studies

> **Scenario · Project or operations?**
> **B** is best. Best. You respect the distinction: operations are sustained with process controls, while a defined improvement (for example automating payroll) is a genuine project with its own end and benefits.

> **Scenario · Delivered — but will it be used?**
> **C** is best. Best. You keep value at the centre, make the trade-off visible and give the decision-maker clear options. This reflects stewardship, stakeholder engagement and value-focused principles.

> **Scenario · Choosing an approach for a bank**
> **C** is best. Best. Each part gets the approach that matches its uncertainty, and an integrated plan manages the dependencies between them.

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Scenario · No charter, but 'just start'

C is best. Best. You respond quickly but secure authority, budget and clarity of purpose — the minimum needed to start responsibly.

Scenario · Study the whole spectrum

B is best. Best. The 60% is split between agile and hybrid, and hybrid work joins predictive and agile parts, so you need both toolkits and the judgement to choose.

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02

MODULE 2 · LEAD

People I: Leading the Project Team

People · Tasks 1–3 (common vision, conflict, leading the team)

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Turn the project vision into team goals, a team charter and clear roles
- Choose leadership styles and delegation levels that empower the team
- Use motivation, emotional intelligence and inclusion to raise team performance
- Resolve conflict and keep ground rules alive
- Lead virtual and hybrid teams with coaching and feedback

LESSONS

1. Shared Vision, Team Expectations & Clear Roles · 28 min
2. Leadership Styles, Empowerment & Team Development · 30 min
3. Motivation, Emotional Intelligence & Inclusive Teams · 27 min
4. Managing Conflict & Ground Rules · 28 min
5. Virtual & Hybrid Teams, Coaching & Feedback · 30 min

MODULE 2 · LEAD · LESSON 1 · 28 MIN

Shared Vision, Team Expectations & Clear Roles

Turn the project vision into team goals, a team charter and unambiguous decision rights

- Translate the project vision into team-level goals and expectations
- Write a team charter with ground rules the team owns
- Build RACI and DACI charts that avoid common errors
- Choose a decision method to suit the stakes

A vision only works when every team member can see how their own work serves it. The project manager's job at team level is to turn the shared vision into **team goals**, **expectations** about how people will work together and **clear roles**, so that nobody has to guess what good looks like or who decides. In agile teams much of this is agreed by the team itself; in predictive projects more of it is set out in plans and role descriptions. In both, people commit more strongly to expectations they helped to write.

From project vision to team commitments

01 Project vision
Why we exist and who benefits

→

02 Team goals
Release, sprint or phase goals that serve the vision

→

03 Team charter
Values, ground rules and ways of working

→

04 Roles and decisions
Who does what and who decides

→

05 Feedback
Reviews and retrospectives check that it all still holds

TEMPLATE **Team charter**

Agree how the team will work together. Write it with the team in the first week and review it when membership or the approach changes.

SECTION	OUR AGREEMENT	HOW WE WILL CHECK IT
Ground rule: meetings	*Cameras optional; one voice at a time; decisions written in the channel within the hour*	*Facilitator checks at the end of each meeting; raised in retrospectives*
Decision rights	*Team decides technical design within architecture standards; product owner decides priority; PM escalates budget changes above 5%*	*Decision log reviewed fortnightly*

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Many project delays are really **decision delays**: nobody knows who decides, so nobody does. Pair clear decision rights with deliberate communication and you remove a huge source of waste.

RACI: one chart, four roles

R

Responsible

Does the work. Can be several people.

A

Accountable

Owns the outcome and approves. **Exactly one per activity.**

C

Consulted

Gives input before the work or decision — two-way.

I

Informed

Kept up to date after the fact — one-way.

TEMPLATE

RACI matrix

Clarify who does what for key deliverables and decisions; check every row has exactly one A.

ACTIVITY / DECISION	SPONSOR	PROJECT MANAGER	PRODUCT OWNER	TECH LEAD	OPS MANAGER
Approve go-live	*A*	*R*	*C*	*C*	*C*
Prioritise backlog	*I*	*C*	*A/R*	*C*	*I*

WATCH OUT · RACI SMELLS

Too many A's on one row (nobody really owns it); no A at all; a person with R everywhere (bottleneck); a column of only I's (why are they on the chart?); and too many C's, which slows every decision.

DEFINITION · DACI FOR DECISIONS

For key decisions, many teams use **DACI**: a **Driver** who runs the decision process, one **Approver** who makes the call, **Contributors** who provide expertise and **Informed** people who hear the outcome. It makes clear that the person organising the decision is not necessarily the one making it.

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Choosing a decision method

Group methods

- **Consensus** – everyone can live with and support it; slow but strong buy-in
- **Consent** – proceed unless someone has a reasoned objection; faster than consensus
- **Majority vote** – quick, but the minority may disengage
- **Fist-of-five** – quick read of support levels

Individual methods

- **Decide with input** – one decider consults, then decides; good default for most work
- **Delegated** – the person closest to the work decides within boundaries
- **Command** – for emergencies when speed matters most

Fist-of-five

FINGERS	MEANING	ACTION
0 (fist)	I object — I cannot support this	Stop; hear the concern
1	Serious concerns	Discuss before proceeding
2	Some concerns I would like to discuss	Discuss; adjust if needed
3	I can live with it	Proceed
4	I support it	Proceed
5	I strongly support it and will champion it	Proceed

SCENARIO · DECIDE · TWO OWNERS, NO DECISION

On a hybrid hospital-records project, the integration lead and the vendor's architect both believe they approve interface changes. Each has rejected the other's proposals twice, and a sprint goal has slipped. What should the project manager do FIRST?

(A) Escalate to the sponsor and ask her to choose the more senior person.

(B) Bring both together with the product owner, agree one accountable approver and the people to consult, and record it in the RACI and decision log.

(C) Let the vendor's architect decide, because the vendor carries the contract risk.

(D) Tell both to take turns approving changes in alternate sprints.

Circle your choice. Answer and feedback: module answer key.

Key terms

RACI
A responsibility assignment matrix showing who is Responsible, Accountable, Consulted and Informed for each activity or decision.

DACI
A decision-rights model naming the Driver who runs the decision, one Approver who decides, Contributors who advise and people who are Informed.

Decision rights
Agreed rules about who may make which decisions and when a decision must be escalated.

Fist-of-five
A quick group technique in which people show zero to five fingers to signal their level of support for a proposal.

Team charter
A short document, written by the team, that records its purpose, values, ground rules, decision rights and ways of working.

Accountable (RACI)
The single person who owns the outcome of an activity or decision and approves it.

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Sprint goal

A short statement of what a sprint aims to achieve, giving the team focus and flexibility in how it gets there.

KNOWLEDGE CHECK

1. Which RACI problem is MOST serious?
   - (A) Two people marked Responsible on one task
   - (B) Several people marked Informed on a decision
   - (C) The sponsor marked Consulted on a deliverable
   - (D) Three people marked Accountable for one decision
2. In a fist-of-five vote, one member shows two fingers. What should happen?
   - (A) Pause to hear and discuss their concerns
   - (B) Proceed immediately, as most members agree
   - (C) Ignore it, because the majority view wins
   - (D) Remove the member from the decision
3. In DACI, who makes the final decision?
   - (A) Driver
   - (B) Approver
   - (C) Contributor
   - (D) Informed
4. Why should a team write its own charter rather than receive one from the project manager?
   - (A) Because team charters have no legal standing if the PM writes them
   - (B) Because it saves the project manager time during the first sprint
   - (C) Because sponsors are not allowed to read team-level documents
   - (D) Because people commit more strongly to expectations they helped to shape
5. A predictive construction team and an agile app team share a project. What is the BEST way to set team-level expectations?
   - (A) Agree common standards, such as responsiveness and escalation, then let each team tailor its own working practices
   - (B) Make the app team follow the construction team's procedures in full
   - (C) Let each team work in isolation and coordinate only at go-live
   - (D) Write one detailed procedure for both teams and enforce it strictly

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MODULE 2 · LEAD · LESSON 2 · 30 MIN

Leadership Styles, Empowerment & Team Development

Serve the team, adapt your style, empower people and build the safety that high performance needs

- Explain servant leadership and apply situational leadership to individual tasks
- Choose a level of delegation that empowers without abandoning
- Recognise team development stages and build psychological safety
- Represent the voice of the team to stakeholders

Plans do not deliver projects; people do. **Lead**, stage 2 of the ZPMP Professional Loop, is where a project manager earns the trust that makes every other stage work. In modern practice, the PM is less a boss and more a **multiplier**: someone who makes a group of capable people more effective together than apart.

Servant leadership in practice

S1

Remove impediments

Clear the blockers the team cannot clear themselves — access, approvals, missing information, outside interference.

S2

Shield focus

Protect the team from constant interruptions and unprioritised requests; channel demand through the backlog or change process.

S3

Grow people

Invest in skills, give stretch opportunities and share credit publicly.

S4

Listen first

Ask before telling. The people closest to the work usually see the problem most clearly.

S5

Hold the purpose

Keep reminding everyone why the work matters and who benefits from it.

INSIGHT · SERVANT DOES NOT MEAN PASSIVE

A servant leader still sets direction, holds standards and makes hard calls. The difference is the motive — the leader's authority is used for the team's success and the project's value, not for personal status.

No single style fits every person or moment. **Situational leadership** (associated with Hersey and Blanchard) matches the leader's mix of direction and support to each person's competence and commitment on a specific task.

Situational leadership: match style to readiness

PERSON'S STATE ON THIS TASK	LEADER STYLE	WHAT IT LOOKS LIKE
New and enthusiastic, low skill	Directing	Clear instructions, close follow-up, frequent check-ins
Some skill, confidence dipping	Coaching	Still directive, plus explanation, encouragement and two-way talk
Capable but hesitant or unsure	Supporting	Less direction; ask questions, build confidence, share decisions
Skilled and committed	Delegating	Agree outcomes, then step back; be available when asked

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Seven levels of delegation

1 **Tell**
I decide and explain the decision.

2 **Sell**
I decide and try to win your support.

3 **Consult**
I ask for your input, then I decide.

4 **Agree**
We decide together.

5 **Advise**
You decide; I give my view first.

6 **Inquire**
You decide and tell me afterwards.

7 **Delegate**
You decide; I do not need to know.

PRO TIP · EMPOWER ON PURPOSE
Agree the delegation level **per decision area** with the team, for example level 7 for coding standards, level 4 for release scope and level 3 for vendor choice, and write it in the team charter. Empowerment fails when people are told they are empowered but their decisions are quietly overturned.

Tuckman's team development stages

01 **Forming**
Polite, unsure; leader provides clarity on goals and roles

→

02 **Storming**
Friction over ideas and approaches; leader coaches and facilitates conflict

→

03 **Norming**
Agreements form; leader supports and reinforces working norms

→

04 **Performing**
Self-managing and productive; leader delegates and removes blockers

→

05 **Adjourning**
Work ends; leader recognises contributions and supports transitions

Traits of high-performing teams

PS **Psychological safety**
People can ask questions, admit mistakes and challenge ideas without fear of humiliation or punishment.

CG **Clear, shared goal**
Everyone can state what success looks like and why it matters.

DT **Dependability**
People do what they say, on time and to standard.

RS **Clear roles & structure**
Who decides what, and how work flows, is understood.

CF **Constructive conflict**
Debate ideas openly, then commit — disagreement is about the work, never the person.

SO **Self-organisation**
The team decides how to do its work within agreed boundaries.

Five habits that build psychological safety

1. **Frame work as learning** Say early: 'This is new; we will get things wrong and that is how we find the right answer.'
2. **Model fallibility** Admit your own mistakes and say 'I might be missing something — what do you see?'

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3. **Ask genuine questions** Invite the quietest voices first; use silent brainstorming.
4. **Respond productively** Thank people who bring bad news; focus on the fix, not the blame.
5. **Run blameless reviews** After incidents, ask what in the system allowed this, not who is at fault.

> **INSIGHT · REPRESENT THE VOICE OF THE TEAM**
> When a steering group pushes for a date the team believes is unrealistic, the project manager carries the team's **evidence**, such as velocity, defect trends and dependencies, into the room and presents options. Speaking for the team builds trust; promising what the team cannot deliver destroys it.

SCENARIO · DECIDE · THE QUIET EXPERT

In stand-ups, a newly joined data engineer never raises blockers. Yesterday you learned she has been stuck for three days waiting for database access. What should the project manager do FIRST?

(A) Remind the whole team at the next stand-up that blockers must be raised immediately.

(B) Report to her line manager that she is not following the process.

(C) Assign her a buddy to monitor her work.

(D) Fix the access, then speak with her privately to understand what made raising it hard and agree an easy way to flag blockers.

Circle your choice. Answer and feedback: module answer key.

Key terms

Servant leadership
A leadership approach that focuses on serving the team — removing obstacles, growing people and enabling them to deliver value.

Tuckman model
A model of team development through forming, storming, norming, performing and adjourning.

Empowerment
Giving people the authority, information and support to make decisions about their own work.

Transformational leadership
Leading by inspiring people with a compelling purpose and encouraging them to grow beyond current expectations.

Psychological safety
A shared belief within a team that it is safe to take interpersonal risks such as speaking up, asking questions or admitting mistakes.

Situational leadership
Matching the mix of direction and support to a person's competence and commitment on a specific task.

Delegation level
An agreed degree of decision authority, from the leader deciding alone to the team deciding without reporting back.

KNOWLEDGE CHECK

6. A team member is highly skilled and motivated on a task. Which situational style fits best?
   (A) Directing (B) Coaching
   (C) Supporting (D) Delegating
7. Team members are openly arguing about the technical approach two weeks after kick-off. Which Tuckman stage is this most likely?
   (A) Forming (B) Storming
   (C) Norming (D) Adjourning

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8. Which behaviour MOST directly builds psychological safety?

   (A) Publicly naming who caused each defect to deter repeats

   (B) Keeping every meeting short and focused

   (C) Thanking someone who reports a problem early

   (D) Using a strict RACI for every decision

9. Which statement about servant leadership is accurate?

   (A) The leader uses authority to enable the team's success

   (B) The leader avoids making decisions for the team

   (C) It only works in Scrum teams with a dedicated coach

   (D) It means doing the team's work for them when needed

10. A team was told it is empowered to choose its testing tools, but the PM has overruled its last three choices without discussion. What is the MOST likely result?

    (A) Higher velocity because decisions are made faster

    (B) Better tool choices, since the PM has more experience

    (C) Loss of trust and fewer decisions taken by the team

    (D) No effect, because tool choice is a minor matter

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MODULE 2 · LEAD · LESSON 3 · 27 MIN

Motivation, Emotional Intelligence & Inclusive Teams

Understand what drives people, lead with self-awareness and make varied skills and perspectives count

- Distinguish intrinsic and extrinsic motivation and apply autonomy, mastery and purpose
- Use hygiene-factor thinking to diagnose dissatisfaction
- Apply the components of emotional intelligence to everyday project moments
- Support a team's varied experiences, skills and perspectives

Project managers often lead people who do not report to them and whose pay they do not set. That makes understanding motivation essential: you must create conditions in which people **want** to do great work.

Two sources of motivation

Extrinsic

- Comes from outside: pay, bonuses, titles, deadlines, praise
- Works well for routine, well-defined tasks
- Can crowd out interest in creative work if overused
- Its effect fades once the reward is expected

Intrinsic

- Comes from the work itself: interest, growth, meaning
- Drives quality and creativity in complex work
- Sustains effort when no one is watching
- Grows when people have choice and see impact

Autonomy, mastery, purpose

AU

Autonomy

Let people choose **how** they do their work — methods, tools, sequence — within agreed outcomes and constraints.

MA

Mastery

Provide challenges slightly beyond current skill, plus time to learn, feedback and visible progress.

PU

Purpose

Connect tasks to the people who benefit: show the farmer using the app, the patient who books in two minutes.

DEFINITION · HYGIENE FACTORS VS MOTIVATORS

A two-factor view (associated with Frederick Herzberg) separates **hygiene factors** — pay, working conditions, policies, job security — whose absence causes dissatisfaction, from **motivators** — achievement, recognition, responsibility, growth — that create real engagement. Fixing hygiene stops complaints; it rarely creates enthusiasm on its own.

Diagnose before you motivate

WHAT YOU OBSERVE	LIKELY ROOT CAUSE	USEFUL RESPONSE
Complaints about laptops, overtime pay, seating	Hygiene problem	Fix or escalate the condition; do not answer with a pep talk
Work is done but with no initiative	Low autonomy or purpose	Delegate decisions; connect work to users
Bored senior engineer	No mastery challenge	Offer a stretch problem or mentoring role
Drop in energy after a reorganisation	Uncertainty and loss of purpose	Explain the why, clarify roles, listen

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Emotional intelligence (EI) is the ability to recognise and manage your own emotions and to read and influence the emotions of others. Popularised by Daniel Goleman, it is often grouped into five components.

Five components of emotional intelligence

SA
Self-awareness
Know your moods, triggers, strengths and limits — e.g. noticing you get curt when a vendor misses dates.

SR
Self-regulation
Pause before reacting; choose the response. 'Let me come back to you in an hour' is a powerful sentence.

MO
Motivation
Drive that comes from inner standards and the goal, staying constructive under pressure.

EM
Empathy
Understand others' perspectives and feelings — including the sponsor's fears and the user's frustrations.

SS
Social skills
Build networks, influence, manage conflict and inspire — EI turned outward into action.

A 90-second reset when emotions spike

1. **Notice** Name the feeling privately: 'I am angry because I feel blamed.'
2. **Breathe and delay** Slow breathing; delay replies to heated messages.
3. **Get curious** Ask: 'What pressure might the other person be under?'
4. **Respond to the problem** Return to facts and next steps.

SCENARIO · DECIDE · THE ANGRY EMAIL

At 11 p.m. a client director emails the whole steering group saying your team is 'incompetent' because a demo crashed. You feel furious. What is the BEST action?

(A) Wait until morning, gather facts, then call the director to listen, acknowledge the impact and share a recovery plan.

(B) Reply-all immediately, pointing out the client's own environment caused the crash.

(C) Ignore the email; it will blow over.

(D) Forward it to your team so they know how the client feels.

Circle your choice. Answer and feedback: module answer key.

Making varied experiences, skills and perspectives count

- Map skills openly, including those outside job titles, and use them when planning work.
- Rotate who facilitates meetings and retrospectives so that quieter voices lead too.
- Share agendas and material in advance so that people working in a second language can prepare.
- Offer more than one way to contribute: speaking, writing, anonymous input or one-to-one.
- Ask how each person prefers to work, instead of assuming from age, gender, culture or role.
- Pair experienced and newer members, in both directions, so that knowledge flows both ways.

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CASE STUDY · UNIVERSITY LMS MIGRATION: MOTIVATION SLUMP

A private university in Sylhet is moving 600 courses to a new learning management system. Six instructional designers were seconded part-time from faculties. After two months, quality is falling: courses are copied over without redesign, and designers skip review meetings. In one-to-ones they say: 'This is admin work — nobody will notice if it is good', and two mention they still have full teaching loads.

1. Which issues here are hygiene problems and which are motivator problems?

2. Propose three actions using autonomy, mastery and purpose.

3. What should the PM escalate, and to whom?

Model answer: module answer key.

Key terms

Intrinsic motivation
Motivation that comes from the satisfaction of the work itself — interest, growth and meaning.

Extrinsic motivation
Motivation driven by outside rewards or pressures such as pay, bonuses, deadlines or praise.

Motivator
A factor such as achievement, recognition or responsibility that creates genuine satisfaction and effort.

Inclusion
Working practices that let every team member contribute fully and be heard, whatever their background.

Emotional intelligence
The ability to recognise, understand and manage one's own emotions and to recognise and influence others' emotions.

Hygiene factor
A condition such as pay, policy or working environment whose absence causes dissatisfaction but whose presence does not by itself motivate.

Empathy
Understanding another person's feelings and perspective and taking them into account.

KNOWLEDGE CHECK

11. Team members complain that their overtime has not been paid for two months. In two-factor terms, this is…

(A) a motivator (achievement) issue
(B) a lack-of-purpose issue
(C) a mastery and growth issue
(D) a hygiene factor issue

12. Which action BEST increases autonomy?

(A) Giving a detailed checklist and method for every task
(B) Letting the team choose its own testing approach within standards
(C) Adding a team bonus for finishing work faster than planned
(D) Holding daily status calls to review each person's tasks

13. A PM notices she gets defensive whenever QA reports defects. Recognising this pattern is an example of…

(A) empathy
(B) social skill
(C) self-awareness
(D) situational leadership

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14. For creative, complex knowledge work, which approach is MOST likely to sustain high-quality effort?

(A) Building autonomy, mastery and purpose
(B) Strict monitoring of hours and output
(C) Public ranking of individual performance
(D) Large bonuses tied to output volume

15. In a hybrid team, two members who work in a second language rarely speak in design meetings but send strong ideas by message afterwards. What should the PM do?

(A) Ask them to speak up more in meetings and set a speaking target
(B) Share agendas early and add a written round to each design meeting
(C) Stop holding live design meetings and decide everything by email
(D) Leave it, because their ideas still reach the team eventually

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MODULE 2 · LEAD · LESSON 4 · 28 MIN

Managing Conflict & Ground Rules

Find the source of a conflict, choose the right mode and keep the team's ground rules alive

- Identify common sources of project conflict and analyse their context
- Choose among the five conflict-handling modes
- Facilitate a conflict conversation to an agreed resolution
- Handle ground-rule breaches fairly and promptly

Conflict is normal on projects: competing priorities, scarce resources, technical opinions, schedules and personalities. Unmanaged, it drains energy. Managed well, it surfaces better ideas. The skill lies in choosing the right **mode** and handling it early, privately and on the facts.

Common sources of conflict and what to analyse

SOURCE	TYPICAL EXAMPLE	CONTEXT TO ANALYSE BEFORE ACTING
Priorities	Two departments want their feature first	Business value of each, strategy, who decides priority
Scarce resources	One specialist needed by two projects	Dates, alternatives, the cost of delay on each side
Schedules	Testers want more time; sales wants the launch date	Fixed versus flexible constraints, risk of each option
Technical opinions	Two architects favour different databases	Evidence, reversibility of the choice, who is accountable
Working styles	Detailed planners versus fast experimenters	Team charter, cultural norms, past friction
Personalities	Sharp remarks in meetings	Whether a ground rule is being broken; private or public setting

Thomas–Kilmann modes: when to use each

MODE	ASSERTIVE / COOPERATIVE	WHEN IT FITS	RISK IF OVERUSED
Collaborating (problem-solving)	High / high	Important issue, time available, need commitment from both sides	Slow; not worth it for trivial issues
Competing (forcing)	High / low	Emergencies, safety, legal or ethical non-negotiables	Resentment, lost ideas, low buy-in
Compromising (reconciling)	Medium / medium	Equal power, time pressure, temporary fix acceptable	Both sides partly unhappy; may mask the real problem
Accommodating (smoothing)	Low / high	Issue matters more to the other side; preserving relationship; you are wrong	Your concerns pile up unheard
Avoiding (withdrawing)	Low / low	Trivial issue, emotions need cooling, someone else is better placed	Problems grow while ignored

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INSIGHT · COLLABORATE IS THE DEFAULT, NOT THE ONLY ANSWER

Situational exam questions usually favour **collaborating / problem-solving** because it addresses root causes. But a safety violation calls for forcing, and a two-minute disagreement over font colours calls for accommodating or avoiding. Match the mode to the stakes and time.

Facilitating a conflict conversation

1. **Meet early and privately** Address it before it hardens; public settings make people defend positions.
2. **Set ground rules** One person speaks at a time; focus on the problem; assume good intent.
3. **Hear each side fully** Ask each person to describe the situation and what matters to them. Summarise back.
4. **Find shared goals and facts** 'We both want the release to be stable by 15 June.' Separate facts from interpretations.
5. **Generate options together** Brainstorm before judging. Look for options that serve both sets of interests.
6. **Agree and follow up** Write down the agreement, owners and a check-in date.

When a ground rule is broken

01	→	02	→	03	→	04	→	05	→	06
Notice Name the behaviour, not the person		**Talk privately** Describe the impact and listen		**Refer to the charter** Remind them of what the team agreed		**Agree a correction** Specific and time-bound		**Escalate if repeated** Follow the agreed process		**Review the rule** Does it still fit the team?

PRO TIP · MAKE THE RULES OF CONFLICT KNOWN

Share how conflicts will be handled with the team **and** with external partners such as vendors and client teams: who mediates, how fast issues are raised and when they go to the sponsor. When everyone knows the process in advance, disagreements stay about the work rather than about fairness.

CASE STUDY · 5G ROLLOUT: CIVIL WORKS VS RADIO PLANNING

A mobile operator is rolling out 5G sites across Dhaka and Narayanganj. The radio planning lead wants to change antenna positions at 40 sites after new drive-test data. The civil works contractor manager refuses: towers are already reinforced to the old design, and every change costs money and two weeks. The two have stopped speaking and copy you on increasingly hostile emails. The regulator's coverage milestone is in 10 weeks.

1. What are the likely interests behind each position?

2. Which conflict mode would you aim for and why?

3. Outline the steps you would take this week.

Model answer: module answer key.

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SCENARIO · DECIDE · TWO SENIORS, ONE TESTER

Two functional managers each claim the same senior tester for the same two weeks. Both are senior to you and are getting heated in a meeting. What should you do FIRST?

(A) Pause the debate, meet both privately to understand the interests and constraints, then bring options to a joint conversation.

(B) Decide yourself, based on whichever project you think matters more.

(C) Escalate straight to the portfolio board.

(D) Suggest they each take the tester for one week to split it evenly.

Circle your choice. Answer and feedback: module answer key.

SCENARIO · DECIDE · THE REPEATED INTERRUPTER

Your distributed team agreed 'one voice at a time; raise a hand in the meeting tool'. A senior engineer in the office keeps talking over remote colleagues, and two of them have stopped contributing. What should you do FIRST?

(A) Remind everyone of the ground rule at the start of the next meeting without naming anyone.

(B) Speak with the engineer privately, describe the specific behaviour and its impact, and agree what will change.

(C) Report the engineer to his line manager for a formal warning.

(D) Switch all meetings to text chat so interruptions cannot happen.

Circle your choice. Answer and feedback: module answer key.

Key terms

Thomas–Kilmann modes
Five conflict-handling approaches — competing, collaborating, compromising, avoiding and accommodating — mapped on assertiveness and cooperativeness.

Ground rules
Explicit expectations about acceptable behaviour that a team agrees and enforces itself.

Collaborating (problem-solving)
A conflict mode that works with both sides to find a solution meeting everyone's interests; slow but durable.

Competing (forcing)
A conflict mode that imposes one view; fitting for emergencies or safety and compliance matters.

Accommodating (smoothing)
A conflict mode that gives way to the other side, emphasising agreement; useful when the issue matters more to them.

Compromising
A conflict mode in which each side gives up something to reach a quick, acceptable agreement.

Avoiding (withdrawing)
A conflict mode that postpones or steps away from the issue; acceptable for trivial matters or to let tempers cool.

KNOWLEDGE CHECK

16. A team member is about to deploy code that bypasses a mandatory security control. Which conflict mode is appropriate?

(A) Accommodating / smoothing
(B) Avoiding
(C) Competing / forcing
(D) Compromising

17. Which mode produces a win-win outcome and long-term commitment, but takes the most time?

(A) Compromising
(B) Collaborating
(C) Accommodating
(D) Avoiding

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18. Two team members disagree mildly about a variable-naming convention. The release is tomorrow. Best mode?

(A) Collaborate through a two-hour design workshop
(B) Escalate the decision to the project sponsor
(C) Force a company-wide naming standard immediately
(D) Accommodate or defer, given the low stakes

19. Before choosing how to resolve a dispute between sales and testers over a launch date, what should the project manager analyse FIRST?

(A) Which of the two managers is more senior in the organisation
(B) Which group complained to the sponsor first
(C) How quickly a compromise can be announced to both groups
(D) Which constraints are fixed, which are flexible, and the risk of each option

20. A team member breaks the agreed ground rule on response times for the third time after two private conversations. What is the BEST next step?

(A) Follow the escalation step the team agreed in its charter
(B) Ignore it to protect the relationship
(C) Rewrite the ground rule so the breach no longer counts
(D) Raise it in front of the whole team at the stand-up

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MODULE 2 · LEAD · LESSON 5 · 30 MIN

Virtual & Hybrid Teams, Coaching & Feedback

Lead people you rarely see, across cultures and time zones, and help them grow

- Design working agreements for virtual and hybrid teams, including async-first practices
- Adapt communication across cultural dimensions without stereotyping
- Use GROW coaching and SBI feedback in one-to-ones

In 2026 most project teams are at least partly distributed — a developer in Khulna, a product owner in Dhaka, a vendor in Kuala Lumpur and a client in Frankfurt. Distance strips away the hallway conversations that used to carry context, so leaders must design what used to happen naturally.

Remote and hybrid essentials

WA

Working agreement

A short, team-written page: core hours, response-time expectations, which channel for what, camera norms, how decisions are recorded.

ASY

Async-first

Default to written updates, recorded demos and shared documents so people in different time zones can contribute; use meetings for what truly needs real time.

OVR

Overlap windows

Protect two to three hours of shared time for collaboration; rotate inconvenient meeting times fairly.

EQ

Hybrid equity

If one person is remote, everyone joins from their own screen so remote people are not second-class participants.

CON

Connection

Deliberate informal time — virtual coffee, show-and-tell, occasional in-person meet-ups at key moments such as kick-off.

Remote team working agreement checklist

☐ Core collaboration hours defined in every member’s time zone

☐ Which tool for urgent, normal and FYI messages is agreed

☐ Expected response times for each channel are written down

☐ Decisions are recorded in one agreed place

☐ Meeting norms: agenda in advance, notes and actions shared within 24 hours

☐ Public holidays and religious observances across locations are on the shared calendar

☐ Inconvenient meeting times rotate fairly

☐ A ‘how to reach me’ and focus-time preference is published by each member

☐ Security rules for devices, networks and shared files are clear

☐ The agreement is reviewed at least every quarter or when membership changes

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WATCH OUT · MEETING TOOLS THAT SUMMARISE AND TRANSLATE

Many collaboration tools now transcribe, summarise and translate meetings automatically. They help distributed teams, but agree as a team **when recording is allowed**, who checks a summary before it is shared, and where transcripts are stored. A wrong summary of a decision travels fast.

Culture shapes how people give feedback, disagree, make decisions and view hierarchy and time. Research frameworks such as Geert Hofstede's dimensions and Erin Meyer's culture map describe national **tendencies** — useful as hypotheses, never as labels for individuals.

Cultural dimensions that show up on projects

DIMENSION	ONE END	OTHER END	WHAT THE PM CAN DO
Power distance	Hierarchy expected; juniors defer	Flat; anyone challenges anyone	Invite input explicitly and privately; use anonymous tools
Communication context	High-context: meaning in what is unsaid	Low-context: meaning in explicit words	Confirm understanding in writing; ask open questions
Feedback style	Indirect, softened	Direct, blunt	Agree team feedback norms; explain intent
Time orientation	Flexible, relationships first	Linear, schedule first	Make deadline importance and reasons explicit
Uncertainty avoidance	Prefers rules and detailed plans	Comfortable with ambiguity	Balance clarity of process with room to experiment

WATCH OUT · GENERALISE CAREFULLY

A Bangladeshi engineer educated in Germany and working for a Singapore firm will not fit a national average. Use dimensions to notice and ask — 'How do you prefer to receive feedback?' — never to assume.

GROW coaching model

G

Goal

What do you want to achieve? 'By the end of this conversation, what would be useful?'

R

Reality

What is happening now? What have you tried? What is getting in the way?

O

Options

What could you do? What else? What would someone you admire do?

W

Will / way forward

What will you do, by when? How will you know it worked? What support do you need?

SBI feedback

MEASURE	FORMULA	WHAT IT TELLS YOU
Situation	`When and where`	'In yesterday's client demo…'

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MEASURE	FORMULA	WHAT IT TELLS YOU
Behaviour	`Observable action — not a judgement`	'...you answered the integration question with specific test data...'
Impact	`Effect on you, the team or the outcome`	'...and the client agreed to sign off the milestone on the spot.' Works for praise and for correction.

PRO TIP · COACH BY ASKING, NOT TELLING

A good coaching ratio is mostly questions. The answer someone finds themselves sticks better than the one you hand them — and it builds their capability for the next problem.

SCENARIO · DECIDE · FEEDBACK ACROSS CULTURES

A talented analyst from a high-power-distance background never challenges the client's architect, even when the analyst's data shows the architect is wrong. What is the BEST approach?

(A) Tell the analyst in the next team meeting that they must speak up more.

(B) Present the analyst's findings yourself without crediting them.

(C) Speak privately, use SBI to recognise the strong analysis, explore what makes challenging difficult (GROW), and agree a supported way — e.g. you introduce the data and invite them to present it.

(D) Accept that this is their culture and do nothing.

Circle your choice. Answer and feedback: module answer key.

Key terms

Working agreement
A team-created set of norms describing how members will collaborate, communicate and make decisions.

SBI feedback
A feedback structure describing the Situation, the observable Behaviour and its Impact.

Virtual team
A team whose members work mainly from different locations and collaborate through digital tools.

Coaching
Helping someone improve by asking questions that lead them to their own insight and action, rather than giving instructions.

GROW model
A coaching structure: Goal, Reality, Options, Will (way forward).

Async-first
A working style in which written, recorded or tool-based updates are the default and live meetings are kept for discussion and decisions.

Power distance
The degree to which people accept and expect unequal distribution of power and deference to seniority.

KNOWLEDGE CHECK

21. A hybrid team holds meetings in a room with four people while two join remotely and struggle to follow. What is the BEST fix?

(A) Have everyone join from their own device for equal participation

(B) Ask remote members to come into the office for meetings

(C) Record the meetings so remote members can catch up afterwards

(D) Hold separate meetings for the in-room and remote members

22. In SBI, which is a behaviour rather than a judgement?

(A) You were unprofessional in front of the client

(B) You arrived 20 minutes after the client call started

(C) You have a negative attitude with clients

(D) You do not care about the project or the client

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23. Which GROW question belongs to the ‘Reality’ stage?

(A) What do you want to achieve?
(B) What will you do by Friday?
(C) What other options are there?
(D) What have you already tried?

24. Why is ‘async-first’ useful for teams spread over several time zones?

(A) It removes the need for any live meetings across time zones
(B) It is cheaper than paying for video-conferencing tools
(C) It lets people contribute without being online at the same time
(D) It hides time-zone differences from managers, clients and partners

25. A distributed team's meeting tool produces automatic summaries. Last week a summary recorded a decision the team had not actually made. What is the BEST fix?

(A) Ban the tool for all meetings
(B) Ask the vendor to improve the tool's accuracy
(C) Accept small errors as the price of speed
(D) Agree that a named person checks each summary before it is shared

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · LEAD

Answer key

Knowledge checks

1. D Each activity needs exactly one Accountable person; multiple A's blur ownership.

2. A Two or fewer signals concerns worth discussing before proceeding.

3. B The Approver makes the call; the Driver runs the process.

4. D Co-created agreements are owned by the team, so members hold each other to them. A charter handed down tends to be ignored.

5. A Shared standards keep the joins working, while each team tailors practices to its own approach. Forcing one style on both, or isolating them, causes friction.

6. D High competence and high commitment call for delegating: agree outcomes and step back.

7. B Friction over ideas and ways of working is typical of storming.

8. C Rewarding candour signals it is safe to speak up.

9. A Servant leaders still lead; they direct their influence toward enabling the team.

10. C Declared empowerment that is quietly withdrawn damages trust and teaches people to wait for the manager. Agree the delegation level and respect it.

11. D Pay is a hygiene factor; its absence causes dissatisfaction.

12. B Autonomy is about choice in how work is done within agreed outcomes.

13. C Recognising one's own emotional triggers is self-awareness.

14. A Intrinsic drivers sustain quality in complex work; heavy extrinsic incentives can backfire.

15. B Offering preparation time and more than one way to contribute brings their ideas into decisions while they are still useful.

16. C Safety, security and legal non-negotiables justify a firm, directive stance.

17. B Collaboration searches for solutions satisfying both parties' interests fully.

18. D Low-stakes issues under time pressure do not justify heavy collaboration; revisit later if needed.

19. D Understanding the context of the conflict, especially fixed versus flexible constraints and risk, comes before choosing a mode or a resolution.

20. A Repeated breaches after private conversations call for the agreed escalation path. Public criticism damages safety, and dropping the rule tells the team rules do not matter.

21. A Equal participation through individual devices removes the in-room advantage.

22. B Behaviour is observable and specific; the others are interpretations.

23. D Reality explores the current situation and what has been tried.

24. C Async communication respects time zones while keeping meetings for collaboration that truly needs real time.

25. D Automatic summaries are useful, but a human owner should verify decisions before they are circulated. Banning the tool loses the benefit; accepting errors spreads them.

Scenarios & case studies

> **Scenario · Two owners, no decision**
> **B** is best. Best. Exactly one person should be accountable for each decision. Agreeing it openly and recording it removes the repeat conflict.

> **Scenario · The quiet expert**
> **D** is best. Correct. Remove the impediment, then build safety one-to-one. The goal is to learn what got in the way, not to blame.

> **Scenario · The angry email**
> **A** is best. Correct. Self-regulation first, then empathy and a fact-based plan — ideally in a conversation, not an email war.

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Case study · University LMS migration: motivation slump
Hygiene: the unchanged teaching load (workload and fairness) — a structural issue the PM should escalate to the deans and the sponsor, asking for workload relief or a revised timeline. Motivators: lack of purpose ('nobody will notice') and no mastery. Actions: (1) **Purpose** — share student feedback and data from pilot courses; invite students to demo days. (2) **Autonomy** — let designers choose which redesign patterns to use within a quality rubric they help write. (3) **Mastery** — run short workshops on active-learning design and showcase excellent courses with credit to their designers. Recognition by the Pro-VC at a faculty meeting costs nothing and signals that the work matters.

Case study · 5G rollout: civil works vs radio planning
Interests — radio planning: meet coverage and quality targets, avoid poor customer experience; civil works: protect margin, avoid unpaid rework, keep crews productive. Shared: hit the regulator milestone. Aim to **collaborate**: the stakes are high and both groups' commitment is needed. Steps: (1) meet each lead privately to hear concerns; (2) joint session with data — which of the 40 sites truly need changes for coverage, and which are nice-to-have? (3) generate options: change only sites critical to the milestone now and batch others later; use objective criteria (contracted change rates) for payment; re-sequence crews to absorb some work; (4) put the agreed approach through change control; (5) agree a weekly sync between both leads. If no agreement is possible before a hard deadline, escalate with options to the sponsor.

Scenario · Two seniors, one tester
A is best. Correct. Understand interests first; often a split schedule, a substitute or a re-sequence solves it. Escalate with options only if they cannot agree.

Scenario · The repeated interrupter
B is best. Best. A prompt, private conversation that refers to the team's own agreement respects the person and protects the team.

Scenario · Feedback across cultures
C is best. Correct. Private, respectful coaching builds confidence and gives a safe path to contribute.

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03

MODULE 3 · ENGAGE

People II: Stakeholders, Expectations, Knowledge & Communication

People · Tasks 4–8 (engage stakeholders, align and manage expectations, knowledge transfer, communication)

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Identify, analyse and engage stakeholders, building trust and influence
- Align stakeholder expectations early and manage them throughout delivery
- Monitor internal and external customer satisfaction and respond to gaps
- Plan and verify knowledge transfer
- Plan communication and reporting that support decisions and governance

LESSONS

1. Identifying, Analysing & Engaging Stakeholders · 30 min
2. Aligning Stakeholder Expectations · 26 min
3. Managing Expectations & Customer Satisfaction · 26 min
4. Knowledge Transfer · 25 min
5. Planning & Managing Communication · 28 min

MODULE 3 · ENGAGE · LESSON 1 · 30 MIN

Identifying, Analysing & Engaging Stakeholders

Find everyone who matters, understand their power and interest, and build the trust that moves them

- Identify stakeholders systematically, including the less obvious ones
- Analyse stakeholders with the power–interest grid and the salience model
- Assess current and desired engagement and plan how to close the gap
- Build trust and influence without formal authority

Projects rarely fail because of technology alone. They fail because someone important was missed, misunderstood or ignored until it was too late. A **stakeholder** is any person, group or organisation that can affect, be affected by, or believe they are affected by the project. Notice the last part: **perception counts**.

Identify stakeholders in four sweeps

1. **Start with the charter** List the sponsor, customers, users, team, suppliers and anyone named in contracts or the business case.
2. **Follow the value chain** Who supplies inputs, who uses outputs, who runs the result, who pays, who regulates?
3. **Ask around** Interview early stakeholders: 'Who else should I speak to? Who might object?'
4. **Look outward** Regulators, unions, community groups, media, partner NGOs, future users and those who lose something from the change.

WATCH OUT · THE QUIET STAKEHOLDERS

Some of the most influential stakeholders never attend your meetings: the IT security officer who can block go-live, the union representative, the regulator's desk officer, the night-shift supervisor whose process you are changing. Find them early.

The **power–interest grid** plots each stakeholder by how much power they have over the project and how interested they are in it. It gives a quick first view of how much attention each group needs.

Power–interest grid: four quadrants

QUADRANT	POWER / INTEREST	TYPICAL STAKEHOLDERS	STRATEGY
Manage closely	High power, high interest	Sponsor, key customer, steering committee	Involve in decisions, consult often, no surprises
Keep satisfied	High power, low interest	Finance director, regulator, CIO	Brief concisely, consult on what touches them, avoid overloading
Keep informed	Low power, high interest	End users, field staff, community groups	Regular, clear updates and channels for feedback
Monitor	Low power, low interest	Peripheral departments, general public	Light-touch updates; watch for changes in position

The **salience model** adds a third dimension. It looks at **power** (ability to impose their will), **legitimacy** (how appropriate their involvement is) and **urgency** (how time-sensitive and critical their claims are). Stakeholders who have all three are **definitive** and need immediate attention.

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Salience model — the seven classes

P

Dormant

Power only. Could matter a lot if they become interested.

L

Discretionary

Legitimacy only. Fair claim but little pressure; good-citizen engagement.

U

Demanding

Urgency only. Noisy but without power or legitimacy; manage, do not ignore.

PL

Dominant

Power and legitimacy. Formal influence — often senior managers or regulators.

PU

Dangerous

Power and urgency without legitimacy. Can disrupt — handle with care.

LU

Dependent

Legitimacy and urgency, but depend on others' power — e.g. affected communities.

PLU

Definitive

All three. Highest priority for engagement.

TEMPLATE Stakeholder register

One living table that captures who your stakeholders are and how you will engage them.

STAKEHOLDER	ROLE / ORGANISATION	INTEREST AND EXPECTATIONS	POWER / INTEREST	CURRENT → DESIRED ENGAGEMENT	ENGAGEMENT APPROACH	OWNER
Head of Nursing	*Hospital operations*	*Ward workflows stay safe during move*	*High / High*	*Neutral → Supportive*	*Monthly design walk-through; nurse champions on each ward*	*PM*
City fire authority	*Regulator*	*Compliance with fire code*	*High / Low*	*Unaware → Supportive*	*Early pre-application meeting; formal submissions at design gate*	*Facilities lead*

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Engagement assessment matrix (C = current, D = desired)

STAKEHOLDER	UNAWARE	RESISTANT	NEUTRAL	SUPPORTIVE	LEADING
Head of Nursing			C	D	
Ward clerks	C		D		
Hospital CEO (sponsor)				C	D
Junior doctors' association		C	D		

DEFINITION · ENGAGEMENT LEVELS

Unaware – does not know about the project. **Resistant** – aware and opposed. **Neutral** – aware, neither for nor against. **Supportive** – aware and in favour. **Leading** – actively championing success. The gap between C and D drives your engagement actions.

A light engagement plan

STAKEHOLDER	GAP (C → D)	KEY MESSAGE	METHOD AND FREQUENCY	OWNER
Junior doctors' association	Resistant → Neutral	Shift handovers will be faster and safer, not monitored more	Two co-design workshops, then monthly Q&A	Clinical lead
Ward clerks	Unaware → Neutral	What changes for you and when; training before go-live	Team briefings, short videos in Bangla, floor walkers at launch	Change manager
Hospital CEO	Supportive → Leading	Visible sponsorship drives adoption	Fortnightly 20-min briefing; CEO opens training days	PM

Building trust and influence

T1

Credibility

Know the subject, bring evidence and admit what you do not know.

T2

Reliability

Do what you said, when you said; small promises kept build large trust.

T3

Openness

Share bad news early and explain the reasons behind decisions.

T4

Their interests

Show how the project serves what each stakeholder cares about, in their language.

T5

Reciprocity

Help others with their goals; influence flows both ways.

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SCENARIO · DECIDE · THE FORGOTTEN DEPARTMENT

Two months into building a new warehouse management system for a Gazipur garment exporter, you learn the customs compliance team was never consulted. They say the system must produce bonded-warehouse reports in a specific format, or the factory may face penalties. What should you do FIRST?

(A) Tell them the scope is already approved and they should raise a change request next year.

(B) Add them to the stakeholder register, meet them quickly to understand the requirement and its urgency, and assess the impact through the change process.

(C) Ask developers to add the report quietly to avoid delaying the project.

(D) Escalate to the sponsor and ask who is to blame for the omission.

Circle your choice. Answer and feedback: module answer key.

Key terms

Stakeholder
Any person, group or organisation that can affect, be affected by, or believe they are affected by the project.

Salience model
A stakeholder model that classifies stakeholders by power, legitimacy and urgency.

Engagement assessment matrix
A table that shows each stakeholder's current and desired engagement level, from unaware to leading.

Influence
The ability to shape others' decisions and actions without relying on formal authority.

Power–interest grid
A two-by-two grid that plots stakeholders by their power over the project and their interest in it, suggesting how closely to engage each.

Stakeholder register
A living record of stakeholders, their interests, influence, engagement levels and planned approach.

Stakeholder engagement plan
The planned actions, messages and methods for moving each stakeholder from current to desired engagement.

KNOWLEDGE CHECK

1. Which stakeholders belong in the 'keep satisfied' quadrant?
   (A) Low power, high interest
   (B) Low power, low interest
   (C) High power, high interest
   (D) High power, low interest
2. In the salience model, a stakeholder with power, legitimacy and urgency is:
   (A) Dormant
   (B) Discretionary
   (C) Definitive
   (D) Dependent
3. When should stakeholder analysis be updated?
   (A) Only at project start, when the register is built
   (B) At gates, after major changes and when positions shift
   (C) Only at closure, as part of the lessons-learned review
   (D) Never; it is a one-time exercise at initiation
4. In an engagement assessment matrix, 'C' and 'D' stand for:
   (A) Current and Desired
   (B) Critical and Dependent
   (C) Customer and Delivery
   (D) Committed and Detached

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5. A powerful head of operations doubts your hybrid project because a previous project missed its dates. Which action will MOST build his trust over time?

   (A) Ask the sponsor to instruct him to support the project

   (B) Send him a detailed plan showing that this project is different

   (C) Invite him to every sprint review and steering meeting

   (D) Make small, specific commitments to him and keep every one, sharing bad news early

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MODULE 3 · ENGAGE · LESSON 2 · 26 MIN

Aligning Stakeholder Expectations

Bring different expectations together into one agreed picture before they collide

- Distinguish aligning expectations from managing them
- Categorise stakeholders and uncover stated and unstated expectations
- Facilitate a discussion that turns conflicting expectations into agreed trade-offs
- Use mentoring to help stakeholders understand what to expect

Stakeholders rarely share one picture of success. The finance director expects savings, the users expect an easier day, the regulator expects compliance and the sponsor expects a launch before the festival season. **Aligning** expectations means bringing these views together, early and at every major decision, into one agreed set that fits the project's objectives. **Managing** expectations, covered in the next lesson, is the ongoing work of keeping what is delivered in line with what was agreed.

Aligning versus managing expectations

Aligning

- Happens early and at major decision points
- Aims for one agreed set of expectations across stakeholders
- Tools: categorisation, expectation mapping, facilitated workshops, trade-off agreements
- Produces agreed success criteria, priorities and boundaries

Managing

- Happens throughout delivery and after release
- Aims to keep outcomes and expectations matched over time
- Tools: demonstrations, satisfaction measures, honest forecasts, change control
- Produces satisfied customers and timely adjustments

DEFINITION · EXPECTATION OR REQUIREMENT?

A **requirement** is a documented condition the result must meet. An **expectation** is a belief about what the project will deliver, how or when, and it is often unstated. Many disputes at acceptance are about expectations that were never written down.

Categorising stakeholders for alignment

CATEGORY	TYPICAL EXPECTATIONS	ALIGNMENT APPROACH
Decision-makers (sponsor, board)	Return, date, reputation	Agree success criteria and fixed versus flexible constraints
Users and customers	Ease, reliability, no extra work	Prototypes, user testing, early releases
Providers (vendors, IT operations)	Clear scope, fair terms, supportable design	Joint planning and interface agreements
Compliance bodies (regulators, auditors)	Evidence that rules are met	Early review of requirements and test evidence
Influencers (unions, community leaders)	Protection of their members' interests	Consultation and visible responses to concerns

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Uncovering what people really expect

1. **Ask about failure** 'What would make you say this project failed?' brings out hidden priorities.
2. **Ask about the first check** 'What will you look at first on go-live day?' shows what success means to them.
3. **Listen for past experience** Expectations are often shaped by the last project, good or bad.
4. **Find the unstated** Note assumptions such as 'of course it will work offline' and confirm them.
5. **Compare across groups** Put expectations side by side to see where they conflict.

A facilitated alignment session

01 Prepare — Map expectations and likely conflicts →
02 Set context — Vision, objectives and constraints →
03 Surface — Silent writing, then share →
04 Cluster — Group and find the conflicts →
05 Trade off — Test each option against objectives →
06 Agree and record — Success criteria, priorities, exclusions

TEMPLATE

Expectation alignment map

Capture each key stakeholder's expectations, find the conflicts and record what was agreed.

STAKEHOLDER	STATED EXPECTATION	UNSTATED ASSUMPTION	CONFLICTS WITH	AGREED POSITION	CONFIRMED BY AND DATE
Estate owners	*Wage payments digital by June*	*All workers have phones*	*Union: no worker left unpaid*	*Phased rollout; cash option kept for six months*	*Sponsor, 12 Feb*

EXAMPLE · ALIGNING A TEA-ESTATE PAYMENT PROJECT

A bank and three tea estates near Sylhet are moving worker wages from cash to mobile wallets. Estate owners expected savings within one quarter; the workers' union expected nobody to lose access to wages; the bank expected full identity checks. In one facilitated session the groups agreed a **phased rollout**, help desks on pay days, and a six-month cash fallback. The agreed position became the release plan's success criteria.

Some expectation gaps are really **capability** gaps. A sponsor new to agile may expect a fixed feature list on a fixed date; a new product owner may expect the team to decide priorities. Mentoring closes these gaps faster than argument. Look for mentoring opportunities, organise them deliberately and act on them.

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Mentoring opportunities that align expectations

- Walk a new sponsor through how release forecasts and burnup charts work before the first steering meeting
- Pair a new product owner with an experienced one for the first few sprints
- Invite client staff to shadow sprint reviews or site inspections
- Review a junior project manager's first status reports together
- Use reverse mentoring: younger staff coaching senior leaders on new digital tools

SCENARIO · DECIDE · FIXED DATE, FLEXIBLE SCOPE?

On an agile wallet project, the sponsor expects every feature in the original list by the Eid launch. The product owner expects scope to flex based on user feedback. Both have told the team their view is 'the plan'. What should the project manager do?

(A) Follow the sponsor's view, because the sponsor funds the project.

(B) Facilitate a session where the sponsor and product owner agree what is fixed, what can flex and what minimum release must ship by Eid, and offer the sponsor a short briefing on agile forecasting.

(C) Let the team decide, since it will do the work.

(D) Plan for the full list and add overtime if the forecast slips.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Think of a project where two stakeholders expected different things. When did the difference first become visible, and what question asked earlier would have revealed it?

Key terms

Expectation
A stakeholder's belief about what a project will deliver, how or when; often unstated.

Unstated assumption
Something a stakeholder takes for granted without saying, such as 'it will work offline'.

Mentoring
A longer-term relationship in which an experienced person helps another build understanding and capability.

Success criteria
Agreed, measurable conditions that show whether the project has achieved what stakeholders expected.

Expectation alignment
Bringing different stakeholders' expectations together into one agreed set that fits the project's objectives.

Trade-off
An agreed choice to give up some of one thing, such as scope, to protect another, such as the date.

Reverse mentoring
Mentoring in which a less senior person coaches a senior colleague, often on new tools or perspectives.

KNOWLEDGE CHECK

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6. Which activity is BEST described as aligning rather than managing expectations?

   (A) Running a satisfaction survey after a release
   (B) Agreeing success criteria with several stakeholder groups before planning
   (C) Demonstrating completed features at a sprint review
   (D) Updating a forecast after a delay

7. A stakeholder says nothing about offline use, but later rejects the app because it does not work without signal. What was missed?

   (A) A documented requirement
   (B) A conflict-handling mode
   (C) An unstated assumption
   (D) A change request

8. Estate owners expect fast savings and the union expects nobody to lose wage access. What should the PM do FIRST?

   (A) Bring both groups together to surface the conflict and agree trade-offs against the objectives
   (B) Side with the estate owners, who are paying
   (C) Ask the union to put its concerns in writing for the steering group
   (D) Plan to satisfy both fully with extra budget

9. A new sponsor keeps asking for exact feature-by-date commitments on an agile project. Which action addresses the root cause?

   (A) Give the commitments to keep her satisfied
   (B) Escalate to the portfolio board
   (C) Stop inviting her to sprint reviews
   (D) Offer a short mentoring session on how agile release forecasts work

10. Which question is MOST useful for uncovering a stakeholder's real priorities?

    (A) What would make you say this project failed?
    (B) Do you approve the current plan?
    (C) Have you read the charter?
    (D) Can you sign off the requirements today?

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MODULE 3 · ENGAGE · LESSON 3 · 26 MIN

Managing Expectations & Customer Satisfaction

Keep outcomes and expectations matched for internal and external customers, and respond when they drift

- Identify internal and external customer expectations
- Translate expectations into measures and acceptance conditions
- Monitor satisfaction and spot drifting expectations early
- Respond when outcomes and expectations diverge

Once expectations are aligned, the work is to keep them aligned. Expectations drift: people see a competitor's app, a new manager arrives, a delay is rumoured before it is reported. **Managing expectations** means keeping what the project delivers and what customers expect in step, and measuring whether customers are satisfied, not assuming it.

Internal and external customers

Internal customers

- Teams inside the organisation who receive the result, such as operations, finance or another project
- Care about supportability, workload and handover
- Express expectations through service levels and readiness criteria
- Satisfaction shown by adoption and fewer escalations

External customers

- Paying clients, citizens, patients or end users outside the organisation
- Care about ease, reliability, price and trust
- Express expectations through contracts, reviews and behaviour
- Satisfaction shown by scores, complaints, repeat use and referrals

The expectation management cycle

01 Identify — What does each customer expect?

→

02 Translate — Turn expectations into acceptance criteria and service levels

→

03 Show early — Demos, prototypes, pilots, site walkthroughs

→

04 Measure — Satisfaction and acceptance data

→

05 Respond — Adjust, explain or change

Simple satisfaction measures

MEASURE	FORMULA	WHAT IT TELLS YOU
Customer satisfaction (CSAT)	`Satisfied responses ÷ total responses × 100`	Share of customers who rate an experience as satisfying, often 4 or 5 on a five-point scale.
Promoter score	`% promoters − % detractors`	The widely used net promoter method: on a 0–10 scale, 9–10 are promoters and 0–6 detractors. Ranges from −100 to +100.
Customer effort	`Average rating of 'how easy was it?'`	Low effort predicts loyalty better than delight for many services.
First-time acceptance rate	`Items accepted at first review ÷ items reviewed × 100`	A project-level signal that the team understands what customers expect.

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Warning signs that expectations are drifting

SIGNAL	WHAT IT MAY MEAN	USEFUL RESPONSE
A stakeholder stops attending reviews	Lost interest or quiet disagreement	Meet one-to-one and ask what has changed
Surprise at a demo or walkthrough	Expectations moved away from the agreed scope	Revisit the agreed criteria together
A stream of 'small' extra requests	Unmet needs or unclear priorities	Route them to the backlog or change control
Escalations that bypass the PM	Low trust in project reporting	Increase transparency; share raw data
Falling satisfaction or acceptance scores	Outcomes are not matching expectations	Find the gap and agree actions

Responding when outcomes and expectations diverge

1. **Acknowledge quickly** Show you have heard the concern; silence feels like dismissal.
2. **Find the type of gap** Is the delivery wrong, the expectation unrealistic or the understanding different?
3. **Bring evidence** Agreed criteria, demo recordings, test results and data.
4. **Offer options** Fix, re-prioritise, change formally or explain why not.
5. **Agree and record** Update the backlog, change log or acceptance criteria.
6. **Follow up** Check satisfaction again after the response.

PRO TIP · SHOW, DO NOT TELL

Expectations stay realistic when customers see the real thing early. Agile teams use sprint reviews and beta releases; predictive projects use mock-ups, material samples, model rooms and walkthroughs. Seeing a sample tile or a working screen settles arguments that documents cannot.

WATCH OUT · MANAGING EXPECTATIONS IS NOT SPIN

Managing expectations never means hiding bad news or lowering ambitions in secret. It means sharing honest forecasts early, explaining trade-offs and letting customers make informed choices.

CASE STUDY · PATIENT PORTAL AT A PRIVATE HOSPITAL IN DHAKA

A hybrid project is launching a patient portal. External customers, the patients, expect lab reports online the same day. Internal customers, the doctors, expect no extra clicks in their workflow. Two months after the pilot, patient satisfaction has fallen from 4.3 to 3.6 out of 5 and doctors complain that report approval now takes longer. The vendor says the system works exactly as specified.

1. What kinds of gap do you see between expectations and outcomes?

2. Which measures would you track for each customer group?

3. What would you do in the next two weeks?

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Model answer: module answer key.

SCENARIO · DECIDE · THE LOBBY THAT DISAPPOINTED

On a predictive office-fit-out contract, the client's facilities head sees the finished lobby and says it looks 'nothing like the rendering'. The work matches the approved specification, which allowed a cheaper stone than the one shown in the marketing rendering. What should the project manager do?

A Point out that the specification was signed and close the discussion.

B Replace the stone at the contractor's cost to keep the client happy.

C Walk through the specification and rendering together, acknowledge the gap, offer costed options through change control, and agree to show material samples before future finishes.

D Ask the architect to explain to the client why renderings are not exact.

Circle your choice. Answer and feedback: module answer key.

TEMPLATE **Customer expectation tracker**

Track each customer group's key expectations and how well outcomes are meeting them.

CUSTOMER GROUP (INTERNAL OR EXTERNAL)	KEY EXPECTATION	MEASURE	TARGET	LATEST READING	ACTION
Patients (external)	*Lab report online the same day*	*Same-day release rate*	*90%*	*71%*	*Redesign approval step; change request raised*

Key terms

Internal customer
A team or department inside the organisation that receives or depends on the project's result.

External customer
A client, citizen, patient or user outside the organisation who receives the project's result.

Customer satisfaction (CSAT)
The share of customers who rate an experience as satisfying on a simple scale.

Net promoter method
A loyalty measure that subtracts the percentage of detractors from the percentage of promoters.

Expectation gap
The difference between what a customer expects and what is delivered or understood.

Demonstration
Showing working results, samples or prototypes to customers so they can give feedback on the real thing.

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First-time acceptance rate

The share of deliverables accepted at their first review, a sign of how well expectations are understood.

KNOWLEDGE CHECK

11. Which is an example of an internal customer?
    - (A) A bank's retail customers using a new app
    - (B) Citizens applying for passports online
    - (C) The operations team that will run the new system
    - (D) A regulator reviewing compliance evidence
12. In a survey, 50% of respondents are promoters, 30% are passives and 20% are detractors. What is the promoter score?
    - (A) +30
    - (B) +50
    - (C) +70
    - (D) +20
13. A stakeholder who used to attend every sprint review has missed the last three. What should the PM do?
    - (A) Remove her from the invitation list
    - (B) Meet her one-to-one to find out what has changed
    - (C) Report her absence to the sponsor
    - (D) Send her the recordings and assume she is satisfied
14. The system meets its specification, but doctors are unhappy because it adds steps to their work. What kind of problem is this MOST likely to be?
    - (A) A defect the vendor must fix free of charge
    - (B) A resource problem
    - (C) A procurement fraud
    - (D) An expectation that was never turned into acceptance criteria
15. Which practice BEST keeps customer expectations realistic on a predictive construction project?
    - (A) Showing material samples and model rooms before finishes are installed
    - (B) Sending monthly percentage-complete reports
    - (C) Avoiding contact with the client until handover
    - (D) Promising upgrades to keep morale high

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · ENGAGE · LESSON 4 · 25 MIN

Knowledge Transfer

Find the knowledge the project depends on, capture it and make sure it moves to the people who need it

- Identify knowledge that is critical to the project
- Choose methods to gather and transfer explicit and tacit knowledge
- Foster an environment in which people share what they know
- Plan and verify knowledge transfer at key handover points

Projects run on knowledge: why a design was chosen, how the legacy system really behaves, which supplier contact gets things done. People change roles, vendors leave and operations teams inherit what the project built. **Knowledge transfer** makes sure the knowledge the project depends on is identified, captured and moved to the people who need it, before it walks out of the door.

Explicit and tacit knowledge

Explicit knowledge

- Can be written down: designs, code, procedures, contracts, decisions
- Transferred through repositories, wikis, runbooks and decision logs
- Risk: it exists but nobody can find it or it is out of date

Tacit knowledge

- Lives in people: judgement, workarounds, relationships, 'how it really works'
- Transferred through pairing, shadowing, storytelling and practice
- Risk: it leaves when the person leaves

Identify knowledge critical to the project

1. **List knowledge areas** Systems, processes, decisions, supplier relationships, regulations and customer insight.
2. **Find concentrations** Ask 'who is the only person who knows this?' for each area.
3. **Rate criticality** What happens if this knowledge is lost tomorrow?
4. **Rate exposure** Is the holder leaving, overloaded, external or part-time?
5. **Decide the response** Capture, spread, transfer or accept, starting with high criticality and high exposure.

DEFINITION · KEY-PERSON RISK

Teams sometimes ask how many people could be unavailable before work stops, often called the **bus factor**. A bus factor of one means a single absence can halt part of the project. It is a risk to record and treat like any other.

Ways to gather and share knowledge

PR

Pairing and ensemble work

Two or more people work on the same task, so skills spread as the work is done.

SH

Shadowing

The receiver watches the expert work, then the roles reverse.

DL

Decision logs

Short records of what was decided, why and what alternatives were rejected.

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RB

Runbooks

Step-by-step guides for operating, supporting and recovering the result.

SS

Show-and-tell sessions

Short regular sessions where team members teach a topic.

CP

Communities of practice

Groups across projects who share know-how on a common discipline.

RW

Recorded walkthroughs

Short screen or site recordings that explain how something works.

Knowledge transfer across the life cycle

MOMENT	WHAT TO TRANSFER	GOOD METHODS
Onboarding a new member	Vision, ways of working, system context	Buddy, onboarding guide, pairing
Vendor or consultant exit	Configuration, design decisions, support know-how	Reverse shadowing, runbooks, contract clause on handover
Between teams or sprints	Interfaces, decisions, open issues	Decision log, joint reviews
Phase transition	Assumptions, risks, lessons so far	Gate review pack, handover meeting
Handover to operations	How to run, support and recover	Runbooks, hypercare, supervised operation

Fostering an environment for knowledge transfer

- Plan capacity for pairing and teaching; knowledge transfer that is not in the plan does not happen
- Add 'documentation updated' and 'knowledge shared' to the definition of done
- Recognise people who teach, not only people who solve problems alone
- Keep tools simple and searchable, with clear access rights for confidential material
- Leaders model asking for help and admitting what they do not know

EXAMPLE · REVERSE SHADOWING AT A GARMENT ERP GO-LIVE

At a Narayanganj knitwear exporter, the ERP vendor's consultants were due to leave four weeks after go-live. For the first two weeks client staff watched the consultants run month-end tasks; for the next two the roles were reversed and the consultants only watched. Each task was signed off when a client staff member completed it alone, with the runbook, without help.

WATCH OUT · A DOCUMENT DUMP IS NOT A TRANSFER

Handing over 400 pages does not transfer knowledge. Transfer is complete only when the receiver can **perform the task** without the original holder. Also check contracts: who owns designs and code, and what handover support the supplier must provide.

TEMPLATE **Knowledge transfer plan**

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Plan who must learn what, from whom, by when, and how you will know it worked.

KNOWLEDGE AREA	CURRENT HOLDER(S)	CRITICALITY (H/M/L)	RECEIVER(S)	METHOD	DONE WHEN
Payment reconciliation module	*Rafiq (only expert)*	*H*	*Nadia and Tom*	*Pairing for two sprints; runbook; recorded walkthrough*	*Nadia runs a month-end reconciliation alone*

SCENARIO · DECIDE · THE ONLY EXPERT IS LEAVING

On an agile team building a mobile-money platform, the only engineer who understands the reconciliation module starts six months' leave in six weeks. The product owner wants the team focused on new features. What should the project manager do?

(A) Ask the engineer to write full documentation in his last week.

(B) Work with the product owner to reserve capacity for pairing on reconciliation work in the next sprints, add runbook updates to the definition of done, and confirm transfer when another engineer completes a reconciliation alone.

(C) Accept the risk and call the engineer if problems arise during leave.

(D) Ask HR to hire a contractor with similar experience.

Circle your choice. Answer and feedback: module answer key.

Key terms

Knowledge transfer
Making sure knowledge the project depends on is identified, captured and moved to the people who need it.

Tacit knowledge
Know-how held in people's experience and judgement that is hard to write down.

Reverse shadowing
A transfer method in which the receiver performs the task while the expert observes and corrects.

Community of practice
A group of people across teams who regularly share knowledge about a common discipline.

Explicit knowledge
Knowledge that can be written down and shared through documents, code or records.

Bus factor
An informal measure of how many people could be unavailable before part of the work stops; a key-person risk.

Runbook
A step-by-step guide for operating, supporting and recovering a system or process.

Decision log
A record of key decisions, their reasons and the alternatives considered.

KNOWLEDGE CHECK

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16. Which is an example of tacit knowledge?

(A) A signed contract with the hardware supplier
(B) The approved network diagram
(C) A senior clerk's sense of which customer files are likely to contain errors
(D) The test scripts stored in the repository

17. When is knowledge transfer from a departing vendor BEST considered complete?

(A) When client staff can perform the tasks without the vendor's help
(B) When the vendor delivers all documentation
(C) When the final invoice is paid
(D) When the vendor runs a one-day training course

18. A project depends on one engineer who alone understands a critical interface. How should this be treated?

(A) As an issue to escalate immediately to the sponsor
(B) As acceptable, because experts are always scarce
(C) As a reason to stop all new work
(D) As a key-person risk with a planned response such as pairing

19. Which action BEST fosters an environment for knowledge transfer on an agile team?

(A) Rewarding the fastest individual problem-solver each sprint
(B) Including knowledge sharing in the definition of done and sprint capacity
(C) Storing all documents in personal folders for security
(D) Asking experts to answer questions only by email

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MODULE 3 · ENGAGE · LESSON 5 · 28 MIN

Planning & Managing Communication

Design a communication strategy, report in ways people read and support governance with honest information

- Define a communication strategy tailored to stakeholder needs
- Choose push, pull and interactive methods and calculate communication channels
- Create reports that match sponsor, stakeholder and governance requirements
- Build feedback loops that show whether communication is working

Communication is the project manager's main tool: most of the job is informing, asking, listening and deciding with others. A **communication strategy** decides deliberately who needs what information, why, in what form and how often, and how you will know it landed. It promotes transparency and collaboration rather than just broadcasting updates.

Building a communication strategy

01 Analyse audiences
Needs, interests, preferred style and language

→

02 Define messages
What each audience must know or decide

→

03 Choose methods
Push, pull or interactive

→

04 Set cadence
Event-driven and regular

→

05 Close the loop
Feedback and adjustment

Communication channels

MEASURE	FORMULA	WHAT IT TELLS YOU
Number of channels	`n × (n − 1) ÷ 2`	n is the number of people. 10 people → 45 channels; 15 people → 105.
Channels added	`new total − old total`	Growing a team from 10 to 15 adds 105 − 45 = 60 channels — complexity grows much faster than headcount.

Push, pull and interactive

Push
- Sent to specific people
- Email, status report, notification
- Good for: updates, formal notices
- Risk: sent ≠ read

Pull
- People access when they need it
- Wiki, dashboard, shared repository
- Good for: large audiences, reference info
- Risk: nobody looks

Interactive
- Real-time, two-way exchange
- Meetings, calls, workshops, chat
- Good for: decisions, conflict, complex or sensitive topics
- Risk: time-consuming

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Promoting transparency and collaboration

- Make the board, backlog, risk register and forecast visible to everyone who needs them
- Use information radiators in team spaces and shared online dashboards for distributed members
- Share raw data alongside summaries so people can check the numbers
- Work in the open: decisions recorded in shared channels, not private messages
- Agree what is confidential and protect it, so transparency does not become a data leak

TEMPLATE **Communication plan**

Decide deliberately who needs what, when, how and from whom.

AUDIENCE	INFORMATION NEEDED	METHOD (PUSH/PULL/INTERACTIVE)	FREQUENCY	OWNER	FEEDBACK LOOP
Steering committee	*Status, forecast, decisions needed, top risks*	*Interactive meeting + pushed one-pager*	*Monthly*	*PM*	*Decisions log reviewed next meeting*
Branch staff (users)	*What changes, when, training dates*	*Push (SMS/email) + pull (FAQ page)*	*Fortnightly, weekly near go-live*	*Change lead*	*Pulse survey*

PRO TIP · BUILD A FEEDBACK LOOP

Every communication channel needs a way to tell whether it works: a two-question pulse survey, read rates, questions asked at reviews or a short 'what was useful?' at the end of steering meetings. Adjust format and frequency based on what you learn, not on habit.

Understanding reporting requirements

AUDIENCE	WHAT THEY NEED	TYPICAL FORMAT AND CADENCE
Sponsor and steering group	Status against objectives, forecasts, decisions needed, top risks	One-page report or dashboard; monthly or at gates
PMO and portfolio governance	Standard metrics, tolerance breaches, benefits outlook	Standard template; monthly

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AUDIENCE	WHAT THEY NEED	TYPICAL FORMAT AND CADENCE
Regulator or auditor	Evidence of compliance and controls	Formal submissions; as required
Team	Priorities, impediments, progress, changes	Board, stand-ups, reviews; daily and per sprint
Customers and users	What is changing for them and when	Release notes, demos, newsletters; per release

A status report people actually read

1. **Bottom line up front** Status colour, trend and one-sentence summary.
2. **Decisions or help needed** What, from whom, by when.
3. **Progress against outcomes** Milestones and value delivered, not a list of meetings held.
4. **Forecast** Dates and cost as ranges, with change since last report.
5. **Top risks and issues** Three to five, with owners and actions.
6. **Keep it to one page** Link to detail for those who want it.

> INSIGHT · REPORTS SERVE GOVERNANCE
> Status reports are not just updates; they are inputs to **decisions**. Align the report with the governance thresholds the board has set, flag clearly when a tolerance is forecast to be breached, and state the decision you need. A report that makes the board's job easy gets faster decisions.

SCENARIO · DECIDE · THE IGNORED REPORT

Your weekly 6-page status email has a 20% open rate, and the sponsor was surprised by a delay you reported three weeks ago. What is the BEST next step?

(A) Send the report twice a week so it gets noticed.

(B) Mark the email as high-priority.

(C) Note in the lessons learned that the sponsor did not read the report.

(D) Ask the sponsor how they prefer to receive information, then switch to a one-page, decision-first summary plus a short monthly conversation for key items.

Circle your choice. Answer and feedback: module answer key.

Key terms

Communication management plan
The part of the project plan that defines who needs what information, when, how and from whom, and how feedback is gathered.

Communication channels
The number of possible two-way links between people, calculated as n × (n – 1) ÷ 2.

Push communication
Information sent to specific recipients, such as emails, reports or notifications.

Pull communication
Information placed where people can access it when they need it, such as wikis and dashboards.

Interactive communication
Real-time, two-way exchange such as meetings, calls and workshops.

Information radiator
A large, visible display of project information, physical or digital, that anyone can see at a glance.

Feedback loop
A built-in way of checking whether communication was received, understood and useful, and adjusting it.

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KNOWLEDGE CHECK

20. A team grows from 6 to 9 people. How many communication channels are added?

(A) 3
(B) 15
(C) 36
(D) 21

21. A sensitive change to staff roles needs to be explained to affected employees. Which method fits best?

(A) Interactive — meet them and allow questions
(B) Pull — post a detailed explanation on the team wiki
(C) Push — send a mass email to all employees
(D) None — say nothing until the go-live date

22. The steering group says it learns about problems too late, although every issue was in the 20-page monthly report. What is the BEST response?

(A) Send the same report weekly instead of monthly
(B) Add an executive summary to page 20
(C) Stop sending reports and rely on informal conversations
(D) Ask members what they need, then lead with status, forecasts and decisions needed on one page

23. Which reporting practice BEST supports governance?

(A) Reporting only good news so the board stays confident
(B) Flagging when a forecast will breach an agreed tolerance and stating the decision needed
(C) Using a different format each month to keep interest
(D) Sending the full risk register with every report

24. A distributed team keeps losing track of decisions made in private chats. Which practice BEST promotes transparency?

(A) Record decisions in a shared channel or decision log visible to the team
(B) Hold a daily two-hour meeting to repeat decisions
(C) Let only the project manager make decisions
(D) Ban private messages entirely

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MODULE 3 · ENGAGE

Answer key

Knowledge checks

1. D High-power stakeholders with low interest should be kept satisfied without overloading them.

2. C Definitive stakeholders combine all three attributes and need the highest attention.

3. B Stakeholder power and interest change; analysis is a living activity.

4. A C shows current engagement; D shows desired engagement. The gap drives your actions.

5. D Trust grows from reliability and openness. Authority or documents may win compliance but not genuine support.

6. B Aligning brings different expectations together into one agreed set, typically early or at major decisions. The other options keep delivery and expectations matched over time, which is managing.

7. C The stakeholder assumed offline use without saying so. Asking about first checks and failure conditions helps surface such assumptions.

8. A Conflicting expectations should be surfaced and resolved through facilitated discussion tied to the objectives, not decided for one side or hidden.

9. D The expectation gap comes from unfamiliarity with agile forecasting. Mentoring builds the understanding that aligns her expectations.

10. A Asking about failure brings out hidden priorities and assumptions; the other questions invite a yes or no.

11. C Internal customers are teams inside the organisation who receive or depend on the result, such as the operations team.

12. A Promoter score = % promoters − % detractors = 50 − 20 = +30. Passives are not counted.

13. B Withdrawal is an early sign of drifting expectations or quiet disagreement. A direct conversation finds out why.

14. D The delivered result matches the specification, so the gap lies in an unstated expectation that should have become an acceptance criterion.

15. A Showing the real thing early, through samples and walkthroughs, lets customers react before work is complete.

16. C Tacit knowledge lives in experience and judgement. The other options are written records, which are explicit knowledge.

17. A Transfer is proven when receivers can perform the work alone. Documents and training are inputs, not proof.

18. D A single point of knowledge is a risk. Spreading knowledge through pairing or documentation reduces it.

19. B Knowledge transfer happens when it is planned and expected as part of done work, not left to spare time.

20. D 6 people: 6×5÷2 = 15. 9 people: 9×8÷2 = 36. Added: 36 − 15 = 21.

21. A Sensitive and complex messages need two-way, real-time communication.

22. D Reports must match the audience's needs. Asking and redesigning around the decisions they make is a feedback loop in action.

23. B Governance relies on reports tied to agreed thresholds and clear decision requests.

24. A Working in the open, with decisions recorded where everyone can see them, keeps a distributed team aligned without extra meetings.

Scenarios & case studies

> **Scenario · The forgotten department**
> **B** is best. Best. You correct the identification gap, understand the real need and route any scope impact through proper change control.

> **Scenario · Fixed date, flexible scope?**
> **B** is best. Best. It aligns expectations openly around trade-offs and uses mentoring to close the capability gap behind the conflict.

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Case study · Patient portal at a private hospital in Dhaka
Two gaps: patients expected **same-day** reports, but approval delays push many to the next day; doctors expected **no extra clicks**, but the approval step adds them. The system meets its specification, so the gap lies in expectations that were not translated into acceptance criteria. Track same-day release rate and patient CSAT for patients, and approval time and clicks per report for doctors. In two weeks: share the data openly with both groups, run a short session with doctors to redesign the approval step (perhaps batch approval), raise a change request to the vendor, tell patients realistic report times in the portal, and measure again after the change.

Scenario · The lobby that disappointed
C is best. Best. It respects the agreed scope, treats the expectation seriously, offers a formal route to change and prevents a repeat.

Scenario · The only expert is leaving
B is best. Best. It treats key-person risk as real work, spreads tacit knowledge through practice and verifies the result.

Scenario · The ignored report
D is best. Correct. Tailor the method to the audience and use interactive communication for important news.

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04

MODULE 4 · PLAN

Process I: Integrated Planning, Scope & Value

Process · Tasks 1–3 (integrated plan and delivery planning, scope, value-based delivery)

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Assess needs, complexity and magnitude and recommend an execution strategy
- Build and maintain an integrated plan across predictive, agile and hybrid work
- Define scope, obtain agreement and break it down with a WBS, backlog or both
- Prioritise and deliver work by value, incrementally where possible
- Track business value and benefits throughout the project

LESSONS

1. The Integrated Project Management Plan & Delivery Planning · 32 min
2. Defining Scope: Requirements, User Stories & Agreement · 28 min
3. Breaking Down Scope: WBS, Backlog & Story Maps · 26 min
4. Value-Based Delivery & Prioritisation · 28 min
5. Tracking Business Value & Benefits · 26 min

MODULE 4 · PLAN · LESSON 1 · 32 MIN

The Integrated Project Management Plan & Delivery Planning

Assess the project, recommend an approach and execution strategy, and build one plan that holds together

- Assess a project's needs, complexity and magnitude before planning
- Identify critical information requirements, including sustainability data
- Recommend an execution strategy and build an integrated plan for any approach
- Review consolidated plans for dependencies, gaps and continued business value

A project plan is not a pile of separate documents. The **integrated project management plan** brings together how scope, schedule, cost, quality, resources, communication, risk, procurement, stakeholders and change will be managed, and makes sure those parts agree with each other. On predictive projects it includes approved **baselines**. On agile projects the same integration happens through lighter artefacts: a product vision, roadmap, release plan, definition of done and working agreements. Hybrid projects need both, joined at milestones.

Assess the project before you plan

1. **Needs** What problem, value and constraints does the charter set? What must be true at the end?
2. **Complexity** How many unknowns, interfaces, stakeholders with different views and regulatory demands are there?
3. **Magnitude** How big is it: budget, duration, number of teams, locations and suppliers?
4. **Organisation** What governance, templates and capabilities already exist and must be used?
5. **Approach** From all of the above, recommend predictive, agile or a hybrid, part by part.

Complexity and magnitude signals

DIMENSION	LOWER	HIGHER	PLANNING IMPLICATION
Technical uncertainty	Proven method	New technology or design	More iterations, prototypes and contingency
Stakeholder diversity	One sponsor, aligned users	Many groups with conflicting goals	More alignment work and engagement planning
Interfaces	Stand-alone result	Many systems, suppliers or teams	Integration milestones and interface agreements
Regulation	Little external control	Licences, audits, safety cases	Compliance plan and evidence built in from the start
Size	One team, a few months	Several teams, years, large budget	Phases, programme structures and scaled coordination

INSIGHT · CRITICAL INFORMATION REQUIREMENTS

Decide early what information the project **must** produce and track, because it is expensive to collect later. Examples: carbon and waste data for a sustainability report, safety statistics for a regulator, data-privacy records for audit, or donor indicators for an NGO grant. Build collection into the plan, the tools and the definition of done.

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What an integrated plan contains (top first)

Direction	`Link to charter and business case` `Life cycle and development approach` `Tailoring decisions and reasons`
Baselines or forecasts	`Scope, schedule and cost baselines` `Or roadmap and release forecasts` `Milestones that join the two`
Management plans	`Scope, schedule, cost, quality, resources` `Communication, risk, procurement, stakeholders` `Change control and compliance`
Agile working artefacts	`Product vision and backlog` `Definition of done` `Working agreements and cadence`
Supporting information	`Requirements and traceability` `Configuration and document control` `Transition and handover approach`

Planning across approaches

Predictive

- Detailed plan and baselines before execution
- Rolling-wave detail for later phases
- Formal change control against baselines

Agile

- Vision, roadmap and release plan; detail each iteration
- Plan adapts through backlog ordering
- Forecasts as ranges from measured velocity or throughput

Hybrid

- Predictive baselines for stable parts
- Release forecasts for adaptive parts
- Integration milestones and one consolidated view

The **execution strategy** answers *how* the work will actually be delivered: which approach for each part, who builds it, how teams are organised, how value reaches users and how many teams must coordinate. Recommend it with reasons, because it drives most later plans.

Execution strategy decisions

DECISION	TYPICAL OPTIONS	CHOOSE BY
Approach per component	Predictive, agile, flow-based	Uncertainty and cost of late change
Sourcing	In-house, vendor, partnership	Capability, speed, cost and risk
Team structure	Feature teams, component teams	Dependencies and skills
Release strategy	Big bang, phased, pilot, incremental	Risk tolerance and readiness to adopt
Coordination at scale	Single team, scaled framework, programme	Number of teams and interdependence

Scaling frameworks at a glance

FRAMEWORK	CORE IDEA	TYPICAL STRUCTURE	OFTEN CHOSEN WHEN	WATCH OUT FOR
SAFe (Scaled Agile Framework)	Comprehensive, configurable model spanning team, program and portfolio levels	Agile release trains of roughly 50–125 people planning together every Planning Interval (often 8–12 weeks)	Large enterprises need alignment with portfolio governance and budgeting	Can become heavy and role-rich if adopted wholesale

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FRAMEWORK	CORE IDEA	TYPICAL STRUCTURE	OFTEN CHOSEN WHEN	WATCH OUT FOR
LeSS (Large-Scale Scrum)	Scrum applied to many teams with as little extra structure as possible	One product, one product owner, one product backlog, several teams in a common sprint	Organisations willing to simplify structure and descale	Requires deep organisational change
Scrum@Scale	Scale-free extension of Scrum through networks of teams	Scrum of Scrums for delivery; a MetaScrum for backlog priorities; executive teams to remove impediments	Organisations wanting to grow Scrum gradually	Relies on strong Scrum at team level
Nexus	Lightweight exoskeleton around 3–9 Scrum teams	A Nexus Integration Team focuses on integration and dependencies	A single product with a few teams and integration pain	Limited guidance beyond about nine teams

Assess the consolidated plan

1. **Check dependencies** Do the schedule, procurement and resource plans agree on dates and people?
2. **Find gaps** Is any deliverable, compliance duty or handover activity missing?
3. **Resolve conflicts** Are two plans competing for the same people, money or environments?
4. **Test realism** Walk through the plan with the people who will do the work.
5. **Re-check value** Is the business case still positive with the planned cost and timing?
6. **Maintain it** Update the plan through change control or backlog refinement as reality changes.

PRO TIP · DECIDE WITH DATA

Plans improve when decisions use evidence: historical durations, velocity, earned value, defect trends and risk data. Many tools now use AI to spot patterns or draft plan sections. Treat that output as a **hypothesis** to check against your data and your team, never as the decision itself.

CASE STUDY · ERP FOR A KNITWEAR EXPORTER IN NARAYANGANJ

A knitwear exporter with three factories and 9,000 workers is implementing a cloud ERP for merchandising, production planning, inventory and finance. The vendor's standard modules cover finance and inventory well, but merchandising and production planning need heavy configuration and a custom mobile app for line supervisors. Buyers in Europe require full traceability data from next autumn's season. Finance staff are experienced and process-driven; line supervisors have low digital literacy. The board wants a single go-live date for simplicity.

1. Which components suit predictive delivery and which suit agile?
2. Is a single big-bang go-live wise? What would you propose?
3. How would you manage the joins between components?
4. What governance information would you give the board?

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Model answer: module answer key.

TEMPLATE

Integrated plan review

Use this at plan approval and at each phase gate to check that the parts of the plan fit together.

PLAN COMPONENT	OWNER	KEY DEPENDENCIES	GAPS OR CONFLICTS FOUND	ACTION AND DUE DATE
Procurement plan	*PM and purchasing*	*Server delivery before integration testing in week 14*	*Lead time quoted as 16 weeks*	*Order early under a letter of intent; add a risk to the register*

Key terms

Integrated project management plan
The single, consolidated plan that defines how the project will be executed, monitored, controlled and closed, combining all subsidiary plans and baselines.

Subsidiary plan
A component of the project plan covering one area, such as risk, quality or procurement.

Execution strategy
The recommended way the work will be delivered: approach per component, sourcing, team structure, release strategy and coordination.

Critical information requirement
Information the project must produce or track, such as regulatory or sustainability data, identified early so it can be collected reliably.

Baseline
An approved version of the scope, schedule or cost plan, changed only through formal change control and used to measure performance.

Rolling-wave planning
Planning near-term work in detail and later work at a higher level, adding detail as it approaches.

Product roadmap
A high-level, time-based view of how a product is expected to evolve, used for planning and alignment.

KNOWLEDGE CHECK

1. Which framework is built around one product backlog and one product owner for several teams, with minimal extra roles?

   (A) SAFe
   (B) Scrum@Scale
   (C) PRINCE2
   (D) LeSS

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2. What should usually come before adopting a scaling framework?

   (A) Hiring more teams to increase delivery capacity
   (B) Reducing dependencies and fixing flow at team level
   (C) Buying a scaling tool licence for all teams
   (D) Writing a long process manual for every role

3. A sponsor wants a donor-funded clinic project to report carbon emissions from construction at closure. When should this requirement be addressed?

   (A) At closure, by estimating emissions from the final accounts
   (B) During planning, by building data collection into the plan and supplier contracts
   (C) Only if the donor asks again during execution
   (D) After the project, by the operations team

4. What is the MAIN purpose of reviewing the consolidated project plan before approval?

   (A) To find dependencies, gaps and conflicts between the parts and confirm the business value still holds
   (B) To make the plan as long and detailed as possible
   (C) To transfer responsibility for the plan to the PMO
   (D) To lock the plan so it never changes during execution

5. On a hybrid project, the PMO wants one view of progress. What BEST integrates the agile and predictive parts?

   (A) Converting the agile backlog into a detailed task-level Gantt chart
   (B) Reporting the two parts separately with no link between them
   (C) Asking the agile team to adopt the predictive change control process for every story
   (D) A milestone roadmap that joins the hardware schedule with release forecasts shown as ranges

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MODULE 4 · PLAN · LESSON 2 · 28 MIN

Defining Scope: Requirements, User Stories & Agreement

Draw out real needs, write requirements that can be built and tested, and get stakeholders to agree the scope

- Choose suitable elicitation techniques for different situations
- Write a scope statement that includes exclusions and acceptance criteria
- Write user stories that meet INVEST with testable acceptance criteria
- Obtain stakeholder agreement on scope in predictive and agile settings

Elicitation means drawing out requirements — not just collecting what people say they want, but discovering what they actually need. People often describe solutions ('add a button') rather than problems ('I cannot find yesterday's orders'). Your job is to get to the problem.

Elicitation techniques: when to use what

TECHNIQUE	BEST FOR	WATCH OUT FOR
Interviews	Depth, sensitive topics, senior stakeholders	Time-consuming; hear one view at a time
Facilitated workshops	Agreement across groups, fast decisions	Dominant voices; needs a skilled facilitator
Observation / job shadowing	Tacit knowledge people cannot describe	People behave differently when watched
Surveys and questionnaires	Many, dispersed respondents	Shallow answers; low response rates
Prototypes and wireframes	Clarifying user interfaces and flows	Users may think the prototype is nearly finished
Document and data analysis	Existing rules, reports, regulations, logs	Documents may be out of date
Focus groups	Attitudes and expectations of a user segment	Groupthink; not a substitute for observing real behaviour
Design thinking / journey mapping	Understanding experience end to end	Can drift without a clear problem statement

DEFINITION · PRODUCT SCOPE AND PROJECT SCOPE

Product scope is the set of features and characteristics of the result. **Project scope** is all the work needed to deliver that result, including management, training and handover. A **scope statement** describes both, so that everyone knows what is in, what is out and how the result will be accepted.

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What a scope statement contains

- Description of the product, service or result
- Key deliverables, including project-management and handover deliverables
- Acceptance criteria for the deliverables
- Explicit exclusions: what is out of scope
- Constraints, such as fixed dates, budgets or regulations
- Assumptions that, if wrong, would change the scope

On adaptive projects, requirements are commonly captured as **user stories**: short statements of a need, told from the user's point of view. A story is a **promise of a conversation**, not a full specification — detail is added just in time through discussion, examples and acceptance criteria.

EXAMPLE · USER STORY FORMAT

As a rural agent, **I want to** capture a customer's ID photo while offline, **so that** I can complete onboarding where there is no mobile signal.

INVEST — the six qualities of a good story

I
Independent
Can be built and released without waiting on another story, as far as possible.

N
Negotiable
Details are open for discussion; it is not a frozen contract.

V
Valuable
Delivers something a user or customer actually cares about.

E
Estimable
The team understands it well enough to size it.

S
Small
Fits comfortably within one iteration — ideally a few days of work.

T
Testable
Clear criteria tell everyone when it is working.

Acceptance criteria set the conditions a story must satisfy to be accepted by the product owner. A popular format comes from behaviour-driven development: **Given** a starting context, **When** an action occurs, **Then** an observable result follows.

EXAMPLE · GIVEN / WHEN / THEN

Scenario: photo saved offline. Given the agent's phone has no network connection, When the agent captures and confirms an ID photo, Then the photo is stored encrypted on the device And the application shows 'Saved — will sync later'.
Scenario: automatic sync. Given a stored photo and the network returns, When the app is open, Then the photo uploads within 2 minutes And the onboarding status changes to 'Submitted'.

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Definition of ready vs definition of done

Definition of ready (team agreement, optional)

- Story is written and understood by the team
- Acceptance criteria are agreed
- Dependencies identified or removed
- Sized and small enough for one iteration
- Designs or data available if needed

Definition of done (quality commitment)

- Code reviewed and merged
- Automated tests written and passing
- Acceptance criteria met and demonstrated
- Security and accessibility checks passed
- Documentation and release notes updated
- Deployed to a production-like environment

Obtaining agreement on scope

1. **Predictive** Review the scope statement and requirements with the sponsor and key stakeholders, resolve conflicts and record formal approval.
2. **Agile** Agree the product goal and roadmap; the product owner orders the backlog; stakeholders accept each increment at reviews.
3. **Hybrid** Approve the fixed parts formally and agree the budget, milestone and product goal within which the backlog may change.
4. **All approaches** Trace each requirement to a business objective and to its acceptance test, so agreement can be checked later.

SCENARIO · DECIDE · TWO LOUD VOICES

In a requirements workshop for a new loan-origination system, two senior credit managers dominate the discussion. Branch officers — the main future users — are silent. What should the facilitator do?

(A) Let the senior managers lead; they have more authority.

(B) Switch to structured techniques — silent brainstorming on cards, round-robin input and small breakout groups — and schedule follow-up interviews or observation with branch officers.

(C) End the workshop and send a survey instead.

(D) Ask the managers to leave the room.

Circle your choice. Answer and feedback: module answer key.

Key terms

Elicitation
Drawing out stakeholders' real needs through techniques such as interviews, observation, workshops and prototypes.

Acceptance criteria
Conditions a deliverable or story must satisfy to be accepted, often written as Given/When/Then.

Scope statement
A description of the project's product and project scope, deliverables, acceptance criteria, exclusions, constraints and assumptions.

Definition of ready
An optional team agreement on the conditions a backlog item should meet before it is taken into a sprint.

User story
A short description of a need from the user's point of view, used as a placeholder for conversation and refined with acceptance criteria.

Definition of done
A shared quality standard that every increment of work must meet before it is considered complete.

Requirements traceability matrix
A table that links each requirement to its source, business objective, design and test, so nothing is lost or added unnoticed.

INVEST
Six qualities of a good user story: Independent, Negotiable, Valuable, Estimable, Small and Testable.

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KNOWLEDGE CHECK

6. Which technique is best to uncover tacit knowledge that users cannot easily describe?

   (A) Observation / job shadowing
   (B) Structured online survey
   (C) Analysis of existing documents
   (D) Email questionnaires

7. In INVEST, the 'N' stands for:

   (A) Necessary
   (B) Negotiable
   (C) Non-functional
   (D) New

8. Which acceptance criterion is best written?

   (A) The search page should load quickly for all users, especially at busy times
   (B) The search should be user-friendly, intuitive and easy for every customer to use
   (C) Search works properly and always returns the right results to the user who asked
   (D) Given 500 concurrent users, When a user searches, Then results show within 2 seconds

9. What is the main purpose of a definition of done?

   (A) To estimate stories consistently across the whole team
   (B) To decide which stories may enter the next sprint
   (C) To set a shared quality standard every increment must meet
   (D) To replace acceptance criteria with one shared list

10. A hybrid project has a fixed regulatory module and an evolving customer app. How should scope agreement be handled?

    (A) Sign off every app story in advance to match the regulatory module
    (B) Formally approve the regulatory scope, and agree the budget, milestone and product goal within which the app backlog may change
    (C) Avoid any formal approval so both parts stay flexible
    (D) Let the vendor define the scope for both parts

11. Why should a scope statement list exclusions explicitly?

    (A) Because exclusions reduce the cost of the project
    (B) Because regulators require exclusions in every scope statement
    (C) Because it removes the need for acceptance criteria
    (D) Because it prevents different assumptions about what will be delivered

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MODULE 4 · PLAN · LESSON 3 · 26 MIN

Breaking Down Scope: WBS, Backlog & Story Maps

Decompose the work with a work breakdown structure, a product backlog and story maps, or a mix of all three

- Build a deliverable-oriented WBS that follows the 100% rule
- Write a WBS dictionary entry and explain the scope baseline
- Split epics and map stories into thin, valuable releases
- Combine a WBS and a backlog on a hybrid project

Whatever the approach, scope must be made visible. Predictive projects usually decompose scope into a **work breakdown structure (WBS)**. Adaptive projects keep an ordered **product backlog**. Hybrid projects often use both: a WBS for the overall project and backlogs inside the parts that evolve.

> DEFINITION · WORK BREAKDOWN STRUCTURE
> A **WBS** is a hierarchical decomposition of the total scope of work into deliverables and smaller components. The lowest level is the **work package**, which can be estimated, scheduled, assigned and controlled.

WBS example — new hospital outpatient wing (top level first)

1. Outpatient wing project	`Total scope`
Level 2: major deliverables	`1.1 Design` `1.2 Structure` `1.3 Building services` `1.4 Fit-out and equipment` `1.5 Commissioning and handover` `1.6 Project management`
Level 3: components (example for 1.3)	`1.3.1 Electrical` `1.3.2 HVAC` `1.3.3 Medical gas` `1.3.4 Fire safety systems`
Work packages (example for 1.3.3)	`Pipework installed` `Outlets fitted` `Pressure tested and certified`

Rules for a strong WBS

- **100% rule:** the children of each element add up to exactly 100% of the parent's work — nothing missing, nothing extra.
- Organise around **deliverables (nouns)**, not activities (verbs). Activities come later in the schedule.
- Include project management work itself as a branch.
- Decompose until work packages can be estimated and owned — a common heuristic is roughly 8 to 80 hours of effort, adjusted to project size.
- Use a numbering scheme (1.3.3) that links to schedule, cost and risk records.
- Build it with the team; people who do the work spot what is missing.

The **WBS dictionary** gives each element the detail the diagram cannot hold. Together, the approved scope statement, the WBS and the WBS dictionary form the **scope baseline** on predictive projects.

TEMPLATE **WBS dictionary entry**

Describe each work package so anyone can understand what it includes, who owns it and how it will be accepted.

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WBS ID AND NAME	DESCRIPTION OF WORK	OWNER	ACCEPTANCE CRITERIA	KEY MILESTONES	ESTIMATED COST / EFFORT	ASSUMPTIONS AND CONSTRAINTS
1.3.3 Medical gas — pressure tested and certified	*Pressure test oxygen, vacuum and medical air lines; obtain certification*	*MEP subcontractor lead*	*All lines pass test at specified pressure; certificate issued by approved inspector*	*Test complete by week 42*	*Tk 6.5 lakh / 120 hours*	*Inspector available within 5 days' notice*

WBS vs product backlog

WBS

- Hierarchy of deliverables
- Aims to capture 100% of known scope early
- Changes go through formal change control
- Basis for estimates, schedule and cost baseline
- Best for stable, well-understood scope

Product backlog

- Ordered list of items (features, stories, fixes)
- Emerges and evolves as learning occurs
- Refined continuously by the product owner and team
- Top items detailed, lower items rough
- Best for evolving products and uncertain needs

PRO TIP · USING BOTH IN A HYBRID

On a hybrid hospital project, the WBS covers the building, equipment and commissioning. One branch — '1.4.5 Patient flow software' — is delivered by an agile team from its own product backlog. The WBS element carries a budget and a milestone; the backlog decides what goes inside it.

Splitting large stories (epics)

Split by...

- Workflow step (capture → verify → approve)
- Business rule (individual vs joint account)
- Data type or channel (photo vs NID scan)
- Happy path first, exceptions later
- Performance level (works first, fast later)

Avoid splitting by...

- Technical layer only (database story, UI story)
- Team member or skill
- Phases like 'design story' and 'test story'
- Anything that delivers no user-visible value alone

TEMPLATE **User story map**

Lay out the user's journey left to right, then stack stories under each step by priority to plan thin, valuable releases.

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USER ACTIVITY (BACKBONE)	USER STEP	RELEASE 1 STORIES (WALKING SKELETON)	RELEASE 2 STORIES	LATER
Open account	*Capture identity*	*Capture ID photo offline*	*Scan NID barcode automatically*	*Face match with liveness check*
Open account	*Verify customer*	*Agent reviews data before submit*	*Auto-check against blacklist*	*Risk-score explanations*

SCENARIO · DECIDE · ACTIVITIES IN THE WBS

A new team member drafts a WBS whose second level reads: 'Plan', 'Design', 'Build', 'Test', 'Deploy'. The level-3 items under each are tasks such as 'Hold meeting' and 'Write code'. What feedback is BEST?

(A) Accept it; any structure is fine as long as it is complete.

(B) Suggest reorganising around deliverables (e.g. 'Mobile app', 'Admin portal', 'Data migration', 'Training', 'Project management') and moving activities into the schedule.

(C) Delete the WBS and use only a Gantt chart.

(D) Add more levels until every task is under one hour.

Circle your choice. Answer and feedback: module answer key.

Key terms

Work breakdown structure (WBS)
A hierarchical, deliverable-oriented decomposition of the total scope of a project.

Work package
The lowest level of a WBS, which can be estimated, scheduled, assigned and controlled.

Scope baseline
The approved scope statement, WBS and WBS dictionary, changed only through formal change control.

Epic
A large body of work that is too big for one iteration and is split into smaller stories.

Product backlog
An ordered, evolving list of everything that might be needed in a product, owned by the product owner.

WBS dictionary
A document that describes each WBS element: its work, owner, acceptance criteria, milestones and estimates.

100% rule
The principle that the children of each WBS element add up to exactly all of the parent's work.

Story map
A layout of the user journey from left to right, with stories stacked under each step by priority to plan releases.

KNOWLEDGE CHECK

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12. What does the 100% rule state?

(A) Every task must be 100% complete before the next starts

(B) Child elements must sum to exactly the work of their parent

(C) All stakeholders must approve the WBS before work begins

(D) The whole budget must be allocated to work packages

13. The lowest level of a WBS is called the:

(A) Activity

(B) Control account

(C) Work package

(D) Milestone

14. On a predictive project, the scope baseline consists of:

(A) The charter, budget and schedule baseline

(B) The product backlog and the sprint backlog

(C) The risk register, issue log and assumption log

(D) Approved scope statement, WBS and WBS dictionary

15. Which statement about the product backlog is accurate?

(A) It is ordered and refined continuously as learning occurs

(B) It is fixed once the sponsor has approved its contents

(C) It contains only technical tasks defined by developers

(D) It is owned and ordered by the sponsor for each release

16. Which is a good way to split a large story?

(A) By workflow step that still delivers user value

(B) By technical layer, such as database, API and interface

(C) By team member, so each person owns one part

(D) Into separate 'design', 'build' and 'test' stories

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MODULE 4 · PLAN · LESSON 4 · 28 MIN

Value-Based Delivery & Prioritisation

Identify the components of value with stakeholders, order work by value and feedback, and deliver it in increments

- Identify the components of value with key stakeholders
- Prioritise work using MoSCoW, value versus effort, Kano and WSJF
- Assess opportunities to deliver value incrementally
- Choose delivery options that demonstrate value early

Value is rarely one number. Work with key stakeholders to name the **components of value** the project should create, then use them to order the work. When feedback arrives, from users, sprint reviews or pilots, the order changes. That is value-based delivery: the most valuable work first, checked against real evidence, delivered in pieces whenever possible.

Components of value

RV
Revenue
New sales, new customers or higher prices.

CS
Cost saving
Less effort, waste, rework or overhead.

RR
Risk reduction
Lower chance or impact of failure, fraud or downtime.

CO
Compliance
Meeting a legal or regulatory duty; often non-negotiable.

CX
Customer experience
Easier, faster or more trusted service.

LV
Learning
Knowledge that reduces uncertainty for later decisions.

SU
Sustainability
Lower environmental impact or stronger social outcomes.

Every project has more ideas than capacity. Prioritisation is how you protect value: doing the most important things first and consciously saying 'not now' to the rest. Without it, teams spread themselves thin and deliver many half-finished things.

MoSCoW prioritisation

CATEGORY	MEANING	EXAMPLE — MOBILE WALLET SAVINGS FEATURE
Must have	Without it the release fails or is illegal/unsafe	Deposits into savings with correct interest calculation
Should have	Important but a workaround exists	Savings goals with progress bar
Could have	Desirable if time allows	Animated celebration when goal is reached
Won't have (this time)	Agreed to exclude from this release	Joint savings with family members

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Value vs effort matrix

High value

- Low effort → **Quick wins**: do first
- High effort → **Big bets**: plan carefully, slice thinly

Low value

- Low effort → **Fill-ins**: do if spare capacity
- High effort → **Money pits**: avoid or drop

Kano model — how features affect satisfaction

B

Basic (must-be)

Expected; absence causes anger, presence is barely noticed. E.g. a banking app that does not lose money.

P

Performance

More is better, roughly in proportion. E.g. faster transfers.

D

Delighters

Unexpected features that create delight; absence is not missed. E.g. smart savings nudges.

I

Indifferent

Users do not care either way — candidates to drop.

R

Reverse

Some users actively dislike it — e.g. heavy gamification for business users.

Weighted shortest job first (WSJF) — in brief

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost of delay	`User/business value + Time criticality + Risk reduction or opportunity enablement`	Relative scores (often on a Fibonacci-like scale) estimating the economic pain of waiting.
WSJF	`Cost of delay ÷ Job size (or duration)`	Higher scores go first: the most value lost per unit of time, delivered fastest. Popularised in scaled agile frameworks.
Worked example	`Feature X: CoD 21, size 8 → 2.6. Feature Y: CoD 13, size 3 → 4.3`	Y goes first, even though X has the higher cost of delay, because it is much smaller.

Ways to deliver value incrementally

01 **Prototype**
Learn whether the idea works

→

02 **Minimum viable product**
Smallest release that tests value with real users

→

03 **Pilot**
Full solution in one site or group

→

04 **Phased rollout**
Wave by wave, learning between waves

→

05 **Regular releases**
Small increments on a steady cadence

Delivery options that demonstrate value

OPTION	WHAT IT PROVES	EXAMPLE
Proof of concept	The technology can work	Testing biometric login on low-cost phones
Pilot ward or branch	The solution works in real operations	E-prescribing in one ward before the whole hospital

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OPTION	WHAT IT PROVES	EXAMPLE
Model unit or show flat	Design and finish meet expectations	One finished apartment before fitting out 200
Beta release	Users adopt and value the features	Savings feature for 5,000 wallet users
Feature toggle	Value without risking everyone	New checkout shown to 10% of shoppers

EXAMPLE · VALUE FIRST IN A WATER-BILLING PROJECT

A city water utility in Rajshahi planned an 18-month billing system replacement. Analysis of value components showed that most complaints were about **surprise bills**. The team released SMS meter-reading alerts in month three, on the old system, cutting billing complaints by a third long before the new system was ready.

CASE STUDY · SCOPE PRESSURE AT A DHAKA FINTECH LAUNCH

A Dhaka fintech start-up is launching a merchant QR-payments app in time for Eid, when transaction volumes peak. The team works in two-week sprints with a fixed launch date eleven weeks away. Investors are pushing for a loyalty-points feature 'for launch'. A sales director has promised one big supermarket chain a custom settlement report. Meanwhile developers have been polishing animations that were never requested. The product owner is overwhelmed and the backlog has 180 items marked 'high priority'.

1. What scope problems can you identify?

2. How would you re-prioritise the backlog for an Eid launch?

3. How should the custom report request be handled?

4. What would you agree with the team about the animations?

Model answer: module answer key.

SCENARIO · DECIDE · FEEDBACK CHANGES THE ORDER

At a sprint review for a farmer-advisory app, extension officers say the planned crop-price feature is 'nice', but what they really need is offline weather alerts, because most villages lose signal during storms. The product owner had planned the price feature for the next two sprints. What should happen?

(A) Keep the plan, because the sprint order was agreed with the sponsor.

(B) The product owner reassesses value with the new evidence and is likely to move offline alerts up the backlog, telling stakeholders why.

(C) Build both features in the next sprint by asking the team to work overtime.

(D) Raise a formal change request to the change control board.

Circle your choice. Answer and feedback: module answer key.

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Key terms

MoSCoW
A prioritisation method that sorts requirements into Must have, Should have, Could have and Won't have this time.

Cost of delay
The value lost for each period that a piece of work is not delivered.

Kano model
A model that classifies features as basic, performance or delighter by how they affect satisfaction.

Pilot
A full solution deployed to a limited site or group to prove it works before wider rollout.

Minimum viable product (MVP)
The smallest release that lets the team test its main value assumption with real users.

Weighted shortest job first (WSJF)
A prioritisation method that divides cost of delay by job size, so that high-value, small items go first.

Value-based prioritisation
Ordering work by the value it creates for stakeholders, adjusted as feedback arrives.

KNOWLEDGE CHECK

17. In MoSCoW, 'W' stands for:
    - (A) Want to have
    - (B) Worth having
    - (C) Won't have this time
    - (D) Will have later for sure

18. A feature is expected by every user; its absence causes strong dissatisfaction. In Kano terms it is:
    - (A) A delighter (attractive) feature
    - (B) A performance (one-dimensional) feature
    - (C) An indifferent feature
    - (D) A basic (must-be) feature

19. Feature A has a cost of delay of 20 and size 10; Feature B has a cost of delay of 9 and size 3. By WSJF, which goes first?
    - (A) A, because its cost of delay is higher
    - (B) B, because its WSJF score is higher
    - (C) Both have equal priority
    - (D) Neither — WSJF cannot compare them

20. A team can release an SMS alert feature in three months or wait 18 months for a full system that includes it. Stakeholders say the alerts address the main complaint. What does value-based delivery suggest?
    - (A) Wait for the full system so everything is released together
    - (B) Cancel the full system, since the alerts solve the problem
    - (C) Release the alerts early if it is feasible, and continue the full system
    - (D) Ask users to vote on whether they want alerts

21. Which delivery option BEST proves that a new process works in real operations before a hospital-wide rollout?
    - (A) A pilot in one ward
    - (B) A proof of concept in the lab
    - (C) A detailed specification
    - (D) A vendor demonstration

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MODULE 4 · PLAN · LESSON 5 · 26 MIN

Tracking Business Value & Benefits

Examine business value throughout the project and make sure a measurement system will show whether benefits arrive

- Trace outputs through outcomes to benefits
- Write benefits register and benefits realisation entries with owners and baselines
- Re-examine business value at gates and releases, including the option to stop
- Verify that a benefits measurement system is in place

Projects deliver **outputs** (a system, a building, a process). Users adopting those outputs create **outcomes** (faster processing, fewer errors). Outcomes produce **benefits** that the organisation values (lower cost, more revenue, better service, lower emissions). Most benefits arrive **after** the project closes — which is why benefit ownership sits with the business, not with the PM.

From output to benefit

01 Output
Online booking system live

→

02 Capability
Patients can book and reschedule by phone or web

→

03 Outcome
Shorter queues, fewer no-shows

→

04 Benefit
More patients seen per doctor-day; higher satisfaction

→

05 Strategic objective
Accessible, efficient district healthcare

Benefits management ensures that the gains promised in the business case actually happen. Most benefits appear **after** the project closes, when operations use the new capability. That is why each benefit needs a named **owner** in the business — usually not the project manager — plus a baseline, a target and a date to measure.

TEMPLATE **Benefits register**

Record each expected benefit so it can be tracked from business case to realisation.

BENEFIT	MEASURE (KPI)	BASELINE	TARGET AND DATE	BENEFIT OWNER	HOW AND WHEN MEASURED
Faster customer onboarding	*Median minutes to open an account*	*18 min*	*6 min by June 2027*	*Head of Retail Operations*	*Monthly report from onboarding system*
Lower support load	*Assisted tickets per 1,000 sign-ups*	*120*	*70 by Sept 2027*	*Customer Service Manager*	*Quarterly ticket analysis*

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OKRs, KPIs and project success criteria

OKR

- Sets direction for a period (often a quarter or year)
- Ambitious, may not be fully reached
- Measures outcomes, not tasks
- Owned by a business team

KPI

- Monitors the health of ongoing operations
- Has normal ranges and alerts
- Continues indefinitely
- Example: system uptime, cost per order

Project success criteria

- Agreed at the start, recorded in the charter
- Define when this project is a success
- Mix of delivery and outcome measures
- Should trace to OKRs and the business case

Examining business value throughout the project

1. **At every gate or quarterly review** Is the business case still valid? Have costs, timing or market conditions changed?
2. **At each release** Are users adopting what was delivered, and are early outcome measures moving?
3. **When a big change is proposed** Does it increase or reduce the expected value?
4. **Compare remaining cost with remaining value** Money already spent is gone; decide on what is still to come.
5. **Recommend** Continue, adjust, pivot or stop, with evidence.

> **INSIGHT · STOPPING CAN BE SUCCESS**
> If the value disappears, for example because a regulation changes or a competitor makes the product unnecessary, recommending that the project **stop** protects the organisation's money. Good governance rewards honest early stops rather than late, expensive failures.

TEMPLATE Benefits realisation plan

Define how each benefit will be measured, by whom and when, and hand it to the business owner at closure.

BENEFIT	MEASURE (KPI)	BASELINE	TARGET & DATE	BENEFIT OWNER	HOW & WHEN MEASURED	DEPENDENCIES / DIS-BENEFITS
Fewer missed appointments	*No-show rate*	*22% (Jan–Mar)*	*12% by December*	*Hospital administrator*	*Monthly report from booking system*	*SMS reminders working; dis-benefit: front-desk retraining time*

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WATCH OUT · DIS-BENEFITS AND BENEFIT LEAKAGE

Some outcomes are negative for someone — extra steps for a department, a role that disappears. Record these **dis-benefits** honestly. Also watch for benefit leakage: time saved that is simply absorbed rather than redeployed to valuable work.

Is a benefits measurement system in place?

☐ Each benefit has a measure, a baseline and a target date

☐ A named business owner has accepted each benefit

☐ The data source exists and someone collects it

☐ Measurement frequency and reporting route are agreed

☐ Early indicators are tracked during the project, not only after it

☐ Tracking after closure is assigned and funded

☐ Dis-benefits and possible benefit leakage are recorded

SCENARIO · DECIDE · THE MISSING BENEFIT

Six months after go-live, the business owner says the expected 20% reduction in processing time has not appeared. You have moved to another project. What is the BEST action?

(A) Explain that benefits are not your responsibility after closure.

(B) Reopen the project and add new features.

(C) Help the business owner analyse the measures and adoption data to find the cause, and suggest corrective actions or a follow-up change.

(D) Lower the target to match current results.

Circle your choice. Answer and feedback: module answer key.

Key terms

Benefits realisation
Measuring and managing the outcomes of a project after delivery to ensure intended benefits are achieved.

Benefit
A measurable improvement that results from an outcome and is valued by the organisation or society.

Benefit owner
The business manager accountable for realising a benefit after the project delivers its outputs.

Benefit leakage
Gains, such as saved time, that are absorbed into other activity instead of producing the intended value.

Dis-benefit
A measurable negative consequence of a change, perceived as adverse by one or more stakeholders.

Benefits register
A list of expected benefits with measures, baselines, targets, owners and how and when each will be measured.

Key performance indicator (KPI)
A measure that tracks how well an ongoing activity or capability is performing.

KNOWLEDGE CHECK

22. Why must a benefit have a baseline?

(A) To make the benefits report more complete

(B) Without a starting value, change cannot be measured

(C) Because auditors require one for every benefit

(D) Because the baseline sets the project budget

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23. ‘Patients can book appointments online’ is best described as…

(A) a benefit
(B) a strategic objective
(C) a dis-benefit
(D) a capability/output

24. Who should normally own a business benefit such as 'lower support costs'?

(A) A business manager who will run the new capability
(B) The project manager who delivered the capability
(C) The vendor that supplied and supports the system
(D) The quality team that verified the deliverables

25. At a phase gate, a new regulation means the product will reach only half its expected market. The project is on time and on budget. What should the project manager recommend?

(A) Continue, because the project is performing to plan
(B) Pause all work until the regulation is withdrawn
(C) Re-examine the business case with updated value and present options, including stopping
(D) Increase scope to make up for the lost market

26. Which shows that a benefits measurement system is really in place?

(A) The business case lists the benefits in detail
(B) A named owner, data source, baseline and reporting route exist for each benefit
(C) The project has a dashboard showing tasks completed
(D) The sponsor has said the benefits are likely

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MODULE 4 · PLAN

Answer key

Knowledge checks

1. D LeSS keeps Scrum's simplicity: one product, one backlog, one product owner, several teams.

2. B Many scaling problems are multiplied team-level problems; fix them first.

3. B Critical information requirements should be identified early so that data collection is built into plans, tools and contracts.

4. A Integration means the parts fit together and still serve the business case; the plan is then maintained as reality changes.

5. D Integration happens at milestones and in one consolidated view, while each part keeps planning artefacts that fit its approach.

6. A Watching people work reveals workarounds and habits they would not think to mention.

7. B Negotiable — details are refined through conversation rather than fixed in advance.

8. D It is specific, observable and testable, using Given/When/Then.

9. C The definition of done is the quality commitment applied to all work; acceptance criteria are story-specific.

10. B Hybrid scope agreement fits each part: formal approval where scope is fixed, and agreed boundaries within which the backlog can evolve.

11. D Stating what is out of scope prevents later disputes about assumed work and helps control scope creep.

12. B The 100% rule ensures the decomposition includes all — and only — the work of the parent element.

13. C Work packages are the lowest WBS level, which can be estimated, scheduled and controlled.

14. D These three approved components together form the scope baseline.

15. A The product backlog evolves; the product owner orders it and the team helps refine it.

16. A Vertical slices along workflow steps keep each piece valuable and testable.

17. C Won't have (this time) records agreed exclusions for the current release.

18. D Basic features are taken for granted; missing them causes anger, having them barely raises satisfaction.

19. B A scores 20 ÷ 10 = 2; B scores 9 ÷ 3 = 3. Higher WSJF goes first.

20. C Delivering a high-value increment early creates benefits sooner while the larger work continues.

21. A A pilot runs the full solution in a real but limited setting, proving it in operations before scaling.

22. B Improvement is measured relative to a baseline.

23. D It is what the project made possible; benefits come from the outcomes of using it.

24. A Benefits usually appear after the project ends, so they are owned by the business manager responsible for the operation.

25. C Business value must be re-examined when conditions change. Being on time and budget does not prove the investment is still worthwhile.

26. B Measurement needs owners, data, baselines and a route for reporting, not just a list of hoped-for benefits.

Scenarios & case studies

> **Case study · ERP for a knitwear exporter in Narayanganj**
> **Predictive:** finance and inventory using standard modules (stable requirements, accounting rules, data migration with clear cut-over). **Agile:** merchandising configuration and the supervisor app — uncertain needs, high user-experience risk; build in sprints with supervisors testing on the floor. **Go-live:** a big bang across three factories is risky. Propose finance and inventory first, then production modules piloted in one factory, then incremental rollout to the other two (pattern H4 combined with H3), with the traceability deadline as a fixed milestone. **Joins:** integration milestones every few sprints, an interface agreement on data structures, a shared release calendar and a cross-team integration lead. **Board:** a milestone roadmap plus release forecasts as ranges, budget burn against the baseline, top risks and readiness measures (training completion, data quality) — not task-level detail.

> **Scenario · Two loud voices**
> **B** is best. Best. Structured facilitation balances voices in the room, and follow-up elicitation reaches users who may not speak up in front of seniors.

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Scenario · Activities in the WBS
B is best. Best. A deliverable-oriented WBS supports the 100% rule, clear ownership and acceptance, while activities belong in the schedule.

Case study · Scope pressure at a Dhaka fintech launch
Problems: scope creep (loyalty points, custom report), gold plating (animations) and a backlog with no real prioritisation. **Re-prioritise:** run a MoSCoW session tied to the launch goal — payments, settlement and security as Musts; use value vs effort or WSJF to order the rest; cap Musts at about 60% of the remaining capacity based on velocity. Loyalty points likely become Should or post-launch, with the trade-off shown to investors using data. **Custom report:** treat it as a change — assess value (one large merchant), effort and risk to the date; the product owner decides with the sponsor; if accepted, something of equal size moves out, and the decision is communicated to sales. **Animations:** discuss gold plating openly in the retrospective, agree that work must trace to backlog items, and channel ideas through the product owner.

Scenario · Feedback changes the order
B is best. Best. Value-based delivery means re-ordering work when feedback shows where value lies, and explaining the change.

Scenario · The missing benefit
C is best. Correct. Benefit realisation is owned by the business, but the former PM can help diagnose — often the cause is adoption or process, not the system.

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05

MODULE 5 · RESOURCE

Process II: Schedule, Resources & Finance

Process · Tasks 4, 6 and 8 (resources, finance, schedule)

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Estimate work with predictive and agile techniques and honest ranges
- Build, baseline and control schedules and release plans
- Plan, acquire and optimise people and physical resources
- Build budgets with the right reserves and plan financial reporting
- Measure and forecast with earned value and escalate variances through governance

LESSONS

1. Estimating Work: Predictive & Agile Techniques · 32 min
2. Building the Schedule: Critical Path, Release Plans & Baselines · 34 min
3. Resources: Plan, Acquire, Optimise & Compress · 30 min
4. Budgets, Reserves & Financial Planning · 32 min
5. Earned Value, Forecasting & Financial Control · 35 min

MODULE 5 · RESOURCE · LESSON 1 · 32 MIN

Estimating Work: Predictive & Agile Techniques

Analogous, parametric, bottom-up, three-point and relative estimating, with honest ranges and real benchmarks

- Choose between analogous, parametric, bottom-up and three-point estimating
- Calculate PERT estimates and a standard deviation
- Size work relatively with story points and planning poker
- Use benchmarks and historical data to keep estimates honest

An estimate is a **forecast**, not a promise. It is a best judgement about duration, effort or cost, given what we know today. Because the future is uncertain, every honest estimate carries a range. Pressure to give a single precise number early is one of the most common sources of project failure.

Four core estimating techniques

TECHNIQUE	HOW IT WORKS	STRENGTHS	LIMITS	TYPICAL USE
Analogous (top-down)	Base the estimate on a similar past project, adjusted for differences	Fast, cheap, needs little detail	Only as good as the similarity	Early business cases, ROM estimates
Parametric	Multiply a unit rate by a quantity (e.g. cost per m², hours per screen)	Scalable and repeatable	Needs reliable historical rates	Construction, rollouts, repeatable work
Bottom-up	Estimate each work package or activity and add them up	Most detailed and defensible	Slow; needs a full WBS	Baselines, contracts, detailed planning
Three-point	Use optimistic, most likely and pessimistic values	Shows uncertainty explicitly	Values can still be biased	Risky or unfamiliar activities

EXAMPLE · PARAMETRIC IN PRACTICE

Floor tiling for a hospital wing covers **1,200 m²**. Historical data shows one crew lays **25 m² per day**. Effort = 1,200 ÷ 25 = **48 crew-days**. With three crews working in parallel, duration ≈ **16 working days**, before allowing for curing, access and inspections.

Three-point estimating formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Triangular mean	`E = (O + M + P) ÷ 3`	Simple average of optimistic (O), most likely (M) and pessimistic (P). Gives equal weight to the extremes.
PERT (beta) mean	`E = (O + 4M + P) ÷ 6`	Weights the most likely value four times. The classic Program Evaluation and Review Technique estimate.
Standard deviation (PERT)	`σ ≈ (P − O) ÷ 6`	A rough measure of spread. A wider gap between O and P means more uncertainty.
Approximate confidence	`E ± 1σ ≈ 68%; E ± 2σ ≈ 95%`	Assumes a roughly normal distribution — a useful approximation, not a guarantee.

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EXAMPLE · WORKED THREE-POINT EXAMPLE

Integrating a payment gateway: O = 4 days, M = 6 days, P = 14 days. Triangular = (4 + 6 + 14) ÷ 3 = **8 days**. PERT = (4 + 24 + 14) ÷ 6 = **7 days**. σ = (14 − 4) ÷ 6 ≈ **1.67 days**. So roughly 68% confidence of 5.3–8.7 days and 95% confidence of 3.7–10.3 days. Report: '**about 7 days, very likely between 4 and 11**'.

Estimate accuracy narrows as you learn (commonly quoted ranges)

ESTIMATE CLASS	WHEN	TYPICAL RANGE
Rough order of magnitude (ROM)	Idea and business case	about −25% to +75%
Budget estimate	After high-level design	about −10% to +25%
Definitive estimate	Detailed design / work packages known	about −5% to +10%

WATCH OUT · BIASES THAT DISTORT ESTIMATES

Anchoring — the first number spoken (often the boss's) pulls estimates toward it. **Optimism bias** — people imagine the smooth path. **Padding** — hidden buffers that get consumed anyway (work expands to fill the time available, and people delay starting when they feel time is plentiful). Counter them with historical data, independent estimates and visible, shared reserves rather than hidden padding.

PRO TIP · USE BENCHMARKS AND REFERENCE CLASSES

Before trusting your own bottom-up numbers, look at how similar projects actually turned out. If ten past ERP rollouts in your organisation overran by an average of 40%, an estimate that assumes no overrun needs a strong reason. This **reference-class** view corrects optimism with evidence. Keep your own estimates and actuals so the next project has better data.

Agile teams often estimate **relatively** rather than in hours: is this item bigger or smaller than that one we already know? People are better at comparing than at predicting absolute durations, and relative sizes stay stable even as the team gets faster.

DEFINITION · STORY POINTS

A **story point** is a unitless, team-specific measure of the overall size of a backlog item — combining effort, complexity and uncertainty. Many teams use a modified Fibonacci scale (1, 2, 3, 5, 8, 13, 20…) so that larger items get deliberately coarse numbers.

Planning poker in five steps

01 Explain — Product owner describes the item and its acceptance criteria

→

02 Clarify — Team asks questions

→

03 Vote privately — Each estimator picks a card

→

04 Reveal and discuss — Highest and lowest explain their reasoning

→

05 Re-vote — Repeat until the team converges

INSIGHT · THE VALUE IS IN THE CONVERSATION

When one developer votes 3 and another votes 13, someone knows something the others do not — a hidden integration, a reusable component, a compliance step. Planning poker surfaces that knowledge before work starts.

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SCENARIO · DECIDE · 'JUST GIVE ME ONE NUMBER'

During a steering meeting, the sponsor of a new ERP rollout asks: "How long will this take? I need one date for the board." Requirements workshops have not started yet. What is the BEST response?

(A) Give an optimistic date to keep the sponsor happy, then adjust later.

(B) Refuse to estimate until all requirements are documented.

(C) Give a range based on comparable rollouts (e.g. 9–15 months), explain the main uncertainties and commit to a narrower forecast after the requirements phase.

(D) Pick the midpoint of your gut feeling and present it as firm.

Circle your choice. Answer and feedback: module answer key.

Key terms

Parametric estimating
Estimating by multiplying a quantity by a unit rate derived from historical data, e.g. cost per square metre.

Story point
A unitless, team-relative measure of the size of a backlog item, combining effort, complexity and uncertainty.

Bottom-up estimating
Estimating each work package or activity in detail and adding them up.

Rough order of magnitude (ROM)
An early, wide-range estimate used for initial decisions, before detail is known.

PERT estimate
A weighted three-point estimate calculated as (Optimistic + 4 × Most likely + Pessimistic) ÷ 6.

Analogous estimating
Estimating from the actual results of a similar past project, adjusted for known differences.

Planning poker
A team technique in which estimators reveal relative sizes at the same time and discuss differences before re-voting.

Reference-class forecasting
Adjusting an estimate using the actual outcomes of a group of similar past projects.

KNOWLEDGE CHECK

1. Which technique is most appropriate for a quick estimate at the business-case stage, using a similar past project?

   (A) Bottom-up
   (B) Analogous
   (C) Parametric
   (D) Monte Carlo

2. O = 2, M = 5, P = 14. What is the PERT estimate?

   (A) 5
   (B) 6
   (C) 7
   (D) 21

3. Which estimating technique uses a statistical relationship such as cost per square metre?

   (A) Parametric
   (B) Analogous
   (C) Bottom-up
   (D) Expert judgement only

4. Why should estimates be communicated as ranges?

   (A) Single-point estimates are not permitted by most methodologies
   (B) Ranges look more professional and credible than single figures in reports
   (C) Sponsors prefer vague answers that cannot be held against the team
   (D) They reflect real uncertainty and help decision-makers judge risk

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5. What is the main benefit of discussion in planning poker?

(A) It produces an exact hour estimate for each task

(B) It lets the manager review and approve the final estimates

(C) It replaces the need for written acceptance criteria

(D) It surfaces hidden knowledge and assumptions early

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MODULE 5 · RESOURCE · LESSON 2 · 34 MIN

Building the Schedule: Critical Path, Release Plans & Baselines

Sequence work, find the critical path, forecast releases and baseline the schedule that fits your approach

- Identify dependency types, leads and lags
- Perform forward and backward passes and calculate total and free float
- Forecast releases from velocity ranges and plan iterations by capacity
- Baseline a schedule and coordinate it with other projects and operations

Once you know the activities, you must decide their **order**. A network diagram shows which activities depend on others. From it you can find the **critical path** — the longest chain of dependent activities, which sets the shortest possible project duration.

Four logical relationships

TYPE	MEANING	EXAMPLE
Finish-to-start (FS)	B cannot start until A finishes — the most common	Install software after the server is delivered
Start-to-start (SS)	B cannot start until A starts	Proofreading can start once writing has started
Finish-to-finish (FF)	B cannot finish until A finishes	Testing cannot finish until development finishes
Start-to-finish (SF)	B cannot finish until A starts — rare	The old shift cannot end until the new shift starts

Nature of dependencies

KIND	DESCRIPTION	EXAMPLE
Mandatory (hard logic)	Required by physics, law or contract	Foundation before walls
Discretionary (soft logic)	Chosen by best practice or preference — can be changed	Finish design before starting build
External	Depends on parties outside the project	Regulator approval, vendor delivery
Internal	Within the project team's control	Testing after coding by the same team

Worked example — clinic booking system go-live (durations in days)

ACTIVITY	DESCRIPTION	DURATION	PREDECESSORS
A	Confirm requirements	3	—
B	Build booking module	4	A
C	Procure tablets	2	A
D	Integrate with patient records	5	B
E	Configure tablets	3	C

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ACTIVITY	DESCRIPTION	DURATION	PREDECESSORS
F	Pilot and go-live	2	D, E

Forward and backward pass

1. **Forward pass** Start the first activity at ES = 0. For each activity, EF = ES + duration. A successor's ES is the **largest** EF of its predecessors.
2. **Project duration** The largest EF at the end of the network is the shortest possible project duration — here 14 days.
3. **Backward pass** Set the last activity's LF = 14. For each activity, LS = LF − duration. A predecessor's LF is the **smallest** LS of its successors.
4. **Float** Total float = LS − ES. Free float = earliest ES of successors − EF.
5. **Critical path** Activities with zero total float form the critical path: A → B → D → F = 3 + 4 + 5 + 2 = 14 days.

Results of the passes

ACTIVITY	DUR.	ES	EF	LS	LF	TOTAL FLOAT	FREE FLOAT
A	3	0	3	0	3	0	0
B	4	3	7	3	7	0	0
C	2	3	5	7	9	4	0
D	5	7	12	7	12	0	0
E	3	5	8	9	12	4	4
F	2	12	14	12	14	0	0

Scheduling formulas (start-at-zero convention)

MEASURE	FORMULA	WHAT IT TELLS YOU
Early finish	`EF = ES + Duration`	Earliest an activity can finish.
Late start	`LS = LF − Duration`	Latest an activity can start without delaying the project.
Total float	`TF = LS − ES = LF − EF`	How long an activity can slip without delaying the project end.
Free float	`FF = (earliest ES of successors) − EF`	How long an activity can slip without delaying any successor's early start.
Critical path	`Longest path through the network (TF = 0 when the end date equals the calculated finish)`	The chain that determines the project duration.

WATCH OUT · WATCH THE NEAR-CRITICAL PATH

A path with very little float is almost critical. If its activities are risky, it may become the critical path overnight. Also watch for **negative float**: it means an imposed deadline is earlier than the network can achieve — a signal to re-plan, compress or renegotiate.

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Velocity and forecasting

MEASURE	FORMULA	WHAT IT TELLS YOU
Velocity	`Sum of points of items that met the definition of done in an iteration`	A team's observed delivery rate. Partly finished items count zero.
Forecast iterations	`Remaining backlog points ÷ velocity`	Use a low and high velocity to produce a range, not one number.
Capacity-adjusted velocity	`Average velocity × (available person-days ÷ usual person-days)`	Adjusts for holidays, leave and training in the coming iteration.

Release planning vs iteration planning

Release planning

- Horizon: several iterations to a few months
- Sets a release goal and target date or scope range
- Uses velocity ranges and a prioritised backlog
- Revisited every iteration as data arrives
- Involves product owner, team and key stakeholders

Iteration planning

- Horizon: one iteration (1–4 weeks)
- Sets an iteration goal
- Team selects items it can complete to the definition of done
- Breaks items into tasks where useful
- Based on recent velocity and actual capacity

TEMPLATE

Release forecast worksheet

Turn backlog size and velocity history into an honest, range-based release forecast.

RELEASE / GOAL	REMAINING POINTS	VELOCITY RANGE (LOW–HIGH)	ITERATIONS NEEDED (RANGE)	FORECAST DATE RANGE	KEY RISKS AND ASSUMPTIONS
R2 – Merchant QR payments	*240*	*28–32*	*8–9*	*16–18 weeks from 2 Feb*	*Payment-switch certification takes ≤ 3 weeks*

Baseline and manage the schedule

1. **Prepare to suit the approach** A network schedule for predictive work, a release plan for agile work, and milestones that join them on hybrid projects.
2. **Coordinate** Check dates against other projects, shared environments, operations' busy periods and holidays such as Eid or year-end.
3. **Baseline** Get the schedule, or the milestone plan, approved as the reference for measuring progress.
4. **Execute the schedule plan** Update progress at an agreed rhythm and keep the critical path or release forecast current.

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5. **Analyse variation** Compare actuals with the baseline or forecast, find causes and act before slips grow.
6. **Change formally** Re-baseline only through change control, never quietly.

SCENARIO · DECIDE · USING FLOAT

In the clinic example, the procurement officer asks to delay activity C (procure tablets) by 3 days to combine it with another order and get a discount. What is the impact, and what should you do?

(A) Refuse — any delay is unacceptable.

(B) Approve it without telling anyone; there is float.

(C) Approve it after confirming the impact: end date unchanged, but E shifts by 3 days and the path A–C–E–F keeps only 1 day of float. Inform the configuration lead and update the schedule and risk log.

(D) Approve it and add 3 days to the project end date to be safe.

Circle your choice. Answer and feedback: module answer key.

Key terms

Critical path
The longest sequence of dependent activities through a schedule network; it determines the shortest possible project duration.

Free float
The amount of time an activity can be delayed without delaying the early start of any successor.

Schedule baseline
The approved version of the schedule used to measure progress, changed only through formal change control.

Lead
An overlap that lets a successor start before its predecessor finishes.

Total float
The amount of time an activity can be delayed without delaying the project finish date.

Velocity
The amount of work, usually in story points, a team completes to the definition of done in one iteration.

Lag
A required waiting time between a predecessor and its successor.

Release plan
A forecast of which backlog items are likely to be delivered in upcoming releases, based on velocity ranges.

KNOWLEDGE CHECK

6. An activity has ES = 5, EF = 9, LS = 8, LF = 12. Its total float is:

(A) 0 days
(B) 3 days
(C) 4 days
(D) 7 days

7. In a forward pass, when an activity has two predecessors finishing on day 7 and day 10, its ES is:

(A) 7
(B) 8.5
(C) 10
(D) 17

8. Negative total float indicates that:

(A) The required finish date is earlier than the network allows
(B) Resources are idle and can be moved to other work
(C) The project is ahead of schedule on its critical path
(D) The critical path has changed to a different set of activities

9. Remaining backlog is 180 points; velocity ranges from 25 to 30. How many iterations are forecast?

(A) 5–6
(B) 6–8
(C) 7–9
(D) 8–10

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10. A hybrid project's schedule shows a go-live in the same week as the bank's year-end closing, when operations staff are unavailable. What should the project manager do before baselining?

(A) Baseline the schedule and deal with the clash nearer the time

(B) Ask operations to postpone year-end closing

(C) Coordinate with operations and adjust the plan or the go-live window before baselining

(D) Remove operations staff from the go-live plan

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · RESOURCE · LESSON 3 · 30 MIN

Resources: Plan, Acquire, Optimise & Compress

Turn requirements into people and equipment plans, secure them, and adjust schedules to real availability

- Define resource needs from scope and plan how to acquire them
- Negotiate for shared resources and protect team capacity
- Distinguish resource levelling from resource smoothing
- Compress a schedule with crashing or fast-tracking and explain the trade-offs

A plan is only as real as the people, equipment and materials behind it. **Resource planning** turns the scope into the skills, roles, equipment, materials and facilities needed, shows when each is needed, and decides how to get them. Agile teams usually work as stable, cross-functional teams, so the question becomes *what capacity do we have?* rather than *who do we assign to each task?*. Predictive projects often assemble people from functional departments and must negotiate for them.

Define and plan resources from requirements

1. **Start from scope** Use the WBS or backlog to list the skills and physical resources each deliverable needs.
2. **Build a resource breakdown** Group needs by type: people by skill, equipment, materials, facilities, licences.
3. **Estimate quantity and timing** How many, when and for how long; note lead times for materials and equipment.
4. **Choose how to acquire** Internal staff, negotiation with functional managers, hiring, contractors or vendors.
5. **Record availability** Resource calendars: holidays, shifts, part-time allocations and shared commitments.
6. **Plan development** Training, onboarding and team-building needed to make people effective.

Ways to acquire resources

SOURCE	BEST WHEN	WATCH OUT FOR
Dedicated internal team	Long-lived product or high-knowledge work	Idle capacity if work is uneven
Shared from functional departments	Specialist skills needed part-time	Competing priorities; negotiate early
Contractors	Short peaks or rare skills	Knowledge leaving at contract end
Vendor-supplied team	Packaged solution or outsourced delivery	Contract terms, control and handover
Physical resources by lease or purchase	Equipment, sites, materials	Lead times, import duties, storage

EXAMPLE · NEGOTIATING FOR A SPECIALIST

A functional manager in a Chattogram power plant refuses to release the only certified protection-relay engineer for your upgrade. Instead of escalating at once, find the manager's interest: his own outage maintenance in the same month. Options include shifting your testing by two weeks, sharing the engineer half-days, or funding a contractor to cover his routine work. Escalate to the sponsor only if no option meets both needs.

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Optimising resource availability

- Keep teams stable; constant reshuffling destroys productivity and knowledge
- Limit multitasking: a person split across four projects completes each one more slowly
- Cross-train so that more than one person can do critical work
- Plan agile work by team capacity per iteration, not by individual task assignment
- Review the resource plan at each phase or release and release resources you no longer need

Schedules also have to be **resource-feasible**. A network may say two activities run in parallel, but if both need the same certified electrician, they cannot. Resource optimisation adjusts the schedule to match real availability.

Resource levelling vs resource smoothing

Resource levelling

- Resolves over-allocation or limited availability
- Activities are delayed until resources are free
- Can change the critical path
- May extend the end date
- Used when resource limits are hard

Resource smoothing

- Evens out peaks and valleys in demand
- Moves activities only within their float
- Does not change the critical path
- End date unchanged
- May not remove all over-allocation

EXAMPLE · LEVELLING VS SMOOTHING ON SITE

Levelling: one licensed medical-gas engineer is needed by two parallel activities. One must wait, so the finish moves out by 4 days. **Smoothing:** painting needs 12 painters in week 6 but only 4 in week 7. Because painting the corridor has 5 days of float, you shift part of it into week 7 and use 8 painters each week — the end date does not move.

Sooner or later someone will ask for the project to finish earlier. Compression is possible, but never free: you pay with **money**, **risk** or **scope**. Your job is to make those trade-offs clear before anyone commits.

Crashing vs fast-tracking

Crashing

- Add resources to critical activities to shorten them
- Overtime, extra crews, faster equipment, express delivery
- Increases cost
- Works best on effort-driven activities
- Diminishing returns — more people do not always mean less time

Fast-tracking

- Overlap activities normally done in sequence
- Change discretionary FS links to SS or add leads
- Little direct cost
- Increases risk of rework
- Only possible where dependencies are discretionary

Crash cost slope

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost slope	`(Crash cost − Normal cost) ÷ (Normal duration − Crash duration)`	Extra cost per day saved. Crash the critical activity with the lowest slope first.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Example	`Activity D: normal 5 d at Tk 100,000; crash 3 d at Tk 160,000`	(160,000 – 100,000) ÷ (5 – 3) = Tk 30,000 per day saved.

SCENARIO · DECIDE · TWO WEEKS EARLIER, PLEASE

Your sponsor asks you to bring forward the launch of a garment buyer's order-tracking portal by two weeks to match a trade fair. The critical path runs through design, build and user acceptance testing. What should you do FIRST?

(A) Immediately approve overtime for the whole team.

(B) Analyse the critical path and present options — fast-track design and build where logic is discretionary, crash specific critical activities, or reduce the scope released for the fair — with cost, risk and quality impacts for each.

(C) Tell the sponsor it is impossible.

(D) Cut user acceptance testing to save two weeks.

Circle your choice. Answer and feedback: module answer key.

Key terms

Crashing
Shortening the schedule by adding resources to critical activities, usually at extra cost.

Fast-tracking
Shortening the schedule by overlapping activities normally done in sequence, at a higher risk of rework.

Resource breakdown structure
A hierarchy of the resources a project needs, grouped by category and type.

Resource calendar
A record of when each resource is available, including shifts, holidays and part-time allocations.

Resource levelling
Adjusting start and finish dates to fit limited resources; may extend the project end date.

Resource smoothing
Adjusting activities within their float to even out resource demand without changing the end date.

Crash cost slope
The extra cost per unit of time saved when an activity is shortened: (crash cost – normal cost) ÷ (normal duration – crash duration).

KNOWLEDGE CHECK

11. Which technique overlaps activities that were planned in sequence?
    (A) Crashing
    (B) Resource levelling
    (C) Fast-tracking
    (D) Smoothing

12. Crash cost Tk 90,000 for 4 days; normal cost Tk 60,000 for 7 days. The cost slope is:
    (A) Tk 7,500/day
    (B) Tk 10,000/day
    (C) Tk 15,000/day
    (D) Tk 30,000/day

13. Which technique may extend the project end date?
    (A) Resource levelling
    (B) Fast-tracking
    (C) Resource smoothing
    (D) Using available float

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14. A functional manager will not release a specialist your project needs next month because his own team has a peak. What should the project manager do FIRST?

(A) Escalate to the sponsor immediately
(B) Hire a contractor without discussing it
(C) Understand the manager's constraints and explore options that meet both needs
(D) Remove the specialist's work from the scope

15. A developer is assigned to four projects at 25% each and all four are late. What is the MOST likely cause?

(A) Context switching and waiting are slowing all four projects
(B) The developer lacks the skills needed
(C) The estimates for the four projects were padded
(D) The projects use different development approaches

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MODULE 5 · RESOURCE · LESSON 4 · 32 MIN

Budgets, Reserves & Financial Planning

Analyse financial needs, build a cost baseline, size reserves and plan how spend will be tracked and reported

- Aggregate estimates into a cost baseline and project budget
- Quantify contingency for risks and distinguish it from management reserve
- Plan spend tracking and financial reporting, including for agile teams
- Anticipate financial challenges such as inflation, exchange rates and funding limits

Cost planning turns estimates into a **time-phased budget** that the organisation can fund and you can control. A sound budget does three things: it covers the planned work, it protects against identified risks, and it shows when money will be needed.

Cost categories worth separating

CATEGORY	EXAMPLES	WHY IT MATTERS
Direct vs indirect	Contractor labour, materials vs office rent, shared IT	Indirect costs are often allocated by formula; agree the rule early
Fixed vs variable	Software licence fee vs cloud usage per transaction	Variable costs grow with scope and volume
One-time vs recurring	Implementation vs annual support and hosting	Business cases fail when recurring costs are forgotten

How a project budget is built (top first)

Project budget	`Cost baseline + management reserve`
Management reserve	`For unknown-unknowns` `Outside the baseline; needs management approval`
Cost baseline	`Approved, time-phased budget` `Control accounts + contingency reserves`
Contingency reserves	`For identified risks (known-unknowns)` `Inside the baseline`
Work package and activity estimates	`Bottom-up costs of the planned work`

Worked example — outpatient wing (Tk crore)

LINE	AMOUNT	RUNNING TOTAL
Sum of work package estimates	4.20	4.20
+ Contingency reserve (identified risks, e.g. soil conditions, import duty changes)	0.40	4.60 = cost baseline
+ Management reserve (unforeseen events)	0.30	4.90 = project budget

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Contingency reserve vs management reserve

Contingency reserve

- Covers identified risks and their responses
- Included in the cost baseline
- Usually controlled by the PM within agreed rules
- Sized from risk analysis (e.g. expected value or simulation)
- Unused amounts can be released as risks pass

Management reserve

- Covers unforeseen, unidentified risks
- Not in the cost baseline but in the total budget
- Released only with management or sponsor approval
- Often set as a policy percentage
- Using it changes the cost baseline via change control

INSIGHT · THE S-CURVE AND FUNDING

Plotting the cumulative cost baseline over time usually produces an **S-curve**: slow spending at the start, a steep middle and a flattening end. Compare it with funding tranches from the finance team or donor; if planned spend exceeds available funding in any period, re-sequence the work — this is called **funding limit reconciliation**.

INSIGHT · FUNDING AGILE WORK

Many organisations fund **persistent teams** rather than fixed scopes. If a team of eight costs Tk 24 lakh per two-week sprint, a release forecast of 8–10 sprints means Tk 1.92–2.40 crore. The budget is fixed per period; the product owner steers what the money buys through the backlog. Financial control then focuses on **value per sprint** and on stopping when value falls.

Plan spend tracking and financial reporting

1. **Set the cost structure** Align WBS or team cost codes with the finance system so actual costs land in the right place.
2. **Agree the reporting rhythm** Monthly for most sponsors; at each gate for governance; as the donor or lender requires.
3. **Choose the measures** Actual versus baseline, forecast at completion, reserve use, cash flow against funding.
4. **Track commitments** Record purchase orders and contracts as committed costs, not only invoices paid.
5. **Define thresholds** Agree the variance that triggers escalation, for example ±5% of a control account.

Anticipating financial challenges

- Inflation in materials and wages over a multi-year project
- Exchange-rate movements on imported equipment or foreign-currency contracts
- Funding released in tranches, by fiscal year or by donor milestone
- Import duties, taxes and VAT that change mid-project
- Cash-flow gaps when spend runs ahead of funding
- Interest costs when the project is financed by loans

Single-number estimates hide risk. **Monte Carlo simulation** runs a schedule or cost model thousands of times, each time drawing random values from the ranges you provide (for example the O–M–P range of each activity). The result is a distribution of possible outcomes and the probability of meeting any date or budget.

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EXAMPLE · READING A SIMULATION

The deterministic plan says the wing will be ready in **13 months**. Simulation shows only about a 20% chance of achieving that. **P50 = 14.5 months**, **P80 = 16 months**, **P90 = 17 months**. The sponsor chooses to commit publicly to 16 months (P80) and to treat 13 months as an internal stretch target. On agile work, a similar simulation can sample past weekly throughput to forecast when a set of backlog items will be finished.

SCENARIO · DECIDE · 'SPARE' MONEY

Halfway through the outpatient wing project, a department head asks you to use Tk 25 lakh of the contingency reserve to add a staff café, saying the money is 'unused anyway'. What should you do?

(A) Agree; the reserve has not been needed so far.

(B) Explain that contingency is reserved for identified risks; treat the café as a change request with its own impact analysis and funding decision by the sponsor.

(C) Use the management reserve instead without telling the sponsor.

(D) Reject the idea outright without passing it on.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · BUDGETING A PRIVATE HOSPITAL WING IN CHATTOGRAM

A private hospital in Chattogram is adding a 60-bed wing. The bottom-up estimate of all work packages is Tk 38 crore. The risk register identifies piling problems on reclaimed land, possible import duty increases on medical equipment and monsoon delays. Expected values of these risks total Tk 3.2 crore. The board's policy sets management reserve at 5% of the cost baseline. Funding arrives in four quarterly tranches of Tk 11 crore each. A Monte Carlo cost simulation shows P50 = Tk 41 crore and P80 = Tk 43.5 crore.

1. Calculate the cost baseline and total project budget.

2. Who controls each reserve?

3. What does the funding schedule mean for planning?

4. At what confidence level would you recommend the board approve the budget, and why?

Model answer: module answer key.

Key terms

Contingency reserve
Budget or time within the baseline set aside for identified risks.

Management reserve
Budget outside the cost baseline for unforeseen work; its use requires management approval.

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Cost baseline
The approved, time-phased budget, including contingency but excluding management reserve, used to measure cost performance.

S-curve
The typical shape of cumulative planned spend over time: slow start, steep middle, flat end.

Monte Carlo simulation
A technique that runs a schedule or cost model thousands of times with random values from estimate ranges to produce probabilities.

Time-phased budget
A budget spread over the periods in which costs are expected, used to plan funding and measure performance.

Funding limit reconciliation
Comparing planned spending with available funding by period and adjusting the schedule if needed.

Committed cost
Money already promised through contracts or purchase orders, even if not yet invoiced or paid.

KNOWLEDGE CHECK

16. Work packages total Tk 2.0 crore, contingency is Tk 0.2 crore and management reserve is Tk 0.1 crore. The cost baseline is:
    - (A) Tk 2.0 crore
    - (B) Tk 2.2 crore
    - (C) Tk 2.3 crore
    - (D) Tk 2.1 crore

17. Management reserve is intended for:
    - (A) Unforeseen work within scope (unknown-unknowns)
    - (B) Identified risks with planned responses (known-unknowns)
    - (C) New features requested by users after sign-off
    - (D) Team bonuses and recognition at project close

18. A Monte Carlo result says 'P80 = 16 months'. This means:
    - (A) The project will take exactly 16 months to complete
    - (B) There is only a 16% chance of finishing on time
    - (C) About 80% of simulated runs finished within 16 months
    - (D) The project will be 80% complete at month 16

19. Comparing the time-phased cost baseline with available funding by period is called:
    - (A) Benefits realisation planning
    - (B) Schedule crashing
    - (C) Resource smoothing analysis
    - (D) Funding limit reconciliation

20. An agile team costs Tk 12 lakh per sprint. The release forecast is 6–8 sprints. What budget range should be reported for the release?
    - (A) Tk 72–96 lakh
    - (B) Tk 12–24 lakh
    - (C) Tk 60–84 lakh
    - (D) Tk 96–120 lakh

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MODULE 5 · RESOURCE · LESSON 5 · 35 MIN

Earned Value, Forecasting & Financial Control

Measure progress in money, read the variances, forecast the finish and work with governance on variations

- Calculate PV, EV, AC and the core variances and indices
- Choose the right EAC formula and compute ETC, VAC and TCPI
- Apply earned value ideas to agile releases
- Monitor financial variation and escalate through governance

'We have spent 42% of the budget and we feel about 40% done' tells a sponsor almost nothing. **Earned value management (EVM)** combines scope, schedule and cost into one measurement system so you can say exactly how much work has been accomplished for the money spent and time elapsed — and forecast where you will end up.

The three base measures

PV

Planned value

The budgeted cost of the work **scheduled** to be done by now. The baseline line on the S-curve.

EV

Earned value

The budgeted cost of the work **actually completed** by now. EV = % complete × BAC (or the sum of the budgets of completed work packages).

AC

Actual cost

What you have **really spent** on the work completed by now.

BAC

Budget at completion

The total approved budget for the work (the cost baseline, without management reserve).

Variances and indices

MEASURE	FORMULA	WHAT IT TELLS YOU
Schedule variance	`SV = EV − PV`	Negative means behind schedule (in value terms).
Cost variance	`CV = EV − AC`	Negative means over budget for the work done.
Schedule performance index	`SPI = EV ÷ PV`	Below 1.0: progressing slower than planned. 0.9 = earning 90 paisa of planned work per taka planned.
Cost performance index	`CPI = EV ÷ AC`	Below 1.0: getting less value than spent. 0.8 = 80 paisa of value per taka spent.

INSIGHT · THE MEMORY TRICK

EV comes first in every variance and index formula. A negative variance or an index below 1.0 is always bad news; positive or above 1.0 is good news.

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Forecasts

MEASURE	FORMULA	WHAT IT TELLS YOU
EAC — typical (CPI continues)	`EAC = BAC ÷ CPI`	Use when current cost performance is expected to continue. The most common default.
EAC — atypical (variance was a one-off)	`EAC = AC + (BAC − EV)`	Use when the overrun had a cause that will not recur; remaining work runs at budget.
EAC — CPI and SPI both matter	`EAC = AC + (BAC − EV) ÷ (CPI × SPI)`	Use when a fixed date is pushing you to spend more to catch up.
EAC — new estimate	`EAC = AC + bottom-up ETC`	Use when the original plan is no longer valid; re-estimate the remaining work.
Estimate to complete	`ETC = EAC − AC`	How much more money is needed.
Variance at completion	`VAC = BAC − EAC`	Negative means expected to finish over budget.
To-complete performance index	`TCPI = (BAC − EV) ÷ (BAC − AC)`	The CPI you must achieve on remaining work to finish on BAC. Use (BAC − EV) ÷ (EAC − AC) if an EAC has been approved.

CASE STUDY · WORKED EXAMPLE: A WAREHOUSE APP IN GAZIPUR

A logistics firm is building a warehouse management app. **BAC = BDT 1,000,000**, planned evenly over 10 months. At the end of month 4, the plan said 40% of the work should be done. The team has actually completed 35% of the work and spent **BDT 420,000**.

1. Calculate PV, EV, SV, CV, SPI and CPI.
2. Calculate EAC (typical), ETC, VAC and TCPI to BAC.
3. Calculate EAC if the overrun is judged atypical, and the CPI × SPI version.
4. Explain the status to the sponsor in two sentences.

Model answer: module answer key.

Reading SPI and CPI together

SPI	CPI	PLAIN-LANGUAGE STATUS	TYPICAL RESPONSE
> 1	> 1	Ahead and under budget	Check estimates were not padded; consider releasing funds
> 1	< 1	Ahead but over budget	Maybe crashing or overtime — is the speed worth the cost?
< 1	> 1	Behind but under budget	Often under-resourced; add capacity if the date matters

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SPI	CPI	PLAIN-LANGUAGE STATUS	TYPICAL RESPONSE
< 1	< 1	Behind and over budget	Root-cause analysis, re-plan, escalate if beyond tolerance

WATCH OUT · EVM TRAPS

1) Measuring % complete by gut feel — use rules such as **0/100**, **50/50** or milestone weights. 2) SPI drifts back to 1.0 at the end of every project because all work eventually gets 'earned' — late in a project, check the critical path too. 3) EVM says nothing about quality: work earned but full of defects is not value.

Earned value for an agile release

MEASURE	FORMULA	WHAT IT TELLS YOU
Percent complete	`Points done ÷ points planned for the release`	Uses the definition of done, so partly finished stories count zero.
Earned value	`EV = % complete × release budget`	Value of the work actually done, in money.
Planned value	`PV = sprints elapsed ÷ sprints planned × release budget`	Budget for the work that should be done by now.
Indices	`CPI = EV ÷ AC; SPI = EV ÷ PV`	Read exactly as in predictive projects.

PRO TIP · VARIANCES BELONG TO GOVERNANCE

When a forecast moves outside an agreed tolerance, do not quietly absorb it. Report the variance, its cause and the options, and let the body with authority decide: release management reserve, reduce scope, extend the date or accept the overrun. Inside tolerance, act within your own authority and record what you did.

TEMPLATE **EVM tracker**

Update at each reporting period to see trends rather than single snapshots.

PERIOD	PV	EV	AC	CPI	SPI	EAC (BAC ÷ CPI)
Month 4	*400,000*	*350,000*	*420,000*	*0.83*	*0.88*	*1,200,000*

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SCENARIO · DECIDE · THE CPI CONVERSATION

At month 6, CPI = 0.78 and the TCPI to BAC is 1.35. The sponsor asks you to 'tighten up and finish on budget'. What is the BEST response?

(A) Agree and promise to finish on budget through extra effort.

(B) Explain that a TCPI of 1.35 is very unlikely to be achieved and present options: approve a revised EAC, reduce scope or change approach.

(C) Stop reporting CPI since it upsets the sponsor.

(D) Cut testing to save money.

Circle your choice. Answer and feedback: module answer key.

Key terms

Earned value (EV)
The budgeted value of the work actually completed to date.

To-complete performance index (TCPI)
The cost efficiency that must be achieved on remaining work to meet a target (BAC or EAC).

Actual cost (AC)
The cost actually incurred for the work performed by a given date.

Estimate at completion (EAC)
The expected total cost of the project when all work is complete, based on performance so far.

Cost performance index (CPI)
EV ÷ AC — the cost efficiency of work performed; below 1.0 means over budget.

Planned value (PV)
The budgeted cost of the work scheduled to be done by a given date.

Schedule performance index (SPI)
EV ÷ PV; below 1.0 means less work has been done than planned.

Budget at completion (BAC)
The total approved budget for the work, the sum of all planned value.

KNOWLEDGE CHECK

21. EV = 90,000, PV = 100,000, AC = 120,000. What is CPI?
    (A) 0.90 (B) 0.75
    (C) 1.11 (D) 1.33

22. Using the same numbers, what is SV?
    (A) −30,000 (B) +10,000
    (C) −10,000 (D) −20,000

23. BAC = 500,000 and CPI = 0.80. Assuming performance continues, what is the EAC?
    (A) 625,000 (B) 400,000
    (C) 580,000 (D) 600,000

24. A one-time cost spike came from a hardware price shock that will not recur. Which EAC formula fits best?
    (A) BAC ÷ CPI (B) BAC × SPI
    (C) AC + (BAC − EV) ÷ (CPI × SPI) (D) AC + (BAC − EV)

25. A release is budgeted at Tk 50 lakh for 200 points. After 4 of 8 planned sprints, 80 points are done and Tk 26 lakh has been spent. What is the CPI?
    (A) About 0.77 (B) About 1.30
    (C) About 0.80 (D) About 1.25

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MODULE 5 · RESOURCE

Answer key

Knowledge checks

1. B Analogous estimating uses comparable past projects and needs little detail — ideal early on.

2. B (2 + 4×5 + 14) ÷ 6 = 36 ÷ 6 = 6.

3. A Parametric estimating multiplies a quantity by a unit rate derived from historical data.

4. D Ranges show the uncertainty that exists, so decisions about reserves and commitments are informed.

5. D Different votes reveal different understanding; discussing them improves shared understanding and the estimate.

6. B TF = LS − ES = 8 − 5 = 3 (or LF − EF = 12 − 9 = 3).

7. C A successor can only start when all predecessors finish, so ES takes the largest EF: 10.

8. A Negative float means a constraint date cannot be met with the current logic and durations.

9. B 180 ÷ 30 = 6; 180 ÷ 25 = 7.2, rounded up to 8. So 6–8 iterations.

10. C Schedules must be coordinated with other projects and operations before they are baselined; known clashes should not be built into the reference plan.

11. C Fast-tracking performs activities in parallel or with overlap, increasing rework risk.

12. B (90,000 − 60,000) ÷ (7 − 4) = 30,000 ÷ 3 = Tk 10,000 per day.

13. A Levelling delays activities to respect resource limits, which can change the critical path and push the end date.

14. C Negotiation starts with understanding the other side's interests. Escalation comes after options have been explored.

15. A Splitting people across many projects adds switching time and delays; limiting multitasking usually speeds up all the work.

16. B Cost baseline = estimates + contingency = 2.2. Management reserve is added on top to reach the total budget of 2.3.

17. A Management reserve covers unidentified risks and requires management approval to use.

18. C P80 is the value at or below which 80% of simulated outcomes fall.

19. D Funding limit reconciliation adjusts the timing of work so planned spending fits within funding limits.

20. A Cost per sprint × forecast sprints: 12 × 6 = 72 and 12 × 8 = 96, so Tk 72–96 lakh.

21. B CPI = EV ÷ AC = 90,000 ÷ 120,000 = 0.75.

22. C SV = EV − PV = 90,000 − 100,000 = −10,000.

23. A EAC = BAC ÷ CPI = 500,000 ÷ 0.80 = 625,000.

24. D When the variance is atypical, assume the remaining work will run at budget rate.

25. A EV = 80 ÷ 200 × 50 = Tk 20 lakh; CPI = EV ÷ AC = 20 ÷ 26 ≈ 0.77, so the release is over budget for the work done.

Scenarios & case studies

> **Scenario · 'Just give me one number'**
> **C** is best. Best. An analogous ROM range is honest, useful for the board and paired with a plan to reduce uncertainty.

> **Scenario · Using float**
> **C** is best. Best. You use float deliberately, understand that it is shared along the path, and communicate the reduced buffer.

> **Scenario · Two weeks earlier, please**
> **B** is best. Best. You base the response on the schedule model and give the sponsor clear trade-offs to decide on.

> **Scenario · 'Spare' money**
> **B** is best. Best. New scope goes through change control, and reserves are used only for their purpose.

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Case study · Budgeting a private hospital wing in Chattogram

Cost baseline = 38 + 3.2 = **Tk 41.2 crore**. **Management reserve** = 5% × 41.2 ≈ Tk 2.06 crore, so **project budget ≈ Tk 43.26 crore**. **Control:** the PM manages contingency within agreed rules for the identified risks; the board or sponsor approves any use of management reserve, which then updates the baseline through change control. **Funding:** total funding is Tk 44 crore, but only Tk 11 crore per quarter — the time-phased S-curve must be checked against each tranche; expensive equipment purchases may need to be re-sequenced or financed. **Confidence:** the baseline (Tk 41.2 crore) sits near P50, so there is roughly a coin-flip chance of staying within it. The total budget (≈ Tk 43.26 crore) is just under P80 (Tk 43.5 crore). Recommend the board approve funding close to P80 and treat the gap as a visible, governed risk — not hide it.

Case study · Worked example: a warehouse app in Gazipur

PV = 40% × 1,000,000 = **400,000**. EV = 35% × 1,000,000 = **350,000**. AC = **420,000**. SV = 350,000 − 400,000 = **−50,000** (behind). CV = 350,000 − 420,000 = **−70,000** (over). SPI = 350/400 = **0.875**. CPI = 350/420 ≈ **0.833**.

EAC typical = 1,000,000 ÷ 0.833 = **1,200,000**. ETC = 1,200,000 − 420,000 = **780,000**. VAC = 1,000,000 − 1,200,000 = **−200,000**. TCPI = (1,000,000 − 350,000) ÷ (1,000,000 − 420,000) = 650/580 ≈ **1.12**.

EAC atypical = 420,000 + 650,000 = **1,070,000**. EAC with CPI × SPI = 420,000 + 650,000 ÷ (0.833 × 0.875 ≈ 0.729) ≈ **1,311,000**.

To the sponsor: ‘We are about 12% behind schedule and getting 83 paisa of value for every taka spent. If this continues we will finish around BDT 1.2 million; finishing on budget would need every remaining taka to deliver 1.12 taka of planned work — far better than our current 0.83 — so we need a decision on scope, approach or budget.’

Scenario · The CPI conversation

B is best. Correct. Use the data to reset expectations and give the sponsor real choices.

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06

MODULE 6 · DELIVER

Process III: Procurement, Quality, Status & Closure

Process · Tasks 5, 7, 9 and 10 (procurement, quality, project status, closure)

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Plan procurement, select contracts, negotiate and manage suppliers
- Plan and optimise quality, including regulatory and sustainability standards
- Run agile delivery and read flow metrics
- Evaluate status with honest metrics, dashboards and well-managed artifacts
- Validate transition readiness and close phases and projects properly

LESSONS

1. Procurement, Contracts & Negotiation · 32 min
2. Quality, Compliance & the Cost of Quality · 30 min
3. Delivery Cadence & Flow: Scrum, Kanban and Flow Metrics · 32 min
4. Evaluating Project Status: Metrics, Dashboards & Artifacts · 30 min
5. Closing a Phase or Project & Transition · 28 min

MODULE 6 · DELIVER · LESSON 1 · 32 MIN

Procurement, Contracts & Negotiation

Plan what to buy, choose the right contract, negotiate on interests and manage suppliers to the finish

- Outline the procurement process from make-or-buy to contract closure
- Select a contract type that fits the scope certainty and risk
- Plan a negotiation strategy using interests, BATNA and walk-away points
- Manage suppliers and evaluate their performance

Few projects do everything in-house. Equipment, software, cloud services, contractors and consultants are bought from outside. **Procurement** is how you decide what to buy, choose the right supplier and contract, and manage the relationship until the work is accepted and closed.

Procurement in six steps

01 Make or buy — Compare in-house vs external on cost, skill, risk, control

→

02 Plan and specify — Statement of work, evaluation criteria, contract type

→

03 Solicit — Request for information, quotation or proposal

→

04 Select and award — Evaluate fairly against published criteria; negotiate

→

05 Manage — Performance, payments, changes, disputes

→

06 Close — Verify deliverables, final payment, lessons, records

DEFINITION · STATEMENT OF WORK (SOW)

The **SOW** describes what the supplier must deliver: scope, deliverables, standards, acceptance criteria, timeline and reporting. A vague SOW is the single biggest cause of supplier disputes.

Contract types and who carries the cost risk

TYPE	HOW PAYMENT WORKS	COST RISK MAINLY ON	BEST WHEN
Firm fixed price (FFP)	Agreed price for defined scope	Seller	Scope is clear and stable
Fixed price incentive fee (FPIF)	Fixed price with bonus or penalty for performance against targets, up to a ceiling	Shared, mostly seller	Clear scope; buyer wants to reward performance
Fixed price with economic price adjustment	Fixed price adjusted for agreed indices (e.g. steel, currency)	Shared	Multi-year contracts with volatile input costs
Time and materials (T&M)	Rates per hour/day plus materials	Buyer	Scope unclear, short engagements, staff augmentation
Cost plus fixed fee (CPFF)	Actual costs plus a fixed fee	Buyer	High uncertainty, research-like work
Cost plus incentive fee (CPIF)	Actual costs plus a fee that varies with performance; overruns and underruns shared	Shared, mostly buyer	Uncertain scope with measurable targets

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TYPE	HOW PAYMENT WORKS	COST RISK MAINLY ON	BEST WHEN
Cost plus award fee (CPAF)	Actual costs plus an award decided by the buyer's evaluation	Buyer	Performance is hard to measure objectively

Agile-friendly contracting options

- **Capped time and materials** — pay for time used, up to an agreed ceiling.
- **Fixed price per iteration or increment** — buyer can stop after any increment if enough value is achieved.
- **Target cost with shared pain and gain** — overruns and savings are split, aligning both sides.
- **Scope-swap clauses** — buyer can exchange not-yet-started backlog items for new items of equal size without renegotiation.
- **Early termination by agreement** — buyer may end early with a modest fee when the remaining value is low.
- **Framework agreements with call-offs** — pre-agreed terms, work ordered in small packages as needs become clear.

Negotiation is structured conflict resolution. Project managers negotiate with vendors, functional managers, sponsors and team members constantly — for people, money, dates and scope.

Positions vs interests

Position (what they say)

- ‘I need two of your testers for the whole of March.’
- ‘The price is final.’
- ‘We cannot go live before Eid.’

Interest (why they say it)

- Their own release has a regulatory deadline mid-March
- The vendor needs predictable cash flow this quarter
- Branch staff will be on leave; nobody to support users

Principled negotiation toolkit

INT

Focus on interests

Ask ‘why does that matter to you?’ until you reach the underlying need.

OPT

Invent options

Expand the pie: phased delivery, different payment terms, shared resources, scope swaps.

CRI

Use objective criteria

Market rates, benchmarks, contract terms, past data — so neither side has to back down to the other’s will.

BAT

Know your BATNA

Your Best Alternative To a Negotiated Agreement — what you will do if no deal is reached. A strong BATNA gives you the confidence to walk away from a bad deal.

ZOPA

Find the ZOPA

The zone of possible agreement lies between each side’s walk-away point. No overlap, no deal — look for new options.

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PRO TIP · PLAN THE NEGOTIATION BEFORE YOU MEET

Write down your **interests**, your **BATNA**, your **walk-away point** and the concessions you can trade, then estimate the same for the other side. The overlap between the two walk-away points is the **zone of possible agreement**. Agree internally who may make which concessions, so the supplier cannot play your team members against each other.

Managing suppliers and contracts

1. **Kick off together** Walk through the statement of work, acceptance criteria, reporting and the change process with the supplier's team.
2. **Design the solution jointly** Where the supplier brings expertise, shape the delivery solution together while keeping ownership of requirements.
3. **Monitor performance** Track delivery, quality and service levels against agreed measures; review them regularly.
4. **Control changes** Route every scope change through the contract's change clause, in writing.
5. **Pay against evidence** Link payments to accepted milestones or increments, not to effort reported.
6. **Verify objectives** Before closure, confirm the agreement's objectives were met and record lessons.

TEMPLATE **Vendor performance scorecard**

Review each key supplier monthly or per release using agreed, objective measures.

CRITERION	MEASURE	TARGET	ACTUAL	RATING (1–5)	ACTION
On-time delivery	*Milestones met on agreed date*	*90%*	*75%*	*2*	*Joint recovery plan; review weekly until two milestones are met on time*

SCENARIO · DECIDE · THE VENDOR'S CHANGE REQUEST

Your firm-fixed-price contract with a software vendor covers a hospital billing system. Users now want a new insurance-claim interface not in the statement of work. The vendor says it will cost extra. The finance manager insists the vendor should absorb it because 'the price is fixed'. What should you do?

(A) Agree with the finance manager and tell the vendor to deliver it at no cost.

(B) Check the SOW to confirm the interface is new scope, then process it through the contract's change procedure — impact, price and schedule — for an authorised decision.

(C) Ask the vendor's developers informally to add it without telling their manager.

(D) Cancel the contract and find a new vendor.

Circle your choice. Answer and feedback: module answer key.

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Key terms

Firm fixed price contract
A contract in which the seller delivers defined scope for an agreed price and carries most of the cost risk.

BATNA
Best Alternative To a Negotiated Agreement: the course of action a party will take if negotiations fail.

Statement of work (SOW)
A description of what a supplier must deliver: scope, deliverables, standards, acceptance criteria and timeline.

Service level agreement (SLA)
An agreement defining the measurable level of service a supplier must provide.

Time and materials contract
A contract that pays agreed rates for time and materials used, with most cost risk on the buyer.

Cost-reimbursable contract
A contract that pays the seller's allowable costs plus a fee, leaving most cost risk with the buyer.

Make-or-buy analysis
Comparing doing work in-house with buying it, on cost, capability, risk and control.

Zone of possible agreement (ZOPA)
The range between the two parties' walk-away points in which a deal is possible.

KNOWLEDGE CHECK

1. Under which contract type does the buyer carry the most cost risk?
   - (A) Firm fixed price
   - (B) Fixed price incentive fee
   - (C) Fixed price with economic price adjustment
   - (D) Cost plus fixed fee
2. Which document defines what a supplier must deliver?
   - (A) Risk management plan
   - (B) Sprint backlog
   - (C) Statement of work
   - (D) Benefits register
3. Which clause lets an agile buyer exchange unstarted backlog items for new ones of equal size?
   - (A) Scope-swap clause
   - (B) Force majeure
   - (C) Liquidated damages
   - (D) Retention clause
4. A vendor’s BATNA becomes weaker. What is the likely effect?
   - (A) The vendor gains more negotiating power
   - (B) The vendor loses negotiating power
   - (C) BATNA has no effect
   - (D) The contract becomes void
5. ‘We need the server room by Friday’ is best described as...
   - (A) an interest
   - (B) a BATNA
   - (C) an objective criterion
   - (D) a position
6. A cloud supplier has missed three of four milestones, but its account manager reports that the team is 'working hard'. What should the project manager do FIRST?
   - (A) Terminate the contract for poor performance
   - (B) Review performance against the agreed measures with the supplier and agree a recovery plan
   - (C) Accept the explanation, since effort is high
   - (D) Withhold all payments until the end of the project

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MODULE 6 · DELIVER · LESSON 2 · 30 MIN

Quality, Compliance & the Cost of Quality

Plan quality in, meet regulatory standards, manage the cost of quality and sustainability, and keep reviewing

- Gather quality requirements, including regulatory and sustainability standards
- Distinguish quality planning, assurance and control
- Classify costs of quality and argue for prevention
- Use quality tools and ongoing reviews to improve deliverables

Quality means the result **meets its requirements and is fit for use**. It is not the same as luxury. A low-cost budget phone can be high quality; an expensive one that crashes is not. Modern quality management prevents problems early instead of inspecting them out at the end.

DEFINITION · QUALITY VS GRADE

Grade is a category for products with the same use but different features (basic vs premium hotel room). **Quality** is how well the result meets its requirements. Low grade may be acceptable; low quality never is.

Three parts of quality management

Plan quality

- Identify standards and requirements
- Set measurable quality metrics
- Define acceptance criteria and definition of done
- Choose reviews, tests and tools

Manage quality (assurance)

- Is the process sound?
- Audits, process reviews, retrospectives
- Design for quality, checklists, training
- Improve the way work is done

Control quality

- Is the product right?
- Inspection, testing, measurement
- Compare results with criteria
- Verify deliverables before acceptance

WATCH OUT · REGULATORY STANDARDS ARE QUALITY REQUIREMENTS

Many deliverables must meet external rules: building and fire codes, medical-device and drug standards, food safety, data-protection law, accessibility rules for public websites. Gather these with the other quality requirements at the start, plan how **evidence** of compliance will be produced, such as test reports, certificates and inspection records, and include them in acceptance criteria.

Cost of quality (CoQ)

MEASURE	FORMULA	WHAT IT TELLS YOU
Cost of quality	`CoQ = Cost of conformance + Cost of non-conformance`	Total cost spent to achieve quality plus the cost of failing to achieve it.
Cost of conformance	`Prevention + Appraisal`	Money spent to avoid failures (training, design reviews) and to find them (testing, inspection).
Cost of non-conformance	`Internal failure + External failure`	Failures found before delivery (rework, scrap) and after delivery (warranty, recalls, lost customers).

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Cost of quality — examples

CATEGORY	SOFTWARE PROJECT	CONSTRUCTION / FACTORY
Prevention	Coding standards, pair programming, training	Method statements, worker training, supplier qualification
Appraisal	Code reviews, automated tests, security scans	Material testing, inspections, quality audits
Internal failure	Bugs fixed before release, rework	Rework of faulty concrete, rejected garments, scrap
External failure	Production outages, data breaches, refunds	Buyer claims, product recalls, lost orders, reputation damage

Quality and sustainability together

- Rework and scrap are both a failure cost and an environmental cost: every defect wastes material and energy
- Specify durability, repairability and energy efficiency as measurable quality requirements
- Include supplier standards on materials, labour and waste in quality audits
- Measure whole-life cost and impact, not only the cost to build

Core quality tools

TOOL	WHAT IT SHOWS	USE WHEN
Cause-and-effect (fishbone) diagram	Possible causes grouped by category	Exploring the root causes of a problem
Pareto chart	Causes ranked by frequency or impact	Choosing which few causes to fix first
Control chart	Process results over time against control limits	Checking whether a process is stable
Check sheet	Tally of occurrences by type	Collecting defect data on the floor
Histogram	Distribution of a measurement	Seeing the spread and shape of results
Scatter diagram	Relationship between two variables	Testing whether one factor drives another
Flowchart	Steps and decision points in a process	Finding where errors or delays enter

Agile quality practices

- A visible **definition of done** applied to every increment
- Acceptance criteria agreed before work starts
- Test automation and continuous integration on every change
- Code review or pairing as standard
- Retrospectives that tackle root causes, not just symptoms
- Small batches, so problems surface early

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Release quality checklist

☐ All acceptance criteria demonstrated to the product owner or customer
☐ Definition of done met for every item in the release
☐ Automated regression tests passing
☐ Security and data-protection checks completed
☐ Performance tested under expected peak load
☐ Accessibility checked for key user journeys
☐ Known defects reviewed and accepted or fixed
☐ User guides, release notes and training materials updated
☐ Rollback plan prepared and tested
☐ Support and operations teams briefed and ready

PRO TIP · REVIEW QUALITY CONTINUOUSLY

Quality is not a gate at the end. Hold short quality reviews at an agreed rhythm, such as each sprint or each month, looking at defect trends, audit findings and customer complaints. Pick one improvement each time, try it, check the result and keep it if it works. Module 8 builds this into a full continuous-improvement habit.

SCENARIO · DECIDE · SKIP THE TESTS?

With two days left in a sprint, developers suggest skipping the regression test suite so they can finish all planned items. The definition of done requires regression tests to pass. What should you encourage?

(A) Skip the tests this once and catch up next sprint.

(B) Keep the definition of done; finish fewer items fully, and discuss with the product owner which items to move back to the backlog.

(C) Mark the items done but add a note that tests are pending.

(D) Extend the sprint by two days.

Circle your choice. Answer and feedback: module answer key.

Key terms

Cost of quality
The total cost of conformance (prevention and appraisal) and non-conformance (internal and external failure).

Grade
A category for products with the same use but different features; low grade can be acceptable, low quality cannot.

Quality assurance
Activities that check whether the processes used to create deliverables are effective.

Quality control
Inspecting and testing deliverables to confirm they meet requirements.

Prevention cost
Money spent to stop defects happening, such as training, standards and design reviews.

Failure cost
The cost of defects found before delivery (internal) or after delivery (external).

Pareto chart
A bar chart that ranks causes by frequency or impact to show the few causes behind most problems.

KNOWLEDGE CHECK

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7. Which is an example of an appraisal cost?
   - (A) Staff training in quality methods
   - (B) Warranty claims from customers
   - (C) Customer refunds after complaints
   - (D) Product testing and inspection
8. A buyer rejects a shipment because of defects found at their warehouse. This is:
   - (A) Prevention and planning cost
   - (B) External failure cost
   - (C) Appraisal cost
   - (D) Internal failure cost
9. Which tool helps choose the few causes responsible for most problems?
   - (A) Pareto chart
   - (B) Scatter diagram
   - (C) Flowchart
   - (D) Histogram
10. Quality assurance mainly focuses on:
    - (A) Inspecting finished deliverables for defects
    - (B) Whether the processes used are effective
    - (C) Setting the quality budget for the project
    - (D) Approving or rejecting change requests
11. A public health app must meet national accessibility rules. When should these rules be addressed?
    - (A) After launch, when users report problems
    - (B) Only if the regulator audits the app
    - (C) As quality requirements from the start, with planned evidence and acceptance criteria
    - (D) During final testing, by a specialist consultant

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MODULE 6 · DELIVER · LESSON 3 · 32 MIN

Delivery Cadence & Flow: Scrum, Kanban and Flow Metrics

Run agile delivery well and read the flow data it produces to assess progress

- Describe the Scrum accountabilities, events and artifacts
- Apply Kanban practices and set WIP limits
- Calculate and interpret lead time, cycle time, throughput and Little's Law
- Read a cumulative flow diagram to assess progress

Scrum is a lightweight framework for teams solving complex problems. It rests on **empiricism** — knowledge comes from experience, so you make work **transparent**, **inspect** it often and **adapt** quickly — and on lean thinking: reduce waste and focus on essentials. A Scrum Team is small, typically ten people or fewer, cross-functional and self-managing.

Three accountabilities

PO

Product Owner

Maximises the value of the product. Owns and orders the product backlog, sets the product goal and is one person, not a committee — though they may represent many stakeholders.

SM

Scrum Master

Accountable for the team's effectiveness. Coaches the team and organisation in Scrum, removes impediments and helps everyone understand empirical working.

DEV

Developers

Everyone who creates the increment — engineers, designers, testers, analysts. They plan the sprint, hold each other accountable and meet the definition of done.

Five events (maximum timeboxes for a one-month sprint; shorter sprints usually need less)

EVENT	PURPOSE	TIMEBOX
The Sprint	Container for all other events; a fixed period in which an increment is created	One month or less
Sprint Planning	Decide why the sprint is valuable (sprint goal), what can be done and how	Up to 8 hours
Daily Scrum	Developers inspect progress toward the sprint goal and adapt the plan for the next day	15 minutes
Sprint Review	Inspect the outcome with stakeholders and adapt the product backlog	Up to 4 hours
Sprint Retrospective	Inspect how the team worked and plan improvements in quality and effectiveness	Up to 3 hours

Artifacts and their commitments

ARTIFACT	WHAT IT IS	COMMITMENT THAT GIVES IT FOCUS
Product Backlog	Ordered, emerging list of what is needed to improve the product	Product Goal — the long-term objective for the product
Sprint Backlog	Sprint goal + selected items + the plan to deliver them	Sprint Goal — the single objective for the sprint
Increment	A usable step toward the product goal that meets the definition of done	Definition of Done — the quality standard for the increment

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One sprint at a glance

01 **Sprint Planning** — Goal and plan → 02 **Daily Scrums** — Daily inspection and adaptation → 03 **Building the increment** — Refinement continues alongside → 04 **Sprint Review** — Stakeholders inspect results → 05 **Retrospective** — Team improves how it works

PRO TIP · WHERE DOES THE PROJECT MANAGER FIT?

Scrum does not define a project manager. In practice, PMs on Scrum projects often coordinate across several teams, manage budgets, contracts, risks and external stakeholders, and protect the team's ability to work. Do not assign tasks to developers or run their Daily Scrum.

SCENARIO · DECIDE · AN URGENT REQUEST MID-SPRINT

On day 4 of a two-week sprint, the head of sales asks a developer directly to add a promotional banner for a campaign starting next week. The developer asks you what to do. What is BEST?

(A) Tell the developer to add it quickly; it is small.

(B) Direct the request to the product owner, who decides whether it is more valuable than current work and, with the developers, whether it fits without endangering the sprint goal.

(C) Cancel the sprint and start a new one including the banner.

(D) Tell the head of sales to wait until the next sprint without discussing it.

Circle your choice. Answer and feedback: module answer key.

Six core Kanban practices

K1 Visualise the work — Show every work item and workflow stage on a board so everyone sees the same picture.

K2 Limit work in progress — Cap how many items can be in each stage at once.

K3 Manage flow — Watch how work moves, spot blockages and act to keep items moving.

K4 Make policies explicit — Write down the rules: what 'ready for test' means, how urgent items are handled.

K5 Implement feedback loops — Regular replenishment, daily stand-ups at the board, delivery and service reviews.

K6 Improve collaboratively — Use data and small experiments to evolve the process together.

DEFINITION · WIP LIMIT

A **work-in-progress limit** caps the number of items allowed in a stage. When a column is full, people **stop starting and start finishing** — they help move existing items forward instead of pulling new ones. Fewer items in progress means less task-switching, faster feedback and shorter waits.

Flow metrics

METRIC	DEFINITION	USE IT TO…
WIP	Number of items started but not finished	Keep load manageable; predict cycle time

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METRIC	DEFINITION	USE IT TO…
Throughput	Number of items finished per unit of time	Forecast how much can be delivered
Cycle time	Elapsed time from when work starts on an item to when it is finished	Measure team responsiveness and predictability
Lead time	Elapsed time from when the request is made (or committed to) until it is delivered	Measure what the customer experiences
Work item age	Time elapsed since an in-progress item was started	Spot items getting stuck before they become late

Flow formulas

MEASURE	FORMULA	WHAT IT TELLS YOU
Little's Law	`Average cycle time = Average WIP ÷ Average throughput`	Holds for a reasonably stable system. Example: 12 items in progress, 6 finished per week → about 2 weeks average cycle time.
Reduce WIP to speed up	`Halve WIP at the same throughput → halve cycle time`	The fastest lever most teams have is starting less work.
Flow efficiency	`Active work time ÷ Total cycle time × 100%`	Often surprisingly low in knowledge work — items spend much of their time waiting, not being worked on.

How to read a CFD

- **Vertical** distance across the in-progress bands on a date = WIP at that moment.
- **Horizontal** distance from where an item enters progress to where it reaches Done ≈ approximate average cycle time.
- A band that **widens** = work is piling up in that stage — a bottleneck downstream of it or slow work in it.
- A **flat** Done line = nothing is being delivered — investigate immediately.
- Bands rising **smoothly and in parallel** = stable, predictable flow.
- A sudden drop in the top (backlog) line = scope removed; a jump = scope added.

SCENARIO · DECIDE · A FULL TEST COLUMN

Your board shows Test at its WIP limit of 3, with two items waiting in 'Develop – done'. Two developers say they have nothing to do because they cannot pull new work. What should the team do?

(A) Raise the Test WIP limit to 6 so developers can keep moving work forward.

(B) Have the idle developers help with testing — pairing with testers, automating checks or fixing issues found — to finish items in Test.

(C) Let the developers start new items anyway and park them.

(D) Send the developers home early.

Circle your choice. Answer and feedback: module answer key.

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Key terms

Product Owner
The accountable person in Scrum who maximises product value and owns the ordering of the product backlog.

Scrum Master
The accountable person in Scrum for team effectiveness, coaching the team and organisation and removing impediments.

Increment
A usable step toward the product goal that meets the definition of done.

WIP limit
A cap on the number of work items allowed in a workflow stage at one time.

Cycle time
The elapsed time from when work starts on an item until it is finished.

Lead time
The elapsed time from when an item is requested or committed to until it is delivered to the customer.

Throughput
The number of work items finished per unit of time.

Cumulative flow diagram
A stacked area chart showing the number of items in each workflow stage over time.

KNOWLEDGE CHECK

12. Who is accountable for ordering the product backlog?
    - (A) Product Owner
    - (B) Scrum Master
    - (C) Developers
    - (D) Project sponsor
13. The commitment for the Increment is the:
    - (A) Sprint Goal
    - (B) Product Goal
    - (C) Definition of Done
    - (D) Acceptance criteria
14. What is the maximum timebox for a Daily Scrum?
    - (A) 5 minutes
    - (B) 15 minutes
    - (C) 30 minutes
    - (D) 1 hour
15. Average WIP is 20 items and average throughput is 10 items per week. By Little's Law, average cycle time is about:
    - (A) 0.5 weeks
    - (B) 2 weeks
    - (C) 10 weeks
    - (D) 200 weeks
16. What is the main purpose of a WIP limit?
    - (A) To keep everyone busy
    - (B) To measure velocity
    - (C) To prevent stakeholders from adding work
    - (D) To reduce multitasking and improve flow
17. Which metric shows how long an unfinished item has been in progress?
    - (A) Throughput
    - (B) Work item age
    - (C) Sprint velocity
    - (D) Lead time

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MODULE 6 · DELIVER · LESSON 4 · 30 MIN

Evaluating Project Status: Metrics, Dashboards & Artifacts

Measure progress honestly, report it so people can act, and keep project artifacts current and accessible

- Read burndown, burnup and velocity charts and avoid common misuse
- Design a status dashboard with agreed RAG criteria
- Forecast with ranges and confidence levels
- Tailor, maintain and reconcile project artifacts and metrics

Agile teams rarely calculate EVM, but they still need to see whether they are on track. Their metrics are lighter, more frequent and more visual — and just as easy to misuse. (Scrum and Kanban mechanics were covered in lesson 6.3; here we focus on judging status with the data they produce, alongside the earned value measures from module 5.)

Burndown vs burnup

Burndown

- Plots work **remaining** against time
- Simple; good for a single sprint
- Hides scope change: if scope grows and the team delivers, the line looks flat
- Question it answers: 'Will we finish what we committed to?'

Burnup

- Plots work **completed** and **total scope** as two lines
- Makes scope growth visible
- Better for releases and stakeholder conversations
- Question it answers: 'When will the lines meet — and is the target moving?'

Velocity: useful, fragile

USE

Planning aid

A team's average completed points per sprint helps it decide how much to pull into the next sprint and roughly how many sprints a backlog needs.

X1

Never compare teams

Story points are relative to each team's own scale. Comparing velocity between teams is meaningless and invites gaming.

X2

Never make it a target

When velocity becomes a goal, estimates inflate and the number rises while delivered value does not.

ALT

Prefer flow measures

Throughput (items finished per week) and cycle time (start to finish per item) are harder to game and work well for forecasting.

A **status dashboard** exists so stakeholders can make decisions. If a reader cannot tell within 30 seconds what is on track, what is not and what they need to do, the dashboard has failed.

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Define RAG criteria in advance

STATUS	SCHEDULE	COST	MEANING FOR READERS
Green	Forecast within 1 week of baseline	EAC within 3% of BAC	On track — no action needed
Amber	Forecast 1–4 weeks late	EAC 3–8% over	At risk — team has a recovery plan; watch
Red	Forecast > 4 weeks late or milestone missed	EAC > 8% over	Off track — decision or help needed from the reader

WATCH OUT · THE WATERMELON PROJECT

Green on the outside, red on the inside. Projects report green for months and then turn red overnight. Causes: subjective RAG, fear of bad news, and reporting activity instead of outcomes. Fix it with objective criteria, forecasts rather than snapshots, and a culture that rewards early warnings.

What a good one-page dashboard shows

1. **Headline**
   Overall status, one-sentence summary and trend arrow (improving, stable, worsening).
2. **Decisions needed**
   What you need from the reader and by when — put this near the top.
3. **Forecast**
   Finish date and cost as ranges with confidence (e.g. P50/P80).
4. **Top risks & issues**
   The three to five that matter, with owners.
5. **Outcomes**
   Leading indicators of value: adoption, OKR key results, quality.

Forecasting honestly with throughput

1. **Gather history** Collect weekly throughput (items finished) for the last 10–15 weeks.
2. **Size the remaining work** Count backlog items, and add a realistic allowance for the scope growth you have historically seen.
3. **Simulate** Randomly sample past weekly throughputs many times to see how many weeks the remaining items take (a simple Monte Carlo).
4. **Report a range** ‘85% likely by 30 June; 50% likely by 9 June.’
5. **Re-forecast often** Update weekly as new data arrives; forecasts are living, not promises carved in stone.

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SCENARIO · DECIDE · THE DEMAND FOR ONE DATE

A marketing director wants ‘one firm date’ for an app launch. Your throughput forecast gives 50% by 12 May and 85% by 2 June. What is the BEST response?

(A) Recommend committing externally to 2 June, plan internally toward mid-May, and agree to update the forecast weekly.

(B) Give 12 May — it is the most likely date.

(C) Refuse to give any date because agile teams do not commit to dates.

(D) Add 50% to the 12 May date to be safe.

Circle your choice. Answer and feedback: module answer key.

INSIGHT · RECONCILE BEFORE YOU REPORT

Metrics drawn from different sources often disagree: timesheets say one thing, the finance ledger another, the board a third. Reconcile them before reporting, explain any difference that remains, and fix the cause. A sponsor who finds two versions of the truth stops trusting both.

Tailoring project artifacts

NEED	PREDICTIVE ARTIFACT	AGILE ARTIFACT	HYBRID CHOICE
Show progress	Schedule and earned value report	Burnup and release forecast	Milestone roadmap plus burnups
Track scope	Scope baseline and change log	Product backlog	Baseline for fixed parts, backlog for evolving parts
Record decisions	Decision register and minutes	Decision log in the team channel	One shared decision log
Track risks and issues	Risk register and issue log	Risks as backlog items; impediment board	One register reviewed at both cadences

Managing artifacts well

- Keep one source of truth for each artifact, with version history
- Name an owner and a review date for each key artifact
- Make artifacts accessible: findable, readable on mobile, usable with screen readers and in the languages people work in
- Control access to confidential material without hiding what people need
- Retire artifacts that nobody uses, and ask regularly whether the set still helps the project

Key terms

Burnup chart
A chart showing completed work and total scope over time, making scope change visible.

Burndown chart
A chart of work remaining against time, used to see whether committed work will finish.

Watermelon reporting
Status that appears green externally while the underlying reality is amber or red.

RAG status
A red, amber or green rating of status, which is useful only when its criteria are defined objectively in advance.

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Probabilistic forecast
A forecast stated as a date or amount with a confidence level, such as '85% likely by 2 June'.

Project artifact
Any document, board, log, chart or model the project creates or maintains to plan, track or record its work.

Reconciliation
Comparing figures from different sources, explaining differences and correcting the cause.

KNOWLEDGE CHECK

18. Which chart BEST reveals that scope is growing faster than the team delivers?

(A) Burnup
(B) Burndown
(C) Pie chart of story types
(D) Velocity bar chart

19. Management proposes ranking teams by velocity to reward the best. What is the main problem?

(A) Velocity is too difficult to calculate consistently across sprints
(B) Points are team-relative, so comparisons mislead and invite inflation
(C) Teams with the highest velocity will burn out sooner
(D) Velocity is only a valid measure for Kanban teams, not Scrum teams

20. What is the MOST effective way to prevent watermelon reporting?

(A) Let each PM define green in a way that suits their project
(B) Ban the colour red so teams focus on solutions, not blame
(C) Report less often so problems have time to be resolved
(D) Use objective RAG criteria and reward early bad news

21. A forecast states '85% likely by 2 June'. This is an example of...

(A) a schedule baseline
(B) a delivery guarantee
(C) a probabilistic forecast
(D) a velocity target for the team

22. Field engineers say they cannot open the project's risk register on their phones and rarely see it. What does this show?

(A) The register should be removed because it is not used
(B) A failure of artifact accessibility that should be fixed
(C) The engineers do not need to see risks
(D) The register should be emailed monthly as a large attachment

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MODULE 6 · DELIVER · LESSON 5 · 28 MIN

Closing a Phase or Project & Transition

Agree closure criteria, validate readiness for transition and finish with accepted deliverables and settled contracts

- Determine criteria for closing a phase or project
- Validate readiness for transition to operations or the next phase
- Complete closure activities, including contracts, finances, resources and lessons
- Close a terminated project with the same discipline

Closure is the part of the project life cycle most often rushed. People are pulled onto new work, the budget code is shut and documents scatter. Yet closure is where you secure acceptance, protect the benefits and capture knowledge for the next project. A project that never formally closes keeps consuming attention and money.

Criteria for closing a phase or project

- Deliverables accepted against agreed criteria by the authorised person
- Transition criteria met, such as support readiness and hypercare exit thresholds
- Contracts closed or transferred, and final invoices settled
- Financial accounts reconciled and the cost code closed
- People released with feedback and recognition
- Lessons captured and records archived
- Benefits tracking handed to a named owner

The ZPMP closing sequence

01 Confirm acceptance
Formal sign-off that deliverables meet the agreed criteria

→

02 Transition
Hand over to operations, support and the business owner

→

03 Close contracts
Verify, settle, pay, release

→

04 Close finances & release people
Final costs, return resources, recognise contributions

→

05 Capture learning & archive
Lessons, final report, records stored

→

06 Celebrate
Mark the end; thank people publicly

DEFINITION · VERIFIED VS ACCEPTED

Quality control checks that a deliverable meets specifications (it is **verified/correct**). Acceptance is the customer or sponsor formally agreeing it meets acceptance criteria (it is **accepted**). Both are needed; only acceptance closes the scope.

Handover to operations

ELEMENT	WHAT ‘DONE’ LOOKS LIKE
Operational owner	A named business owner has accepted responsibility for the product and its benefits

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ELEMENT	WHAT 'DONE' LOOKS LIKE
Support model	Service desk, escalation paths and support hours agreed; tickets routed correctly
Hypercare	An agreed period (often 2–6 weeks) of extra project support after go-live, with exit criteria
Knowledge transfer	Runbooks, configuration records, known issues list, training for support staff
Warranties & licences	Vendor warranties, licence renewals and support contracts transferred to the operational owner
Measures	Benefit and performance KPIs set up and reporting to the business owner

Validate readiness for transition

1. **Agree readiness criteria early** Define with operations what 'ready to receive' means, months before go-live.
2. **Rehearse** Run a dress rehearsal or pilot of support processes, including incident handling.
3. **Check people** Support staff trained, runbooks current, escalation paths agreed.
4. **Check data and systems** Migration verified, monitoring in place, backups tested.
5. **Decide formally** Hold a go or no-go review against the criteria and record the decision.

> **INSIGHT · CLOSURE IN AGILE AND PRODUCT SETTINGS**
> Agile teams close each **release** with acceptance at the sprint review and a retrospective, and the product often continues with a stable team. Even so, a project or funding period still ends: contracts close, the budget is reconciled, temporary staff move on and the final lessons are shared beyond the team.

Closing a contract

1. **Verify deliverables** Confirm everything contracted was delivered and accepted; check against the statement of work.
2. **Resolve open items** Settle claims, disputes and change requests — negotiation first, escalation or alternative dispute resolution if needed.
3. **Final payment** Release retention money and final invoices according to the terms.
4. **Procurement audit** Review the procurement process for lessons: what worked in selection, contract type and management.
5. **Formal closure notice** Written notice to the seller that the contract is complete; update vendor performance records.

> **WATCH OUT · EARLY TERMINATION STILL NEEDS CLOSURE**
> If a project is cancelled — strategy changed, business case collapsed — close it with the same discipline: document why, what was completed, what can be reused, how contracts end (termination clauses), and what was learned. Stopping a failing project early is a **good governance outcome**, not a failure of the PM.

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Project closure checklist

☐ Final deliverables formally accepted in writing by the authorised person

☐ Operational owner, support model and hypercare exit criteria agreed

☐ All contracts closed, claims settled and final payments made

☐ Open risks and issues closed or transferred to an owner in operations

☐ Final costs reconciled and project budget codes closed

☐ Team members released with feedback to their line managers

☐ Lessons learned captured with owners for follow-up actions

☐ Benefits realisation plan handed to the business owner

☐ Project records archived according to retention policy and searchable

☐ Final report issued and sponsor sign-off on project closure obtained

☐ Team and stakeholders thanked and successes celebrated

CASE STUDY · ERP GO-LIVE AT A GARMENT FACTORY IN ASHULIA

A knitwear exporter implemented an ERP system for planning, inventory and buyer compliance. Go-live happened in October. In December the PM was moved to a new project. In February: the vendor has not received final payment because nobody confirmed acceptance; 45 support tickets have no owner; the factory's IT staff do not know the vendor warranty expires in April; and buyers are asking for compliance reports that the system can produce but no one knows how to run.

1. Which closure activities were missed?

2. What should be done now, and by whom?

3. How would you prevent this on the next project?

Model answer: module answer key.

SCENARIO · DECIDE · THE KEEN SPONSOR

The sponsor says: 'The system is live and users seem happy. Let's skip the formal acceptance meeting — just move the team to the next project tomorrow.' What should the project manager do?

(A) Agree; users are happy so acceptance is implied.

(B) Explain the risks of skipping formal acceptance and propose a short, focused sign-off session this week alongside a staged team release.

(C) Refuse to release anyone until all documentation is perfect.

(D) Send an email stating that silence will be treated as acceptance.

Circle your choice. Answer and feedback: module answer key.

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Key terms

Formal acceptance
Documented confirmation by an authorised stakeholder that deliverables meet agreed acceptance criteria.

Contract closure
Completing all contract obligations, settling claims and payments and formally ending the contractual relationship.

Operational readiness
The state in which the receiving organisation has the people, processes, tools and data to run the result.

Go or no-go review
A formal decision point that checks readiness criteria before a release or handover proceeds.

Hypercare
A defined period of intensified support immediately after go-live, ending when agreed stability criteria are met.

Closure criteria
Agreed conditions that must be met before a phase or project is formally closed.

Transition
Moving responsibility for a deliverable from the project to operations, support or the next phase.

KNOWLEDGE CHECK

23. A project is cancelled halfway because strategy changed. What should the PM do?

(A) Stop all work immediately and release the team without records
(B) Perform closure: document status, close contracts, capture lessons
(C) Continue until the original end date to protect the investment
(D) Hand all materials to the PMO without notes and move to new work

24. Which activity confirms that the customer agrees deliverables meet acceptance criteria?

(A) Formal acceptance
(B) Quality control
(C) Procurement audit
(D) Hypercare

25. Why should open risks be transferred at closure?

(A) So the risk register can be deleted at handover
(B) To justify increasing the final project budget
(C) It is not necessary once the project has closed
(D) Some risks continue into operations and need an owner

26. A procurement audit at closure mainly aims to...

(A) identify procurement lessons for future use
(B) reopen negotiation on the final contract price
(C) penalise the vendor for any missed obligations
(D) approve the final payment to the supplier

27. Two weeks before go-live, the service desk has not been trained and no runbook exists. The sponsor wants to keep the date. What should the project manager do?

(A) Go live and train the service desk afterwards
(B) Present the failed readiness criteria at a go or no-go review with options and their risks
(C) Delay go-live indefinitely until everything is perfect
(D) Ask the development team to act as the service desk permanently

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · DELIVER

Answer key

Knowledge checks

1. D In cost-reimbursable contracts the buyer pays actual costs, so overruns fall mainly on the buyer.

2. C The statement of work describes scope, deliverables, standards and acceptance criteria for the supplier.

3. A A scope-swap clause keeps price and timeline while allowing priorities to change.

4. B A weaker alternative makes walking away more costly, reducing leverage.

5. D It is a stated demand. The interest is the reason behind it.

6. B Supplier management relies on objective measures. Reviewing them together and agreeing a recovery plan comes before contractual remedies.

7. D Appraisal costs are spent finding defects: testing, inspections and audits.

8. B Defects found after delivery to the customer are external failure costs — usually the most expensive.

9. A A Pareto chart ranks causes so the team focuses on the vital few.

10. B Assurance looks at the process; control checks the product.

11. C Regulatory standards are quality requirements. Gathering them early and planning evidence avoids costly late rework.

12. A The product owner orders the backlog to maximise value.

13. C Each artifact has a commitment; for the increment it is the definition of done.

14. B The Daily Scrum is timeboxed to 15 minutes, regardless of sprint length.

15. B 20 ÷ 10 = 2 weeks.

16. D Limiting WIP reduces task switching and queues, which shortens cycle time.

17. B Work item age tracks items still in progress, so you can act before they become late.

18. A A burnup shows total scope as its own line, so growth is visible.

19. B Velocity is a team's internal planning aid, not a cross-team productivity measure.

20. D Objective thresholds plus psychological safety around bad news keep status honest.

21. C It expresses a date with a confidence level based on data.

22. B Artifacts must be accessible to the people who need them. Making the register usable on mobile, or providing a suitable view, fixes the problem.

23. B Terminated projects still need orderly closure.

24. A Acceptance is the customer's formal agreement; quality control is the internal check.

25. D Operational risks continue after the project and must be owned by operations.

26. A It reviews the procurement process to improve future procurement.

27. B Readiness for transition should be validated against agreed criteria, and the decision made formally with clear options, not assumed.

Scenarios & case studies

> **Scenario · The vendor's change request**
> **B** is best. Best. The contract and SOW define the baseline; genuine new scope is handled through the agreed change mechanism.

> **Scenario · Skip the tests?**
> **B** is best. Best. Quality standards are not negotiable within the sprint — scope is. Fewer done items beat more half-done ones.

> **Scenario · An urgent request mid-sprint**
> **B** is best. Best. The product owner owns priorities; the developers own the sprint plan. Together they can adjust scope while protecting the goal.

> **Scenario · A full test column**
> **B** is best. Best. 'Stop starting, start finishing': swarming on the bottleneck improves flow and builds shared skills.

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Scenario · The demand for one date
A is best. Correct. Match the commitment's confidence to the cost of missing it, and keep the forecast live.

Case study · ERP go-live at a garment factory in Ashulia
Missed: formal acceptance, a defined support model and operational owner, hypercare with exit criteria, transfer of warranties, knowledge transfer to IT and compliance staff, and contract closure. Now: the sponsor appoints an operational owner; the vendor and factory hold an acceptance review against the original criteria and agree any punch-list; tickets are triaged and assigned; warranty and support renewal decided before April; compliance-report training run quickly. Prevention: include closure tasks in the schedule and budget from day one, make sponsor sign-off on closure a gate, and do not release the PM until the closure checklist is complete.

Scenario · The keen sponsor
B is best. Correct. Keep closure lightweight but formal: acceptance protects the client, the vendor and the team.

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07

MODULE 7 · GOVERN

Business Environment I: Governance, Compliance, Change & Risk

Business Environment · Tasks 1, 2, 3 and 5 (governance, compliance, change control, risk)

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Establish project governance with success metrics, escalation paths and ethical standards
- Plan and manage compliance and prove it with evidence
- Control change in predictive, agile and hybrid settings
- Identify, analyse, respond to and communicate risk, including security and sustainability risks

LESSONS

1. Project Governance, Process Assets & Ethics · 32 min
2. Planning & Managing Compliance · 30 min
3. Managing & Controlling Change · 28 min
4. Identifying & Analysing Risk · 32 min
5. Responding to Risk & Communicating It · 30 min

MODULE 7 · GOVERN · LESSON 1 · 32 MIN

Project Governance, Process Assets & Ethics

Set the structure, rules, success metrics and escalation paths a project runs within, and act with integrity

- Describe the governance structure, rules and reporting a project needs
- Use organisational process assets to establish governance quickly
- Define success metrics, escalation paths and tolerance thresholds
- Apply professional values and a structured check to ethical dilemmas

A single well-run project is good. An organisation that consistently picks the **right** projects and runs them well is far better. Governance is the framework of authority, decision rights and oversight that makes that consistency possible — without drowning teams in bureaucracy.

A typical project governance structure (top first)

Portfolio or investment board	`Selects and funds projects` `Sets risk appetite and strategic priorities`
Steering committee	`Directs the project at gates` `Decides on issues beyond tolerance`
Sponsor	`Owns the business case` `Champions the project and removes organisational blocks`
Project manager	`Manages within agreed tolerances` `Reports, escalates and integrates`
Team and product owner	`Deliver the work` `Decide within their delegated authority`

DEFINITION · ORGANISATIONAL PROCESS ASSETS (OPAS)

OPAs are the plans, policies, procedures, templates, standards and knowledge bases an organisation already has, such as a stage-gate procedure, a risk-scoring template, a code of conduct or a lessons-learned library. Use them to establish governance quickly and consistently, tailoring rather than reinventing. Conditions outside the project team's control, such as laws, market conditions or company culture, are often called **enterprise environmental factors**.

PMO types

TYPE	DEGREE OF CONTROL	WHAT IT DOES	BEST FIT
Supportive	Low	Templates, training, best practice, lessons-learned library; a consultative role	Organisations new to formal PM or with diverse, autonomous units
Controlling	Moderate	Requires compliance with methods, reporting and reviews; audits and assures	Organisations needing consistency and visibility, e.g. regulated sectors
Directive	High	Directly manages projects; PMs report to the PMO	Strategic or high-risk portfolios needing central control

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Stage-gate governance

01 **Idea / mandate**
Is the problem worth exploring?

→

02 **Gate 1: business case**
Is it viable and aligned? Fund discovery

→

03 **Gate 2: plan approved**
Baseline, risks, funding for delivery

→

04 **Gate 3: ready to launch**
Quality, readiness, change plan

→

05 **Gate 4: closure & benefits**
Accept, hand over, track benefits

DEFINITION · TOLERANCES KEEP GOVERNANCE LIGHT

Boards set **tolerances** — e.g. ±10% cost, ±3 weeks — within which the PM manages without escalation. Only forecasts beyond tolerance come back to the board. This is management by exception: senior time goes where it is needed.

Escalation paths and thresholds

VARIANCE OR EVENT	PROJECT MANAGER ACTS WHEN…	SPONSOR DECIDES WHEN…	STEERING COMMITTEE OR BOARD DECIDES WHEN…
Cost forecast	Within ±5% of budget	Between 5% and 10%	Beyond 10% or reserve release needed
Milestone date	Slip under 2 weeks	Slip of 2–6 weeks	Slip over 6 weeks or regulatory date at risk
Scope	Backlog reordering within the product goal	Change to a major deliverable	Change to objectives or the business case
Risk	Score below 12 with response in place	Score 12 or above	Beyond the organisation's risk appetite

Defining success metrics

- Separate **project** success (delivered as agreed) from **product** success (value created over time)
- Agree a few metrics per objective, each with a baseline, target and data source
- Combine leading measures, such as adoption in the first month, with lagging ones, such as cost savings after a year
- Include stakeholder satisfaction and sustainability measures where they matter
- Record the metrics in the charter or governance plan so they cannot drift quietly

Project managers hold information, money and influence. Sponsors, clients and teams rely on what they report and how they decide. The professional community broadly agrees on four core values — **responsibility, respect, fairness and honesty** — and these are reflected, for example, in the code of ethics published by PMI. Here is what they mean in day-to-day project life, in ZPMP's own words.

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Four values in action

RSP

Responsibility

Own your decisions and their consequences. Take only work you are competent to do. Keep commitments. Report errors — including your own — promptly and fix them.

RES

Respect

Treat people with dignity. Listen to other views. Honour other cultures and norms. Never abuse power or speak about others behind their backs to harm them.

FAI

Fairness

Decide impartially. Disclose and manage conflicts of interest. Give equal access to information and opportunity in procurement and staffing.

HON

Honesty

Seek and tell the truth. Report status accurately, even when it is unwelcome. Do not mislead by omission, selective data or silence.

A five-step ethical decision check

1. **Recognise** Is there an ethical issue? Who could be harmed or treated unfairly?
2. **Get the facts** What do you actually know? What laws, policies, contracts and codes apply?
3. **Consider options** List possible actions, including ‘raise it’, ‘decline’ and ‘seek advice’.
4. **Test them** Would I be comfortable if this appeared in the news, or if my mentor saw it? Is it legal? Fair to all? Honest?
5. **Act and reflect** Act, document your reasoning, and raise concerns through proper channels if needed.

SCENARIO · DECIDE · THE GREEN REPORT

Two days before a board meeting, your sponsor asks you to change the project’s status from red to amber ‘to avoid panic’, promising to fix things next month. What should you do?

A Change it to amber — the sponsor is accountable for the project.

B Report the sponsor to the ethics hotline immediately.

C Keep the status red, and offer to help the sponsor present a credible recovery plan and decisions needed alongside it.

D Skip the board meeting to avoid presenting.

Circle your choice. Answer and feedback: module answer key.

Key terms

PMO

Project (or portfolio/programme) management office: a function that standardises governance, supports or directs projects and improves delivery capability.

Conflict of interest

A situation in which personal interests could improperly influence, or appear to influence, professional judgement.

Stage gate

A decision point between phases where governance reviews progress and decides whether to continue, change or stop.

Project governance

The framework of authority, decision rights, rules and oversight within which a project is directed and controlled.

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Organisational process assets (OPAs)
An organisation's existing plans, policies, procedures, templates and knowledge that projects can use and tailor.

Escalation path
The agreed route by which an issue or variance moves to a higher level of authority for decision.

Tolerance
The permitted deviation from a plan, such as ±5% cost, within which the project manager can act without escalating.

Management by exception
A governance style in which senior people are involved only when agreed tolerances are forecast to be exceeded.

KNOWLEDGE CHECK

1. A PMO provides templates and coaching but does not require their use. Which type is it?
   - (A) Directive
   - (B) Controlling
   - (C) Portfolio
   - (D) Supportive
2. A project forecast is 4% over budget; the board's tolerance is ±10%. What should the PM do?
   - (A) Escalate to the board immediately for a decision
   - (B) Pause non-urgent work until the board has reviewed it
   - (C) Manage within the project and report normally
   - (D) Request a new cost baseline from the board
3. You discover your company's bid evaluation includes a firm owned by your cousin. What should you do FIRST?
   - (A) Evaluate the bid fairly and say nothing about it
   - (B) Disclose the relationship and step aside from it
   - (C) Mark the firm down to avoid any suspicion of favour
   - (D) Ask your cousin to withdraw the bid quietly before scoring
4. A sponsor asks you to omit a major risk from a report to investors. Which value is primarily at stake?
   - (A) Honesty
   - (B) Respect
   - (C) Fairness
   - (D) Responsibility only for the sponsor
5. A new project manager must set up governance for a project in an organisation with an established PMO. What should she do FIRST?
   - (A) Design a new governance framework from scratch
   - (B) Review the organisation's process assets, such as gate procedures and templates, and tailor them
   - (C) Ask the team to propose their own governance rules
   - (D) Wait for the steering committee to define governance

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · GOVERN · LESSON 2 · 30 MIN

Planning & Managing Compliance

Confirm what the project must comply with, protect it from threats and prove compliance with evidence

- Confirm and classify a project's compliance requirements
- Identify threats to compliance and the consequences of non-compliance
- Choose methods that build compliance into delivery
- Measure the extent to which the project is compliant

Every project operates inside rules it did not write: laws, regulations, licences, safety codes, data-protection duties, donor conditions, sustainability commitments and internal policies. **Compliance management** confirms which rules apply, builds them into the work and produces evidence that they were met. Non-compliance can stop a project overnight, so it belongs in planning from the first week, not in a panic before go-live.

Compliance categories

CATEGORY	EXAMPLES	TYPICAL EVIDENCE
Legal and regulatory	Central bank circulars, telecom licences, labour law	Approvals, licences, regulator correspondence
Security and data privacy	Data-protection law, security standards, customer consent	Risk assessments, penetration test reports, consent records
Health and safety	Site safety, fire codes, electrical safety	Inspection certificates, permits to work, incident logs
Environmental and sustainability	Environmental clearances, emissions and waste reporting	Impact assessments, monitoring data, waste manifests
Financial and procurement	Public procurement rules, anti-bribery law, donor conditions	Tender records, conflict declarations, audit reports
Standards and accessibility	Building codes, medical-device standards, accessibility rules	Test reports, conformity certificates
Internal policy	Architecture standards, records retention, brand rules	Design review sign-offs, policy checklists

Confirm compliance requirements

1. **Find the sources** Laws and regulators, contracts, donor or lender agreements, industry standards and internal policies.
2. **Consult the specialists** Legal, compliance, security, safety and sustainability teams, early.
3. **Classify** Group obligations by category and rate how severe a breach would be.
4. **Assign owners** One named owner for each obligation or group of obligations.
5. **Map to the work** Link each obligation to deliverables, backlog items, acceptance criteria and tests.
6. **Plan the evidence** Decide what proof will show compliance, who produces it and where it is kept.

TEMPLATE **Compliance register**

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Track each compliance obligation from identification to verified evidence.

OBLIGATION	SOURCE	CATEGORY	APPLIES TO	OWNER	EVIDENCE	STATUS
Customer consent before storing ID photos	*Data-protection law; central bank e-KYC guideline*	*Data privacy*	*Agent onboarding app*	*Compliance officer*	*Consent screen test report; consent log sample*	*Evidence under review*

Threats to compliance

THREAT	EXAMPLE	METHOD THAT HELPS
Schedule pressure	Inspections skipped to meet a date	Compliance steps built into gates and the definition of done
Supplier non-compliance	Subcontractor without valid safety permits	Contract clauses, supplier audits, permit checks
Unclear ownership	Everyone assumes legal has checked it	Named owners in the compliance register
Rules change mid-project	New data-localisation rule	Regulatory horizon scanning; change control
Unapproved tools	Customer data pasted into an unapproved AI tool	Approved-tools list, training, data classification
Poor records	Evidence scattered in personal inboxes	One evidence repository with version control

Methods that support compliance

CBD
Compliance by design
Write obligations into requirements, stories and acceptance criteria from the start.

GAT
Gates and checklists
Make compliance evidence a condition for passing gates or releasing.

AUT
Automated controls
Security scans, access checks and data-masking that run on every change.

TRN
Training
Short, role-specific training for team members and suppliers.

AUD
Reviews and audits
Internal reviews before external audits find the gaps.

SUP
Supplier controls
Contract clauses, certificates and audit rights.

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Consequences of non-compliance

Legal and financial

- Fines and penalties
- Contract damages or termination
- Loss of licence or funding

Operational

- Stop-work orders
- Delayed or blocked go-live
- Expensive rework and retesting

People and reputation

- Harm to customers, workers or communities
- Loss of trust with regulators and the public
- Personal liability for managers

EXAMPLE · COMPLIANCE IN SPRINTS

A mobile financial services firm in Dhaka was building a new savings feature under central bank rules. Every story touching customer data carried a privacy acceptance criterion, automated checks ran on each build, and the compliance officer joined the sprint review once a month. When the regulator reviewed the feature, the evidence already existed and the approval came without late rework.

Measuring compliance

MEASURE	FORMULA	WHAT IT TELLS YOU
Compliance coverage	`Obligations with verified evidence ÷ applicable obligations × 100`	How much of the project's compliance duty is proven, not just assumed.
Open findings	`Count of audit or review findings not yet closed, by severity`	Where the project is exposed now.
Time to close findings	`Average days from finding to verified fix`	How quickly the project responds to gaps.

SCENARIO · DECIDE · POSTPONE THE CONSENT SCREEN?

Two sprints before launch, the product owner proposes moving the customer consent screen to a later release so that a marketing feature can ship on time. The consent screen is required by data-protection law before any customer data is stored. What should the project manager do?

(A) Agree, because the product owner owns backlog priority.

(B) Explain that the consent screen is a compliance obligation that must ship with any data collection, involve the compliance officer, and help the product owner find what else can move.

(C) Launch without the screen and add a disclaimer to the app store listing.

(D) Escalate to the board and pause the release.

Circle your choice. Answer and feedback: module answer key.

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Compliance readiness checklist

☐ All compliance sources identified and specialists consulted
☐ Obligations classified and recorded in the compliance register
☐ Each obligation has a named owner
☐ Obligations mapped to deliverables, stories or acceptance criteria
☐ Evidence planned, collected and stored in one place
☐ Suppliers bound by matching contract clauses
☐ Compliance coverage and open findings reported to governance

Key terms

Compliance
Meeting the laws, regulations, standards, contracts and policies that apply to a project and its result.

Compliance register
A record of obligations, their sources, owners, evidence and status.

Non-compliance
Failure to meet an applicable obligation, which can lead to penalties, stoppages or harm.

Regulatory horizon scanning
Regularly checking for new or changing rules that may affect the project.

Compliance obligation
A specific requirement the project must meet, such as obtaining consent or a safety permit.

Compliance by design
Building obligations into requirements, design and acceptance criteria from the start rather than checking at the end.

Audit trail
A record that shows who did what, when and why, used as evidence in reviews and audits.

KNOWLEDGE CHECK

6. A hospital annex must pass a fire-safety inspection before occupancy. Into which compliance category does this fall?
   - (A) Health and safety
   - (B) Internal policy
   - (C) Financial and procurement
   - (D) Data privacy

7. What is the BEST way to make sure compliance obligations are not forgotten in an agile project?
   - (A) Run a compliance review just before go-live
   - (B) Ask the compliance team to approve every story
   - (C) Leave compliance to the vendor
   - (D) Write obligations into stories, acceptance criteria and the definition of done

8. A subcontractor arrives on site without valid safety permits, and the schedule is tight. What should the project manager do?
   - (A) Let them start and collect permits later
   - (B) Stop the unpermitted work and resolve the permits under the contract terms
   - (C) Report the subcontractor to the regulator immediately
   - (D) Ask the site team to supervise them closely instead

9. Out of 40 applicable obligations, 30 have verified evidence. What is the compliance coverage?
   - (A) 30%
   - (B) 40%
   - (C) 75%
   - (D) 133%

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10. Which is a threat to compliance that an approved-tools list and data classification help prevent?

(A) A delayed hardware shipment

(B) Team members pasting customer data into unapproved AI tools

(C) A change in the sponsor

(D) A slow sprint review

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · GOVERN · LESSON 3 · 28 MIN

Managing & Controlling Change

Run change control that fits the approach, communicate decisions and keep the plan and documents current

- Recognise scope creep and gold plating
- Execute integrated change control on predictive work
- Absorb change through the backlog on agile work, with clear limits on hybrid projects
- Communicate change status, implement approved changes and update documentation

Change is normal; uncontrolled change is not. Good change control lets the project say yes to the right changes quickly, say no or not now to the rest, and keep everyone working from the same, current picture. Predictive work uses formal change requests against baselines. Agile work absorbs most change through the product backlog. Hybrid projects need both, with clear rules about which path a change takes.

Scope creep vs gold plating vs healthy change

Scope creep

- Uncontrolled growth in scope
- Requests absorbed without impact analysis
- Usually from stakeholders
- Leads to delays, overruns, burnout

Gold plating

- Team adds extras nobody asked for
- Well-meant but unapproved
- Adds cost and risk
- Hides real progress

Healthy change

- Visible, analysed and decided
- Linked to value or necessity
- Trade-offs made explicitly
- Baselines or backlog updated transparently

Integrated change control (predictive)

01 **Request** — Change request logged →

02 **Analyse** — Impact on scope, time, cost, quality, risk, benefits →

03 **Decide** — PM, sponsor or change control board approves, rejects or defers →

04 **Update** — Baselines, plans and documents revised →

05 **Communicate** — Stakeholders and team informed; change implemented

Two ways to absorb change

Integrated change control

- Used when scope, schedule and cost are baselined
- Formal request and impact analysis
- Decision by authorised body (e.g. CCB)
- Protects contracts and commitments
- Can feel slow; tailor thresholds to keep it fast

Backlog refinement

- Used for evolving product scope
- New items added to the backlog by the product owner
- Ordered against everything else; something else may move down
- Change is expected and cheap if made before work starts
- Iteration goal protected during the iteration

INSIGHT · HYBRID CHANGE THRESHOLDS

Many hybrid projects let the product owner reorder the backlog freely **within** an agreed budget and milestone, but send anything that moves a milestone, increases total budget or affects a contract to formal change control. Write the thresholds down in the plan.

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TEMPLATE

Change log

Record every change request from submission to implementation, so that its status is visible to all.

ID AND DESCRIPTION	REQUESTED BY	IMPACT (SCOPE, TIME, COST, RISK, VALUE)	DECISION, BY WHOM AND DATE	STATUS COMMUNICATED TO	DOCUMENTS UPDATED
CR-014 Add claims-integration interface	*Insurance partner via sponsor*	*+3 weeks, +Tk 18 lakh, new data-sharing risk, enables 12% more insured patients*	*Approved by CCB, 4 Mar*	*Requester, team, finance, vendor*	*Scope baseline, schedule, contract amendment, risk register*

After the decision: communicate, implement, update

1. **Tell the requester and stakeholders** Share the decision and the reason, including rejections and deferrals.
2. **Update baselines or the backlog** Revise scope, schedule and cost baselines, or reorder the backlog.
3. **Update documents** Plans, registers, contracts, requirements, test cases and training material.
4. **Implement and verify** Carry out the change and confirm it was done as approved.
5. **Close the request** Record the outcome in the change log.

DEFINITION · CONFIGURATION MANAGEMENT

Configuration management keeps track of the versions of deliverables and their documentation, so that everyone builds, tests and approves the same thing. Version control for code, controlled drawing registers for construction and a single source for requirements are all forms of it.

PRO TIP · MAKE CHANGE STATUS VISIBLE

Stakeholders worry most about changes they cannot see. Publish the change log or a simple board showing each request as submitted, under analysis, decided or implemented. Fewer people will chase you, and fewer side-door requests will reach the team.

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SCENARIO · DECIDE · WHICH PATH FOR THIS CHANGE?

On a hybrid hospital project, the product owner may reorder the patient-flow software backlog freely within its agreed budget and milestone. The hospital's insurance partner now asks for a claims-integration feature that the team estimates would push the software milestone back by three weeks and add 6% to the software budget. What should the project manager do?

(A) Let the product owner add it to the backlog; backlog changes are the product owner's call.

(B) Raise it through integrated change control with an impact analysis, so the sponsor or change board can decide on the milestone and budget change.

(C) Reject it because the scope baseline is fixed.

(D) Ask the team to absorb it with overtime so the milestone does not move.

Circle your choice. Answer and feedback: module answer key.

SCENARIO · DECIDE · THE SIDE-DOOR REQUEST

On a predictive warehouse project with an approved scope baseline, a department head asks a developer directly to add a new stock report. The developer estimates two days and is happy to do it. What should the project manager do?

(A) Allow it, because two days is a small change.

(B) Thank the department head, log the request, analyse its impact and take it through the agreed change control process.

(C) Refuse, because the scope is already approved.

(D) Add the report and update the baseline afterwards.

Circle your choice. Answer and feedback: module answer key.

Key terms

Scope creep
Uncontrolled expansion of scope without adjusting time, cost or resources and without proper approval.

Change request
A formal proposal to modify a baseline, deliverable, plan or contract.

Integrated change control
Reviewing every change for its combined effect on scope, schedule, cost, quality, risk and benefits before deciding.

Change log
A record of change requests, their impacts, decisions and status.

Gold plating
Adding features or quality that were not requested, usually by the team, without approval.

Change control board (CCB)
A group with authority to review, approve, defer or reject change requests.

Configuration management
Controlling versions of deliverables and their documentation so that everyone works from the same approved version.

KNOWLEDGE CHECK

11. A developer adds an extra dashboard nobody requested. This is:

(A) Gold plating
(B) Change control
(C) Scope creep
(D) Progressive elaboration

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12. On an agile team, a stakeholder asks for a new feature mid-project. What normally happens?

(A) It is added to the current sprint immediately by the team

(B) The product owner adds it to the backlog and orders it by value

(C) It must be rejected because the scope is already agreed

(D) The PM rewrites the charter and asks the sponsor to re-approve it

13. A change request for a new interface has been approved by the change control board. What should happen NEXT?

(A) Start building the interface immediately without updating documents

(B) Wait for the sponsor to inform the team

(C) Communicate the decision, update baselines and documents, then implement and verify the change

(D) Ask the requester to submit the request again in writing

14. Stakeholders keep asking the team directly about the status of their change requests. Which practice BEST helps?

(A) Publishing the change log with the status of each request

(B) Asking stakeholders to stop requesting changes

(C) Holding a weekly meeting for each requester

(D) Approving all requests to reduce follow-ups

15. On a hybrid project, the product owner may reorder the app backlog within its budget and milestone. A requested change would move the milestone by four weeks. Which path should it take?

(A) The product owner decides, as with any backlog change

(B) The team decides during sprint planning

(C) No path, because milestones cannot change

(D) Formal change control, because it affects an agreed milestone

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MODULE 7 · GOVERN · LESSON 4 · 32 MIN

Identifying & Analysing Risk

Name uncertainty precisely, then rank it quickly and quantify the risks that matter

- Distinguish risks, issues and opportunities using a single clear test
- Write cause–event–effect risk statements and widen identification with a pre-mortem and RBS
- Prioritise risks with defined probability and impact scales
- Calculate expected monetary value and read Monte Carlo outputs

Every project is a bet on the future. **Risk management**, part of the Govern stage of the ZPMP Professional Loop, is the discipline of watching that bet, adjusting it and telling the truth about it. It starts with identification and analysis: the uncertain events, threats and opportunities, that could push outcomes off course, ranked so attention goes where it matters.

Risk, issue or opportunity? The ZPMP time test

R

Risk (threat)

An **uncertain** future event that would hurt at least one objective if it happened. It has a probability below 100%. You manage it *before* it happens.

O

Opportunity

An uncertain future event that would **help** an objective — finish earlier, cost less, deliver more value. Positive risk deserves the same attention as negative risk.

I

Issue

Something that **has already happened** or is certain and is affecting the project now. You resolve it, you do not ‘mitigate’ it.

A

Assumption

Something treated as true for planning without proof. Every assumption hides a risk: ‘what if this is false?’

INSIGHT · ONE QUESTION SORTS THEM ALL

Ask: **‘Has it happened yet?’** If no and it would hurt — risk. If no and it would help — opportunity. If yes — issue. A risk that materialises moves from the risk register to the issue log, and the risk entry is closed with a note.

A well-formed risk statement

MEASURE	FORMULA	WHAT IT TELLS YOU
Cause–event–effect	`Because of <cause>, <event> may occur, which would lead to <effect on objective>.`	Forces precision. The cause is a fact today; the event is uncertain; the effect names cost, time, scope, quality or benefit.
Example	`Because our only certified network engineer is due for parental leave in March, cut-over testing may slip, which would delay go-live by 3–4 weeks.`	Now the team can see who owns it, what would trigger it and which responses make sense.

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Ways to find risks (use more than one)

PM

Pre-mortem

Imagine it is a year from now and the project has failed badly. Each person silently writes why. Then share and cluster. Hindsight imagined in advance unlocks honest concerns people would not otherwise voice.

RBS

Risk breakdown structure

A hierarchy of risk sources — e.g. Technical, External, Organisational, Project management — each split into sub-categories. Walk through it to spot blind areas.

AL

Assumptions & constraints log

Review every assumption and ask what happens if it is wrong. Track who validates it and by when.

SH

Stakeholder & expert interviews

Operations staff, suppliers and finance see risks the core team cannot. Ask: 'What usually goes wrong with projects like this?'

DR

Document & data review

Lessons-learned registers, audit findings, past schedules, contracts and incident logs from similar work.

SW

SWOT & prompt lists

Strengths, weaknesses, opportunities, threats — or checklists such as PESTLE — to cover the external environment.

A starter risk breakdown structure

LEVEL 1	LEVEL 2 EXAMPLES	TYPICAL QUESTIONS
Technical	Requirements, technology maturity, integration, performance, data migration	Have we built this before? How many interfaces?
External	Suppliers, regulation, market, weather, currency, politics	What could change outside our control?
Organisational	Funding, priorities, shared resources, other projects	Could our people or budget be pulled away?
Project management	Estimating, planning, communication, control	How confident are we in our estimates?
People & change	Adoption, skills, morale, key-person dependency	Will users actually change how they work?

Once you have a list of risks, you cannot treat them all equally. Analysis answers two questions: **which risks deserve attention first**, and **how much exposure does the project carry overall?** Qualitative analysis is fast and covers every risk. Quantitative analysis is slower and is reserved for the risks and decisions that drive outcomes.

Qualitative vs quantitative analysis

Qualitative

- Ranks individual risks using scales (e.g. 1–5)
- Fast, cheap, done on every project
- Output: prioritised risk list, heat map
- Weakness: subjective unless scales are defined

Quantitative

- Models combined effect of risks on cost/schedule
- Needs data, tools and skill
- Output: probability of hitting dates/budgets, EMV, sensitivity
- Use on large, complex or high-stakes projects

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Define the scales before you score

SCORE	PROBABILITY	SCHEDULE IMPACT	COST IMPACT
1 Very low	< 10%	< 1 week	< 1% of budget
2 Low	10–30%	1–2 weeks	1–3%
3 Medium	31–50%	2–4 weeks	3–5%
4 High	51–70%	1–2 months	5–10%
5 Very high	> 70%	> 2 months	> 10%

Multiply probability by impact to get a risk score (1–25). Plot each risk on a 5×5 grid: the top-right zone (scores of roughly 15+) gets immediate, owned responses; the middle zone gets watched; the bottom-left zone may be accepted and monitored.

Expected monetary value

MEASURE	FORMULA	WHAT IT TELLS YOU
EMV of one risk	`EMV = Probability × Impact`	Threats carry negative impact, opportunities positive. A 30% chance of a −50,000 loss has EMV −15,000.
Project risk exposure	`Σ EMV of all quantified risks`	Netting threats and opportunities gives a rational starting point for the contingency reserve.
Decision node	`Choose the branch with the best EMV`	In a decision tree, each option's EMV = its direct cost + Σ (probability × outcome) of what follows.

CASE STUDY · CHATTOGRAM PORT: SHOULD WE REHEARSE THE CUT-OVER?

A container terminal at Chattogram is replacing its gate and yard-planning software. The vendor proposes an optional full cut-over rehearsal on a weekend, costing **USD 40,000**. Without a rehearsal, the team estimates a **35%** chance that go-live fails and the old system must be restored, causing berth delays and penalties of about **USD 300,000**. With a rehearsal, the chance of failure drops to **10%** (the same USD 300,000 impact if it happens).

1. Draw the decision tree and calculate the EMV of each option.

2. Which option should the project choose, and by how much does it improve expected cost?

3. What non-monetary factors should the sponsor also weigh?

Model answer: module answer key.

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Reading a Monte Carlo output

S

S-curve

Cumulative probability of finishing by each date (or within each cost). Read across from 80% to find the P80 date.

P50

P50 / P80

P50: half of simulated outcomes finish by this date. P80: four in five do. Commit externally to a higher-confidence value; manage internally to a stretch value.

TOR

Tornado chart

Ranks which inputs most influence the result. Spend risk effort on the longest bars.

CI

Criticality index

How often an activity appears on the critical path across simulations — a task at 70% deserves more attention than its single-point float suggests.

Key terms

Risk
An uncertain event or condition that, if it occurs, affects at least one project objective — negatively (threat) or positively (opportunity).

Risk breakdown structure (RBS)
A hierarchical grouping of risk sources used as a prompt to identify risks systematically.

Opportunity
An uncertain event that would benefit at least one project objective if it happened.

Probability–impact matrix
A grid that combines probability and impact scores to rank risks.

Pre-mortem
A workshop in which the team imagines the project has failed and works backwards to identify the reasons, surfacing risks early.

Expected monetary value (EMV)
Probability multiplied by monetary impact; summed across risks or branches to compare options and size reserves.

Risk score
Probability score multiplied by impact score, used to rank risks on a heat map.

P80
The value, such as a date or cost, that 80% of simulated outcomes meet or beat.

KNOWLEDGE CHECK

16. A key supplier has just informed you their factory is closed for two weeks. How should this be recorded?

A) As an issue in the issue log
B) As a new risk with high probability
C) As an assumption to validate
D) As an opportunity to renegotiate

17. Which risk statement is best formed?

A) Vendor risk: the vendor might be poor at support, which could cause problems for users
B) There is a risk of delays, cost overruns and quality problems across the whole project
C) The vendor has no local office and is new to us, so the support contract is at risk
D) With no local vendor office, fixes may take over 48 hours and breach the 24-hour SLA

18. A risk has a 40% probability and an impact of −80,000. Its EMV is...

A) −32,000
B) −48,000
C) −80,000
D) +32,000

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19. Which is the strongest reason to define probability and impact scales before scoring?

(A) It makes the heat map clearer and more colourful

(B) It is required by law in most regulated industries and sectors

(C) It reduces subjective, inconsistent scoring between people

(D) It removes the need for later quantitative risk analysis

20. A Monte Carlo S-curve shows the P80 finish date as 20 March. This means...

(A) the project is certain to finish on 20 March as planned

(B) 80% of simulated outcomes finished on or before 20 March

(C) there is an 80% chance that the project finishes after 20 March

(D) 20 March is the end date of the critical path schedule

21. A new regulation might allow your team to reuse an approved design, saving six weeks. This is...

(A) an issue

(B) a constraint

(C) a residual risk

(D) an opportunity

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MODULE 7 · GOVERN · LESSON 5 · 30 MIN

Responding to Risk & Communicating It

Plan risk management, choose and own responses, cover security, AI and sustainability risks and report risk clearly

- Develop a risk management plan with appetite and thresholds
- Select response strategies for threats and opportunities and assign owners and triggers
- Include security, AI and sustainability risks in the register
- Communicate the status and impact of risks to different audiences

Develop a risk management plan

1. **Method** How risks will be identified, analysed and reviewed, and how often.
2. **Roles** Who owns the process, who owns individual risks, who approves responses.
3. **Scales** Agreed definitions of probability and impact for this project.
4. **Appetite and thresholds** How much risk the organisation will accept and the scores that trigger escalation.
5. **Budget and reserves** How contingency is sized and released.
6. **Reporting** What risk information goes to whom, and when.

DEFINITION · RISK APPETITE AND RISK THRESHOLD

Risk appetite is how much uncertainty an organisation is willing to accept in pursuit of its goals; a hospital has little appetite for patient-safety risk but may accept more on a new booking app. A **risk threshold** turns appetite into a number, for example 'any risk scoring 15 or more, or any data-breach risk, goes to the steering committee'.

Analysis tells you which risks matter. Response planning decides **what you will actually do** — and who will do it. A response without an owner, a budget and a date is just a hope.

Response strategies

For threats

- **Escalate** – outside the project's authority; hand to the right level and confirm they accept it
- **Avoid** – change the plan to remove the threat (e.g. drop a risky feature)
- **Transfer** – shift the financial impact to a third party (insurance, fixed-price contract, warranty)
- **Mitigate** – reduce probability and/or impact (prototype, extra testing, second supplier)
- **Accept** – active (set aside contingency) or passive (deal with it if it happens)

For opportunities

- **Escalate** – beyond project authority; pass to programme/portfolio
- **Exploit** – act to make it certain (assign your best people to finish early)
- **Share** – partner with a party better placed to capture it (joint venture, gain-share)
- **Enhance** – raise probability and/or impact (add resources to a promising path)
- **Accept** – take the benefit if it comes, without active pursuit

DEFINITION · TRANSFER MOVES THE MONEY, NOT THE RISK

Buying insurance or signing a fixed-price contract shifts the financial consequence, but the event can still delay your project and hurt your reputation. Transfer almost always leaves residual risk you must still watch.

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Every response needs these five parts

OWN

Risk owner

One named person who monitors the risk and makes sure the response happens. Not a department, not 'the team'.

TRG

Trigger

An observable warning sign that the risk is about to occur or has occurred — e.g. 'supplier misses the second milestone'.

ACT

Response actions

Specific tasks, with dates and cost, added to the schedule and budget.

RES

Residual risk

The exposure that remains after the response. If it is still above tolerance, plan more.

SEC

Secondary risk

A new risk created by the response itself — e.g. a second supplier introduces integration risk.

PRO TIP · FALLBACK PLANS

For high-impact risks, prepare a **fallback plan** ('Plan B') to use if the primary response fails, and decide in advance what trigger activates it. Decisions made calmly beforehand are better than decisions made in a crisis.

TEMPLATE **Risk register**

Track every identified risk from first sighting to closure. Review the top risks at every status meeting.

ID / STATEMENT (CAUSE–EVENT–EFFECT)	P × I = SCORE	OWNER	TRIGGER	STRATEGY & ACTIONS	RESIDUAL / SECONDARY	STATUS
R07 Because only one engineer holds the PLC licence, the cut-over may slip if he is unavailable, delaying go-live 3 weeks	*3 × 4 = 12*	*Farhana (ops lead)*	*Engineer leave request or illness*	*Mitigate: certify a second engineer by 15 May (BDT 85k)*	*Residual 1 × 4 = 4; secondary: training pulls engineer off build for 1 week*	*Open – action on track*

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WATCH OUT · SECURITY AND SUSTAINABILITY RISKS BELONG IN THE REGISTER

Poor IT security is a project risk, not only an IT department matter: weak access control on a test environment holding real customer data, or ransomware on the project's file share. Sustainability risks belong there too: flooding at a site during the monsoon, a supplier found using child labour, or an environmental clearance withdrawn. Give them owners, triggers and responses like any other risk.

Distinctive risks of AI projects

DAT

Data

Not enough, poor quality, unrepresentative or legally restricted data — often discovered late.

PRF

Performance uncertainty

You cannot promise accuracy upfront; plan experiments, success thresholds and stop/go decisions.

BIA

Bias & fairness

Models can reproduce historic discrimination, e.g. in credit scoring or hiring.

DRF

Drift

Real-world data changes over time and accuracy decays; monitoring and retraining must be budgeted.

EXP

Explainability

Regulators and users may need to know why a decision was made.

SEC

Security & misuse

Prompt injection, data leakage and misuse of generated content in gen-AI products.

REG

Regulation

Risk-based laws such as the EU AI Act and sector rules impose duties depending on use; check obligations in every market served.

Communicating risk status

AUDIENCE	WHAT THEY NEED	FORMAT
Steering committee	Top risks, trend, exposure against reserves, decisions needed	One page with a heat map and a short narrative
Sponsor	Risks near threshold, responses that need support	Regular one-to-one briefing
Team	Risks affecting this sprint or phase, triggers to watch	Risk board or review at planning
Suppliers and partners	Shared risks and joint responses	Joint risk reviews

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SCENARIO · DECIDE · THE TRIGGERED RISK

Your risk register lists ‘key subcontractor insolvency’ with a fallback plan to activate a pre-qualified alternative. Today a news report says the subcontractor has entered administration. What should the project manager do FIRST?

(A) Call a team meeting to brainstorm what to do.

(B) Ask the sponsor for permission to look for a new subcontractor.

(C) Implement the agreed fallback plan and update the issue log and risk register.

(D) Wait to confirm the news through official channels before doing anything.

Circle your choice. Answer and feedback: module answer key.

Key terms

Residual risk
The risk that remains after responses have been implemented.

Risk management plan
The part of the project plan that defines how risks will be identified, analysed, responded to, monitored and reported.

Risk threshold
A measurable level of risk exposure above which a risk must be escalated or treated.

Trigger
An observable warning sign that a risk is about to occur or has begun to occur.

Secondary risk
A new risk that arises as a direct result of implementing a risk response.

Risk appetite
The amount of uncertainty an organisation is willing to accept in pursuit of its objectives.

Risk owner
The one named person who monitors a risk and makes sure its response is carried out.

Fallback plan
A prepared plan B to use if the primary risk response fails.

KNOWLEDGE CHECK

22. A team buys an insurance policy covering equipment damage during shipment. Which strategy is this?

(A) Transfer (B) Avoid
(C) Mitigate (D) Exploit

23. To make sure a likely early-finish opportunity actually happens, the PM assigns the most experienced engineers to the task. This is...

(A) Enhance (B) Exploit
(C) Accept (D) Share

24. A response to one risk creates a new integration risk. The new risk is called...

(A) residual risk (B) an emerging issue
(C) a trigger (D) secondary risk

25. A threat is outside the project’s authority but within the programme’s. What should the PM do?

(A) Accept it passively and note it in the register (B) Ignore it, as it is outside the project’s authority
(C) Escalate it to the programme manager and confirm ownership (D) Avoid it by recommending that the project is cancelled

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26. A test environment holds copies of real customer records with weak access controls. How should the project treat this?

(A) As an IT department matter outside the project's scope

(B) As acceptable, since it is only a test environment

(C) As a project risk with an owner and a response, such as masking the data

(D) As an issue to raise only if a breach occurs

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MODULE 7 · GOVERN

Answer key

Knowledge checks

1. D Supportive PMOs provide resources with low control.

2. C Within tolerance, the PM manages; management by exception keeps the board for larger deviations.

3. B Disclosure and recusal manage the conflict transparently.

4. A Omitting material information misleads — an honesty issue.

5. B OPAs let governance be established quickly and consistently; the PM tailors them to the project rather than reinventing them.

6. A Fire codes and occupancy inspections are health and safety obligations.

7. D Compliance by design builds obligations into the work itself, so evidence is produced as the product is built.

8. B Working without permits is non-compliance with possible harm and stop-work consequences. The work must not proceed until the obligation is met.

9. C Coverage = 30 ÷ 40 × 100 = 75%. The remaining 25% is exposure that needs evidence or action.

10. B Unapproved tools can breach data-protection duties. Clear rules on approved tools and data classes reduce this threat.

11. A Gold plating is the team adding unrequested extras; scope creep usually comes from unmanaged stakeholder requests.

12. B Backlog refinement absorbs change: the product owner orders new items by value while the current iteration's goal is protected.

13. C Approved changes are communicated, reflected in baselines and documentation, implemented and verified.

14. A Visible change status reduces chasing and side-door requests while keeping the process transparent.

15. D Changes within the agreed boundaries are handled in the backlog; anything that moves a milestone, budget or contract goes to formal change control.

16. A It has already happened, so it is an issue. Risks are uncertain future events; this one is certain and current.

17. D It names a present cause, an uncertain event and a specific effect on an objective. The others are vague, or leave out the cause, the event or a measurable effect.

18. A 0.40 × −80,000 = −32,000. Threats carry negative values.

19. C Shared definitions turn opinions into comparable scores.

20. B P80 is the value that 80% of iterations meet or beat.

21. D It is uncertain and would benefit the schedule — a positive risk, or opportunity.

22. A Insurance shifts the financial impact to a third party: transfer.

23. B Exploit aims to make the opportunity certain. Enhance would only increase its likelihood or size.

24. D A risk caused by a response is a secondary risk.

25. C Escalation is correct when the response belongs at a higher level; the PM confirms acceptance and records it.

26. C Security weaknesses in project environments are project risks; they need an owner and a response before they become issues.

Scenarios & case studies

> **Scenario · The green report**
> **C** is best. Correct. Honest status plus a constructive path forward serves the sponsor and the organisation.

> **Scenario · Postpone the consent screen?**
> **B** is best. Best. It treats the obligation as non-negotiable, makes the consequences visible and still supports the product owner in re-planning.

> **Scenario · Which path for this change?**
> **B** is best. Best. Changes that move milestones or budgets beyond agreed thresholds belong in formal change control, even on the agile part of a hybrid project.

> **Scenario · The side-door request**
> **B** is best. Best. Every change to a baselined scope goes through change control, however small, and the requester sees the status.

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Case study · Chattogram port: should we rehearse the cut-over?
Rehearse: −40,000 + (0.10 × −300,000) = **−70,000**. No rehearsal: 0 + (0.35 × −300,000) = **−105,000**. Rehearsing is better by **USD 35,000** in expected value, so choose the rehearsal. Beyond money: reputation with shipping lines, safety in the yard, staff confidence with the new system, and the fact that the rehearsal also trains people. EMV supports the decision; it does not replace judgement, especially for low-probability, catastrophic outcomes.

Scenario · The triggered risk
C is best. Correct. The trigger has occurred; execute the pre-planned response, record the new issue and close or update the risk.

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08

MODULE 8 · ADAPT

Business Environment II: Impediments, Improvement, Change & the Outside World

Business Environment · Tasks 4, 6, 7 and 8 (impediments and issues, continuous improvement, organisational change, external environment), plus capstone and exam readiness

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Remove impediments and resolve issues with the right interventions
- Use lessons learned and continuous improvement to update process assets
- Assess culture and support organisational change
- Evaluate external changes, including regulation, markets, AI and sustainability
- Integrate the whole loop in a capstone and prepare for situational questions

LESSONS

1. Removing Impediments & Managing Issues · 28 min
2. Continuous Improvement & Lessons Learned · 28 min
3. Supporting Organisational Change · 30 min
4. The External Environment: Regulation, Markets, AI & Sustainability · 32 min
5. The Whole Loop: Capstone & Exam Readiness · 40 min

MODULE 8 · ADAPT · LESSON 1 · 28 MIN

Removing Impediments & Managing Issues

Spot what blocks the team, choose the right intervention and resolve issues with the people who can fix them

- Distinguish impediments, issues and risks
- Evaluate, prioritise and make impediments visible
- Choose an intervention strategy and escalate with options
- Run an issue log and collaborate with stakeholders on resolution

Teams lose more time to waiting than to hard work: waiting for access, decisions, environments, approvals or another team. An **impediment** is anything that slows or blocks the team. An **issue** is a current problem that is already affecting a project objective. Removing impediments and resolving issues quickly is one of the most visible ways a project manager adds value, in agile and predictive work alike.

Impediment, issue or risk?

Impediment

- Something blocking or slowing the team now
- Often small and frequent
- Raised at stand-ups and on the board
- Example: no access to the test server

Issue

- A current problem affecting scope, time, cost, quality or benefits
- Recorded in the issue log with an owner
- Example: customs is holding the imported lifts

Risk

- An uncertain future event
- Recorded in the risk register
- Becomes an issue when it happens
- Example: customs might hold the lifts

When a risk becomes an issue

01 **Risk in register** — Owner watches the trigger →

02 **Trigger observed** — Warning sign appears →

03 **Event occurs** — It is now an issue →

04 **Move to issue log** — Activate the planned response →

05 **Close the risk entry** — Note what happened

Evaluate and prioritise impediments

1. **Impact** Which sprint goal, milestone or critical path does it threaten?
2. **Reach** How many people or items are blocked?
3. **Age** How long has it been open? Old blockers signal a system problem.
4. **Resolver** Who can actually remove it: the team, the PM, another team, a manager?
5. **Priority** Highlight the highest-impact blockers at the top of the board and in reports.

Impediment escalation ladder

LEVEL	WHO ACTS	TYPICAL TIMEFRAME	EXAMPLE
1 — Team	Developers themselves	Same day	Pair to resolve a tricky merge conflict
2 — Team coach or PM	Scrum Master, team lead or PM	1–2 days	Get test devices or environment access

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LEVEL	WHO ACTS	TYPICAL TIMEFRAME	EXAMPLE
3 — Product owner or functional manager	PO, line manager, vendor manager	Within the iteration	Clarify conflicting requirements, free a shared specialist
4 — Sponsor or steering group	Sponsor, steering committee	When a milestone or goal is threatened	Resolve funding, cross-department priority clashes

Intervention strategies

FIX
Remove it
Resolve directly, for example by arranging access or making a decision within your authority.

WRK
Work around
Find a temporary path while the root cause is fixed.

NEG
Negotiate
Agree priorities or timing with the other team or manager involved.

ESC
Escalate
Take it to the person with authority, with facts and options.

PRV
Prevent
Change the process so the same blocker does not return.

RPL
Re-plan
Accept the delay and adjust the plan openly if it cannot be removed.

PRO TIP · MAKE BLOCKERS VISIBLE AND KEEP REASSESSING
Mark blocked items on the board, show how many days each has been blocked and review them every day. Ask at each stand-up whether yesterday's interventions worked. A blocker that is still open after its target date needs a different strategy, not the same reminder again.

Issue lifecycle

01 **Log** — Describe impact, date raised, who raised it →
02 **Assess** — Severity, urgency, objectives affected →
03 **Assign** — One owner and a target resolution date →
04 **Resolve or escalate** — Within tolerance: act. Beyond tolerance: escalate with options →
05 **Close & learn** — Record the fix and feed lessons learned

Escalate with options, not just problems

ESCALATION ELEMENT	WHAT TO INCLUDE
What happened	One-sentence description and when it was discovered
Impact	Effect on scope, schedule, cost, quality, benefits — with numbers
Tolerance	Which agreed threshold it breaches (e.g. > 2 weeks or > 5% cost)
Options	2–3 realistic options with trade-offs
Recommendation	Your preferred option and why
Decision needed by	The date after which options close

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EXAMPLE · LIFTS HELD AT THE PORT

On a predictive hospital-construction project in Chattogram, imported lifts were held at customs over a documentation error, threatening the critical path. The PM logged the issue, gave it one owner, and within a day brought the sponsor three options: pay for expedited clearance through an agent, re-sequence interior works to use the float elsewhere, or accept a two-week slip. The sponsor chose the first two together, and the milestone held.

TEMPLATE **Impediment and issue log**

Track every blocker and issue to resolution, with one owner each.

ID AND DESCRIPTION	TYPE (IMPEDIMENT OR ISSUE)	IMPACT	OWNER	INTERVENTION	RAISED AND TARGET DATES	STATUS
I-07 Test environment not available	*Impediment*	*Blocks 4 stories; sprint goal at risk*	*PM*	*Escalated to infrastructure lead with options*	*Raised 3 May; target 6 May*	*Open, day 3*

SCENARIO · DECIDE · THREE DAYS WITHOUT A TEST ENVIRONMENT

An agile team has been waiting three days for a test environment from the infrastructure team. The Scrum Master has asked twice by message. Four stories are blocked and the sprint goal is at risk. What should the project manager do?

(A) Wait, since the infrastructure team knows about the request.

(B) Meet the infrastructure lead with the impact data, agree a date or a temporary environment, and help the team pull other ready work meanwhile.

(C) Ask the developers to build their own environment on personal cloud accounts.

(D) Report the infrastructure team to the sponsor in the next steering meeting.

Circle your choice. Answer and feedback: module answer key.

Key terms

Impediment
Anything that slows or blocks the team's progress.

Issue
A current problem or condition that is affecting a project objective and needs resolution.

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Issue log
A record of issues with their impact, owner, actions, target dates and status.

Workaround
A temporary solution that lets work continue while the root cause is fixed.

Impediment board
A visible list or board of current blockers, their owners and ages, reviewed regularly.

Blocker age
How long an impediment has remained unresolved; a rising age signals a need for a different intervention.

Escalation
Passing an issue to a higher authority, with facts and options, when it is beyond the current level's power to resolve.

KNOWLEDGE CHECK

1. A blocker threatens a release milestone and the team and PM cannot resolve it. What is the appropriate next step?
   - (A) Wait and raise it with the team at the next retrospective
   - (B) Keep it from the board until a fix is found
   - (C) Escalate promptly to the right decision-maker with options
   - (D) Remove the blocked item from the definition of done
2. Which element of an escalation is MOST often missing and most valuable to decision makers?
   - (A) The project logo and reporting template
   - (B) A list of everyone who has been informed
   - (C) The full risk register as an attachment
   - (D) Realistic options with a recommendation
3. A risk in the register, 'supplier may deliver servers late', has just happened: the supplier confirms a three-week delay. What should the project manager do?
   - (A) Leave it in the risk register and raise its probability to 100%
   - (B) Record it in the issue log, activate the planned response and close the risk entry
   - (C) Wait to see whether the delay affects the critical path
   - (D) Ask the supplier to submit a new risk statement
4. Which impediment should the team tackle FIRST?
   - (A) One that has been open for one day and blocks a low-priority item
   - (B) A request for new monitors for the team room
   - (C) A minor tooling annoyance raised by one developer
   - (D) One that blocks three stories needed for the sprint goal
5. A blocker has passed its target date after two reminders to the other team. What does this suggest?
   - (A) A different intervention, such as escalation with options, is needed
   - (B) The team should stop raising it
   - (C) More reminders should be sent
   - (D) The blocker should be removed from the board

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MODULE 8 · ADAPT · LESSON 2 · 28 MIN

Continuous Improvement & Lessons Learned

Turn experience into changed behaviour and keep the organisation's process assets current

- Facilitate retrospectives that produce owned improvement actions
- Use lessons learned from past and current projects
- Apply a simple improvement cycle and measure whether changes work
- Update organisational process assets so improvements last

Most organisations have folders full of lessons learned that nobody reads. The ZPMP view is simple: **a lesson is only learned when behaviour changes**. Until then it is merely a lesson observed.

Retrospective vs lessons-learned review

Retrospective

- Frequent — end of each iteration or phase
- Owned by the team, about how the team works
- Output: 1–3 improvement actions for the next cycle
- Learning applied immediately

Lessons-learned review

- At milestones and closure (ideally also during the project)
- Broader: process, governance, vendors, stakeholders
- Output: register entries and recommendations for the organisation
- Learning must travel to other projects

Retrospective formats

SSC

Start / Stop / Continue

Quick and clear; good for new teams.

4L

4Ls

Liked, Learned, Lacked, Longed for — balances positives and gaps.

SB

Sailboat

Wind (what pushes us), anchors (what holds us back), rocks (risks ahead), island (goal).

TL

Timeline

Plot events and the team’s energy over time; great after long phases or incidents.

A retrospective that changes something

1. **Set the stage** Remind everyone of the shared assumption: people did their best with what they knew at the time. Check in briefly.
2. **Gather data** Use a format to collect facts and feelings silently first.
3. **Generate insights** Cluster and ask ‘why?’ to find root causes, not symptoms.
4. **Decide actions** Choose at most one to three actions, each with an owner and a date.
5. **Close and follow up** Put actions in the backlog; start the next retro by reviewing them.

TEMPLATE **Retrospective board**

Capture one retrospective’s output so actions are visible and tracked.

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WHAT WENT WELL	WHAT DID NOT	ROOT CAUSE	ACTION	OWNER	DUE / REVIEW
Pairing on data migration scripts	*Late test data from the bank*	*No named contact in client data team*	*Agree a data contact and a weekly test-data drop*	*Rina*	*Next sprint review*

TEMPLATE **Lessons-learned register**

Record lessons so that they change templates, checklists and decisions on future projects.

DATE / PHASE	WHAT HAPPENED	IMPACT	LESSON (DO THIS NEXT TIME)	ACTION TO EMBED IT	OWNER	STATUS
Build	*Integration testing found 60% of defects in the last 2 weeks*	*3-week delay*	*Start integration tests from sprint 2 using service stubs*	*Add to PMO delivery checklist and test strategy template*	*QA lead*	*Done – template v2.1*

PRO TIP · FROM OBSERVED TO APPLIED

Every lesson needs an **embedding action**: change a template, a checklist, a contract clause, a training module or a gate criterion. And at every project kick-off, review the register for similar projects — pull, not just push.

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A simple improvement cycle

01 Plan
Pick one problem; find its root cause; design a small change

→

02 Do
Try it on a small scale

→

03 Check
Measure the effect against a baseline

→

04 Act
Keep, adapt or drop it; standardise what works

INSIGHT · SMALL STEPS, OFTEN

Continuous improvement works best as many small changes rather than rare big programmes, an idea often called **kaizen**. A team that improves one thing every sprint for a year makes twenty-five improvements. Measure each one, so you keep what works and drop what does not.

Updating organisational process assets

- Templates: add the section that was missing on this project
- Checklists: add the check that would have caught the problem
- Estimating data: record actual durations, costs and velocity for future reference
- Risk lists: add new risk categories and triggers that appeared
- Contract clauses: strengthen terms that caused disputes
- Gate criteria: add the readiness test that was skipped

Keep the improvement process itself improving

1. **Review the retrospective format** Change it when it stops producing useful actions.
2. **Track action completion** Report how many improvement actions were completed and what changed.
3. **Share across teams** Use communities of practice and the PMO to spread what works.
4. **Retire stale actions** Close actions that no longer matter, so the list stays credible.

SCENARIO · DECIDE · THE SAME PROBLEM, SPRINT AFTER SPRINT

For four sprints the team's retrospective has raised 'late requirements changes from the business'. Each time the action is 'remind stakeholders', and nothing changes. What should the project manager do?

(A) Stop recording the item, since nothing can be done.

(B) Help the team find the root cause, agree one measurable experiment with an owner, such as a weekly refinement session with the business, and check its effect next sprint.

(C) Escalate to the sponsor and ask her to forbid late changes.

(D) Extend the sprint length to absorb the changes.

Circle your choice. Answer and feedback: module answer key.

Key terms

Retrospective
A regular team meeting to inspect how the team worked and agree concrete improvements for the next cycle.

Plan-do-check-act (PDCA)
A four-step improvement cycle: plan a change, try it, check the result, then standardise or adjust.

Continuous improvement
An ongoing effort to improve products, services or processes through many small, measured changes.

Kaizen
A philosophy of continuous, incremental improvement involving everyone.

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Lessons-learned register
A record of what happened, its impact and the action taken so the lesson changes future practice.

OPA update
Changing an organisational template, checklist, data set or policy so that a lesson is applied on future projects.

Root cause analysis
Techniques, such as asking 'why?' repeatedly, used to find the underlying cause of a problem.

KNOWLEDGE CHECK

6. Which shows that a lesson has been truly learned?
   - (A) It is stored in the organisation's archive
   - (B) A practice has changed and is used on new projects
   - (C) It was discussed in detail at the closure meeting
   - (D) The sponsor has read the final report
7. A retrospective produces 14 improvement ideas. What is the BEST next step?
   - (A) Assign all 14 ideas to the PM to implement
   - (B) Email them all to the sponsor for a decision
   - (C) Pick one to three actions with owners and dates
   - (D) Wait until the next retrospective to discuss
8. During closure, the team finds that late involvement of the security team caused a three-week delay. What BEST ensures the lesson is applied?
   - (A) Record it in the final report
   - (B) Email it to all project managers
   - (C) Mention it at the closing party
   - (D) Update the planning template and gate checklist to require security review in the first phase
9. A team tries a new code-review rule for two sprints. What should happen NEXT in a plan-do-check-act cycle?
   - (A) Measure defect rates against the earlier baseline and decide whether to keep, adapt or drop it
   - (B) Make the rule permanent across the organisation
   - (C) Try a different rule immediately
   - (D) Ask the sponsor whether she likes the rule
10. At kick-off of a new hospital booking project, the PMO's register holds lessons from three similar projects. What should the PM do?
    - (A) Read them only if problems arise
    - (B) Review them with the team and build relevant actions into the plan and risk register
    - (C) Archive them, since every project is different
    - (D) Send them to the sponsor for approval

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MODULE 8 · ADAPT · LESSON 3 · 30 MIN

Supporting Organisational Change

Assess the culture, help people adopt the change and respond when the organisation changes around the project

- Assess organisational culture and its effect on the project
- Apply ADKAR to diagnose where individuals are stuck
- Use Kotter's eight steps to plan an organisational change effort
- Evaluate the impact of organisational changes on the project and act

A new system can go live perfectly and still fail if nobody uses it properly. **Organisational change management (OCM)** is the people side of delivering value: helping individuals and groups move from the current way of working to the new one. Project management delivers the solution; change management makes sure it is **adopted**.

Assessing organisational culture

DIMENSION	WHAT TO LOOK FOR	IMPLICATION FOR THE PROJECT
Decision-making	Consensus, top-down or delegated?	How long approvals take; who must be in the room
Attitude to failure	Are mistakes punished or discussed?	Whether people will report problems early
Hierarchy	Can juniors challenge seniors?	How to gather honest input
Pace and stability	Frequent reorganisations or long-stable structures?	Change fatigue and readiness
History of change	Did past changes succeed or fail?	The level of scepticism to expect

ADKAR: the individual journey

A
Awareness
of why the change is needed. ‘Why are we doing this, and why now?’

D
Desire
to support and take part. ‘What is in it for me — and what happens if we do not?’

K
Knowledge
of how to change. Training, guides, examples of the new process.

Ab
Ability
to actually perform the new way. Practice, coaching, time, the right tools.

R
Reinforcement
to make it stick. Recognition, measurement, removing the old way.

PRO TIP · FIND THE FIRST BARRIER

ADKAR is sequential. If a branch manager lacks **desire**, more training (knowledge) will not help. Survey or interview, find the earliest element that is weak, and fix that first.

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ADKAR focuses on individuals. **John Kotter's** eight-step model focuses on leading change across an organisation. In our own words, the steps are:

Kotter's eight steps, summarised

1. **Create urgency** Help people see, with evidence, why the status quo is risky.
2. **Build a guiding coalition** Assemble influential, credible leaders from across the organisation.
3. **Form a vision and strategy** A simple picture of the future and how to get there.
4. **Communicate and involve** Share the vision constantly through many channels; recruit volunteers.
5. **Remove barriers** Change structures, systems and processes that block the new way.
6. **Generate short-term wins** Plan visible, early successes and celebrate them.
7. **Sustain acceleration** Use credibility from wins to tackle bigger changes; do not declare victory early.
8. **Anchor in culture** Connect new behaviours to success; build them into hiring, promotion and routines.

Why people resist — and what helps

ROOT OF RESISTANCE	WHAT YOU MIGHT HEAR	HELPFUL RESPONSE
Not understanding why	'The old system works fine.'	Share data and stories on the problem; let them see it themselves
Loss of status, control or skill	'I have been the expert for 12 years.'	Involve them as super-users or trainers; honour their expertise
Fear of job loss	'Is this automation going to replace us?'	Be honest about impacts; explain redeployment and reskilling plans
Change fatigue	'Another initiative?'	Sequence changes; stop something old; show quick wins
Past broken promises	'Last time they said it would be easier.'	Under-promise; pilot first; let peers report results

CASE STUDY · DIGITAL ACCOUNT OPENING AT A COMMERCIAL BANK

A Bangladeshi commercial bank is introducing tablet-based digital account opening at 120 branches, replacing paper forms. The pilot at 10 branches is technically successful, but after six weeks only 30% of new accounts are opened digitally. Interviews show: many officers did not know why the change was happening beyond 'head office says so'; senior officers worry that faster processing will be used to cut staff; several were trained once, two weeks before the tablets arrived; and branch targets still reward the number of accounts, not the channel used.

1. Diagnose the situation with ADKAR. Where is the first barrier?

2. What would you change before rolling out to the other 110 branches?

3. Which Kotter steps were skipped?

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Model answer: module answer key.

TEMPLATE **Change impact assessment**

Identify who is affected, how much and what support they need before go-live.

STAKEHOLDER GROUP	WHAT CHANGES FOR THEM	IMPACT (H/M/L)	ADKAR GAP	SUPPORT ACTIONS	CHANGE CHAMPION
Branch customer-service officers	*Tablet process replaces paper forms; new KYC checks*	*H*	*Desire (job-security fears)*	*Leadership briefing on reskilling; super-user buddies; scorecard update*	*Senior officer, Agrabad branch*

WATCH OUT · WHEN THE ORGANISATION CHANGES AROUND THE PROJECT

Restructures, mergers, a new chief executive or a new strategy can change a project's sponsor, stakeholders, priorities and business case overnight. When it happens, re-check the sponsor's commitment, re-map stakeholders, re-test the business case and agree with governance whether the project should continue, change or stop. Do not assume the old mandate still holds.

SCENARIO · DECIDE · THE VOCAL SCEPTIC

A respected senior supervisor openly criticises the new warehouse scanning process in front of her team, and adoption on her shift is lowest. What should the project manager do FIRST?

(A) Meet her privately to understand her concerns, then involve her in fixing them — possibly as a change champion.

(B) Report her to her manager for undermining the project.

(C) Exclude her shift from the rollout for now.

(D) Run another training session for her shift.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Recall a change you were asked to adopt at work. Using ADKAR, which element was weakest for you, and what would have helped most?

Key terms

Organisational change management (OCM)
Structured methods for helping people move from current to future ways of working so that project outputs are adopted and benefits realised.

Kotter's eight steps
A model for leading organisational change from creating urgency to anchoring the change in the culture.

Organisational culture
The shared values, beliefs and habits that shape how people in an organisation behave.

Change impact assessment
An analysis of who is affected by a change, how much, and what support they need.

ADKAR
A model of individual change: Awareness, Desire, Knowledge, Ability, Reinforcement.

Change champion
A respected person in the affected group who promotes and supports the change locally.

Change saturation
The point at which people are asked to absorb more change than they can handle, so adoption drops.

KNOWLEDGE CHECK

11. Users have been trained and understand the new process, but still prefer the old spreadsheet because their bonus depends on speed. Which ADKAR element needs attention?
    - (A) Awareness of the need for change
    - (B) Desire / Reinforcement
    - (C) Knowledge of the new process
    - (D) Ability to use the new tools
12. Which Kotter step is MOST concerned with celebrating visible early successes?
    - (A) Build a guiding coalition
    - (B) Create a sense of urgency
    - (C) Anchor changes in the culture
    - (D) Generate short-term wins
13. The BEST way to treat resistance is to...
    - (A) overcome it with stronger mandates from leadership
    - (B) ignore it until go-live, when it usually fades
    - (C) explore it as information and address root causes
    - (D) remove persistent resisters from the project team
14. Training delivered three months before people get the new system is a problem mainly for which ADKAR element?
    - (A) Knowledge and Ability
    - (B) Desire and Reinforcement
    - (C) Awareness and Desire
    - (D) Awareness and Reinforcement
15. Halfway through a project, the organisation merges with a competitor and the sponsor leaves. What should the project manager do FIRST?
    - (A) Continue as planned, since the charter is still signed
    - (B) Stop all work until the merger is complete
    - (C) Confirm a new sponsor and re-check stakeholders and the business case with governance
    - (D) Ask the team to speed up before priorities change

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MODULE 8 · ADAPT · LESSON 4 · 32 MIN

The External Environment: Regulation, Markets, AI & Sustainability

Survey changes outside the project, assess their impact on scope and backlog, and keep watching

- Survey changes in regulation, technology, markets and geopolitics
- Assess and prioritise their impact on scope, backlog and business case
- Use AI tools responsibly and respond to AI as a source of change
- Build sustainability into project decisions

Projects do not run in a sealed room. A new regulation, a competitor's launch, an exchange-rate shock, a trade dispute or a new generation of AI tools can change what a project should deliver, sometimes in weeks. Project managers **survey** the external environment, **assess** what changes mean for scope and the backlog, and **keep reviewing** throughout the project, not just at the start.

Scanning the external environment

AREA	EXAMPLE CHANGE	POSSIBLE PROJECT IMPACT
Political and geopolitical	Trade restrictions on a component's country of origin	Re-sourcing; schedule and cost changes
Economic and market	Currency devaluation; a competitor launches first	Budget pressure; reprioritised features
Social	Customers expect mobile-first, local-language service	New backlog items; changed design
Technological, including AI	New AI capability makes a planned feature obsolete	Scope removed or redesigned
Legal and regulatory	New data-protection or AI rules	Compliance work added; go-live conditions
Environmental	Stricter emissions reporting; climate-driven flooding	New data requirements; site and design changes

From external change to the backlog

1. **Survey** Watch regulators, industry news, suppliers, customers and competitors; make it someone's job.
2. **Assess relevance** Does it affect our scope, users, suppliers, costs or business case?
3. **Estimate impact** How much, how soon, and is it mandatory?
4. **Prioritise** Rank against existing backlog items or planned scope by value and urgency.
5. **Act** Add backlog items, raise change requests or record risks, and inform governance.
6. **Keep reviewing** Make external change a standing item at quarterly planning and gate reviews.

TEMPLATE **External change radar**

Track external changes and what the project is doing about each one.

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CHANGE	SOURCE AND DATE	AREA	IMPACT ON SCOPE OR BACKLOG	URGENCY	ACTION AND OWNER
Data-localisation rule for customer data	*Regulator notice, 10 Jan*	*Legal and regulatory*	*Hosting must move in-country; 3 new backlog epics*	*High: 9-month deadline*	*Change request raised; compliance officer and tech lead*

INSIGHT · AI AS A CHANGE IN THE ENVIRONMENT

Artificial intelligence changes projects in two ways. It is a **tool**: drafting, summarising, analysing and forecasting. And it is an **external force**: customers expect smarter services, competitors ship AI features, new laws regulate AI use, and some planned features become obsolete. Review both: how your team uses AI, and how AI changes what your project should deliver.

Where an AI assistant helps (and where to be careful)

PM TASK	GOOD USE	WATCH OUT FOR
Charter and plan drafts	First draft of objectives, scope statements, assumptions from your notes	Invented constraints or targets; generic text that fits no real project
Risk brainstorming	Generate candidate risks by RBS category to widen thinking	Plausible but irrelevant risks; missing local context
Status summaries	Turn raw updates and tracker exports into a one-page draft	Softening bad news; mis-stated numbers
Meeting notes	Summarise transcripts into decisions, actions and owners	Wrong attribution; confidential remarks captured; consent to record
Schedule what-ifs	Explain trade-offs and suggest scenarios to test	Arithmetic and date errors — run the real numbers in your scheduling tool
Stakeholder messages	Adapt one message for different audiences and reading levels	Tone that does not fit the relationship; over-promising

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Data-privacy rules for AI use

APP

Approved tools only

Use the AI tools your organisation has approved under an enterprise agreement. Consumer accounts may retain or use your inputs in ways your contracts forbid.

CLS

Classify before you paste

Public and internal data may be fine; confidential, client, personal or regulated data usually is not — unless the tool is explicitly approved for that class.

MIN

Minimise and anonymise

Remove names, phone numbers, account numbers and contract values; use placeholders such as ‘Vendor A’.

SEC

No secrets ever

Never paste passwords, access keys or security configurations.

REC

Consent for recording

Tell participants before an AI note-taker records or transcribes a meeting, and follow local law and company policy.

DEFINITION · USEFUL REFERENCES FOR AI GOVERNANCE

Organisations increasingly align AI work with frameworks such as the **NIST AI Risk Management Framework** and the management-system standard **ISO/IEC 42001**. A PM does not need to be an expert, but should ask early: ‘Which AI policy and risk process applies to this project?’

ESG in project decisions

E

Environmental

Energy and carbon over the life cycle, materials, waste, water and pollution, climate resilience (e.g. flood-proofing sites).

S

Social

Labour standards in the supply chain, health and safety, community impact, accessibility and inclusion of users.

G

Governance

Anti-corruption, transparency in procurement, data protection, ethical AI, honest reporting.

Practical ways to build sustainability into a project

- Add a sustainability objective or key result to the charter where relevant
- Include carbon and lifecycle cost — not just purchase price — in option appraisal
- Use supplier criteria for labour standards and environmental performance
- Choose energy-efficient hosting and right-size infrastructure
- Plan for reuse, recycling or safe disposal of equipment and materials
- Report ESG measures alongside cost and schedule at stage gates

CASE STUDY · SOLAR IRRIGATION PUMPS IN NORTHERN BANGLADESH

An NGO-funded programme will install 300 solar irrigation pumps across Rangpur and Dinajpur over three years, run as three projects: procurement and installation, a farmer payment-and-service scheme, and a maintenance training project. The board has limited funds and two other proposals: rooftop solar for clinics, and a mobile weather-advisory service.

1. Why is this better run as a programme than three separate projects?

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2. Suggest four weighted criteria the portfolio board could use to compare the three proposals.

3. Name two environmental and two social risks the programme should manage.

Model answer: module answer key.

SCENARIO · DECIDE · A NEW RULE MID-PROJECT

Six months into a cloud-hosted customer app, the regulator announces that customer data must be stored in-country within nine months. The product owner is focused on new features for a festival campaign. What should the project manager do FIRST?

(A) Ignore it until the rule is finalised in law.

(B) Assess the impact on scope, backlog, costs and the business case with the product owner and compliance officer, then present prioritised options to governance.

(C) Move all hosting immediately, pausing feature work.

(D) Add a risk to the register and continue as planned.

Circle your choice. Answer and feedback: module answer key.

Key terms

Generative AI copilot
An AI assistant that generates text, summaries, code or other content from natural-language prompts to support a person's work.

Data minimisation
Sharing only the data strictly needed for a task, with personal or confidential details removed or masked.

Model drift
The decline in an AI model's performance over time as real-world data diverges from its training data.

PESTLE analysis
A checklist for scanning the external environment: political, economic, social, technological, legal and environmental factors.

Hallucination
Output from a generative model that is fluent and plausible but factually wrong or invented.

ESG
Environmental, social and governance considerations used to assess the wider impact and responsibility of projects and organisations.

Environmental scanning
Regularly surveying political, economic, social, technological, legal and environmental changes that may affect a project.

KNOWLEDGE CHECK

16. Which data is generally SAFEST to share with an approved AI assistant?

(A) Anonymised meeting notes with internal-only information

(B) A customer list with names and phone numbers

(C) Administrator passwords for the test environment

(D) A signed contract with pricing, in a public chatbot

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17. Comparing options by lifecycle energy cost rather than purchase price alone is an example of...

(A) social sustainability and community benefit
(B) gold plating of the specification
(C) environmental and financial sustainability
(D) scope creep driven by stakeholders

18. An AI model for loan approvals performed well at launch but accuracy falls after a year. This is most likely...

(A) scope creep
(B) a hallucination
(C) a lagging indicator
(D) model drift

19. A competitor launches an AI feature that makes one of your planned features far less valuable. What should the product owner and PM do?

(A) Continue, because the feature is in the approved roadmap
(B) Cancel the whole project
(C) Copy the competitor's feature exactly
(D) Reassess the backlog's value in light of the change and reprioritise with stakeholders

20. Which practice BEST makes sure external changes keep being noticed during a long project?

(A) A standing review of external changes at quarterly planning and gate reviews
(B) A one-off environmental scan at initiation
(C) Waiting for the sponsor to report changes
(D) Reading industry news when the project is in trouble

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MODULE 8 · ADAPT · LESSON 5 · 40 MIN

The Whole Loop: Capstone & Exam Readiness

Watch all eight stages work together on one project, plan your capstone and prepare for situational questions

- Trace a project through all eight stages of the ZPMP Professional Loop
- Plan your ZPMP capstone and its deliverables
- Apply a reliable method to situational-judgement questions
- Recognise predictive, agile and hybrid context cues

Real projects do not separate neatly into modules. A risk spotted in Govern changes the plan; a stakeholder concern in Engage reshapes scope; a lesson in Adapt improves the next Frame. Here is one mini-project followed through the whole **ZPMP Professional Loop**.

EXAMPLE · THE MINI-PROJECT

Cumilla District Hospital — SMS and web appointment booking. Patients queue from dawn for outpatient tickets; 22% of booked follow-ups are missed. A 6-month project with a budget of BDT 4.5 million and a team of 7 (PM, product owner from hospital administration, 3 developers, 1 tester, 1 change and training lead) will deliver online and SMS booking with reminders for four outpatient departments.

Walking the loop

STAGE	KEY QUESTION	WHAT THE TEAM DID	KEY ARTEFACT
Frame	Is this worth doing, and do we share the vision?	Business case for fewer queues and no-shows; hybrid approach; vision co-written with doctors and front desk	Business case, charter, vision
Lead	Is the team set up to succeed?	Team charter and RACI; conflict over slot rules resolved collaboratively	Team charter, RACI
Engage	Who cares, and what do they expect?	Stakeholder map; alignment session agreed SMS-first for older patients; knowledge transfer plan for the hospital IT officer	Stakeholder register, alignment map, communication plan
Plan	What exactly, and in what order of value?	Integrated plan joining the SMS-gateway contract with a prioritised backlog	Integrated plan, backlog
Resource	How long, who and how much?	Velocity-based release forecast; P80 at 6.5 months; budget with contingency	Release plan, budget
Deliver	Are we building it well and on track?	Sprints with usability tests; vendor scorecard; burnup and dashboard	Increments, dashboard
Govern	Are we compliant and in control?	Patient-data consent built into stories; change control for the national ID integration; risk register	Compliance register, change log, risk register
Adapt	What blocks us, what is changing, and what did we learn?	Impediment board; front-desk adoption plan; new national ID rule assessed; retrospectives and benefits plan	Impediment log, change impact assessment, lessons register

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SCENARIO · DECIDE · A MID-PROJECT CURVEBALL

In month 4, the health directorate announces that all hospital systems must integrate with a new national patient ID within 6 months. The product owner wants to add it now. The P80 forecast already sits at the deadline. What should the PM do FIRST?

(A) Add the integration to the current sprint to show responsiveness.

(B) Tell the directorate the project cannot comply.

(C) Analyse the requirement and its impact with the product owner and team, then present options (e.g. phase 2 release before the mandate date) through change control.

(D) Escalate to the sponsor to request more budget.

Circle your choice. Answer and feedback: module answer key.

Your **ZPMP capstone** asks you to apply the whole loop to a real or realistic project of your choice, ideally from your workplace with confidential details removed. You will produce a compact pack of artefacts and a short reflective report showing how you handled people, process and the business environment.

Capstone deliverables by stage

1. **Frame** One-page business case, charter and shared vision with measurable success criteria.
2. **Lead** Team charter with ground rules and a RACI for key decisions.
3. **Engage** Stakeholder register, expectation alignment map, knowledge transfer plan and communication plan.
4. **Plan** Integrated plan summary and a prioritised backlog or WBS with acceptance criteria.
5. **Resource** Schedule or release forecast, resource plan and budget with reserves.
6. **Deliver** Quality approach, supplier approach and a simulated status dashboard.
7. **Govern** Governance and escalation thresholds, compliance register, change log and top-10 risk register.
8. **Adapt** Impediment log, change-adoption plan, external change radar and lessons with OPA updates.

TEMPLATE **Capstone planner**

Plan your capstone: one row per stage of the Professional Loop.

STAGE	ARTEFACTS I WILL PRODUCE	DATA OR INPUT I NEED	WHO CAN REVIEW IT	TARGET DATE
Govern	*Compliance register and top-10 risk register*	*Regulatory list from compliance team; risk workshop notes*	*Line manager and compliance officer*	*Week 7*

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STAGE	ARTEFACTS I WILL PRODUCE	DATA OR INPUT I NEED	WHO CAN REVIEW IT	TARGET DATE

Many PM certifications — including ZPMP's final exam — use **situational-judgement** questions: a short scenario and four plausible actions. Usually more than one option is reasonable; you must find the **best** one for the context.

The ZPMP 5-R method for situational questions

1. **Read the last line first** Spot the qualifier: FIRST, NEXT, BEST, MOST likely, LEAST, EXCEPT.
2. **Recognise the context** Agile, predictive or hybrid? Who is involved? Which domain and which stage of the Professional Loop?
3. **Reduce** Eliminate options that are unethical, passive, blame-focused or bypass the process.
4. **Rank** Between the remaining two, prefer the action that addresses the root cause, gathers facts or involves the right people directly.
5. **Re-check** Does your pick answer the actual question — e.g. FIRST rather than eventually?

Context cues

Agile cues

- Product owner, sprint, backlog, iteration review
- Self-organising team, servant leader
- Best answers: collaborate with the product owner on priorities, let the team decide how, inspect and adapt, frequent feedback

Predictive cues

- Baseline, change control board, WBS, signed contract
- Phase gates, formal approvals
- Best answers: assess impact, follow the change management plan, update baselines only after approval

Universal cues

- Ethics, safety, conflict, stakeholder concern
- Best answers: understand first, talk directly and privately, act within your authority, escalate with options when needed

Answer patterns to be suspicious of

PATTERN	WHY IT IS USUALLY WEAKER
Escalate to the sponsor immediately	Skips analysis and actions within the PM's own authority
Ignore it / wait and see	Passive; problems grow
Fire, punish or publicly blame someone	Damages safety; rarely solves root causes
Implement the change straight away because the client asked	Bypasses impact analysis and change control
Add more people immediately	May not fix the cause; can slow a late project further
Absolute words: always, never, all	Real practice is contextual

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Exam-day habits

- Pace yourself: note the time per question and flag hard ones to return to
- Answer every question — an unanswered question cannot score
- Do not overthink: choose the best answer from the options given, not the one you wish were there
- Watch for qualifiers: FIRST and NEXT are about sequence, BEST is about overall quality
- Use breaks to reset attention

Key terms

Situational-judgement question
An exam item presenting a realistic scenario and asking for the best action among plausible alternatives.

Capstone
A culminating project in which learners apply the whole ZPMP Professional Loop to a real or realistic project and present the resulting artefacts and reflections.

Qualifier
A word in a question such as FIRST, NEXT, BEST or EXCEPT that defines what kind of answer is wanted.

Distractor
A plausible but weaker option in a multiple-choice question.

Context cue
A detail in a scenario, such as 'sprint' or 'baseline', that signals the approach and shapes the best answer.

Integration
Combining decisions across scope, schedule, cost, people, risk and value so they work together.

KNOWLEDGE CHECK

21. A question asks what the PM should do FIRST after a team member reports a new risk. Which option is most likely best?
    - (A) Escalate it to the sponsor straight away
    - (B) Assess and record it in the risk register
    - (C) Implement a response immediately to be safe
    - (D) Inform all stakeholders by email

22. Which option pattern is usually weakest in situational questions?
    - (A) Review the relevant plan and baseline before acting
    - (B) Meet the stakeholder privately to understand their concern
    - (C) Ignore the issue and wait for the next status meeting
    - (D) Collaborate with the product owner on the backlog

23. Which cue suggests a predictive context?
    - (A) Sprint review with stakeholders
    - (B) Product backlog refinement
    - (C) Self-organising, cross-functional team
    - (D) Change control board approval

24. A mandatory new requirement arrives mid-project. What is the FIRST step?
    - (A) Add it to the plan immediately, since it is mandatory anyway
    - (B) Analyse its impact and present options via change control
    - (C) Refuse it until the current phase has been completed
    - (D) Request extra budget and time from the sponsor to cover it

25. In the walk-through, the hospital's new national ID rule was assessed for its effect on the backlog. Which ZPMP stage does this MOST represent?
    - (A) Adapt
    - (B) Lead
    - (C) Resource
    - (D) Frame

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · ADAPT

Answer key

Knowledge checks

1. C Impediments that threaten goals should move up the escalation ladder quickly with clear options.

2. D Leaders decide faster and better when given options, trade-offs and a clear recommendation.

3. B When a risk occurs it becomes an issue. Move it to the issue log, apply the planned response and note the outcome against the risk.

4. D Prioritise impediments by impact on goals and how much work they block.

5. A Continual reassessment means changing the intervention when the current one is not working.

6. B A lesson is learned when it changes behaviour or practice.

7. C Focused, owned actions are more likely to happen.

8. D A lesson is learned when practice changes. Updating the organisation's templates and gate criteria builds it into future projects.

9. A After 'Do' comes 'Check': measure the effect before standardising or abandoning the change.

10. B Lessons are valuable when pulled into new projects at the start, not only pushed out at the end of old ones.

11. B They know how but lack motivation, and the reward system reinforces the old way.

12. D Short-term wins build credibility and momentum.

13. C Resistance often reveals real issues; understanding it leads to better solutions and adoption.

14. A Knowledge fades without use; just-in-time training plus practice builds ability.

15. C Organisational change can remove the project's mandate. Confirm sponsorship and re-test the business case before carrying on.

16. A Anonymised internal data in an approved tool is the low-risk option.

17. C Lifecycle costing captures operating energy and environmental impact.

18. D Changing real-world data degrades model performance over time.

19. D External changes should be assessed for their impact on scope and the backlog, then reprioritised by value.

20. A The external environment must be reviewed continually; a standing agenda item makes it routine.

21. B Analyse and document before acting or escalating.

22. C Passive waiting lets problems grow.

23. D Formal change control boards are characteristic of predictive governance.

24. B Impact analysis and options precede decisions.

25. A Evaluating changes in the external environment and their impact on scope is part of Adapt (Business Environment Task 8).

Scenarios & case studies

> **Scenario · Three days without a test environment**
> **B** is best. Best. It escalates with facts and options to the person who can act, and limits the damage while the blocker is removed.

> **Scenario · The same problem, sprint after sprint**
> **B** is best. Best. A recurring problem needs root-cause analysis and a measurable change, not repeated reminders.

> **Case study · Digital account opening at a commercial bank**
> ADKAR: **Awareness** is weak (no compelling why), **Desire** is undermined by job-loss fear and by targets that do not reward the change, **Knowledge** decayed because training came too early, **Ability** lacked practice time, **Reinforcement** is absent. The first barrier is awareness/desire, so more training alone will fail. Before rollout: leaders explain the customer and compliance reasons (faster opening, fewer errors, better KYC data); HR and leadership give an honest message on staffing and reskilling; train just-in-time with practice sessions and branch super-users; add digital-channel share to branch scorecards; publish pilot-branch wins. Kotter steps skipped: urgency, guiding coalition beyond head office, communicating the vision, removing barriers (targets) and short-term wins.

> **Scenario · The vocal sceptic**
> **A** is best. Correct. Her influence is an asset; if her concerns are addressed, she can bring the team along.

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Case study · Solar irrigation pumps in northern Bangladesh
Programme: the benefit — higher, sustainable farmer incomes and lower diesel use — only appears when pumps, payment scheme and maintenance skills work together; a programme manages those dependencies and benefits. Criteria (example): impact on income/livelihoods 35%, emissions reduction 25%, cost per beneficiary 25%, delivery risk 15%. Environmental: groundwater over-extraction when water becomes cheap; end-of-life disposal of panels and batteries. Social: unequal access (larger landowners capturing benefits), and safety/theft issues at remote sites — plus fair labour practices among installers.

Scenario · A new rule mid-project
B is best. Best. It turns an external change into an assessed, prioritised response and involves the right people.

Scenario · A mid-project curveball
C is best. Correct. Understand the mandate, assess impact on scope, schedule, cost and risk, and bring options to the right decision makers.

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REFERENCE

Glossary

1

100% rule The principle that the children of each WBS element add up to exactly all of the parent's work. (Plan)

A

Acceptance criteria Conditions a deliverable or story must satisfy to be accepted, often written as Given/When/Then. (Plan)

Accommodating (smoothing) A conflict mode that gives way to the other side, emphasising agreement; useful when the issue matters more to them. (Lead)

Accountable (RACI) The single person who owns the outcome of an activity or decision and approves it. (Lead)

Actual cost (AC) The cost actually incurred for the work performed by a given date. (Resource)

Adaptive approach An approach that plans in short cycles and refines scope through frequent feedback; agile methods are the most common form. (Frame)

ADKAR A model of individual change: Awareness, Desire, Knowledge, Ability, Reinforcement. (Adapt)

Analogous estimating Estimating from the actual results of a similar past project, adjusted for known differences. (Resource)

Async-first A working style in which written, recorded or tool-based updates are the default and live meetings are kept for discussion and decisions. (Lead)

Audit trail A record that shows who did what, when and why, used as evidence in reviews and audits. (Govern)

Avoiding (withdrawing) A conflict mode that postpones or steps away from the issue; acceptable for trivial matters or to let tempers cool. (Lead)

B

Baseline An approved version of the scope, schedule or cost plan, changed only through formal change control and used to measure performance. (Plan)

BATNA Best Alternative To a Negotiated Agreement: the course of action a party will take if negotiations fail. (Deliver)

Benefit A measurable improvement that results from an outcome and is valued by the organisation or society. (Plan)

Benefit leakage Gains, such as saved time, that are absorbed into other activity instead of producing the intended value. (Plan)

Benefit owner The business manager accountable for realising a benefit after the project delivers its outputs. (Plan)

Benefits realisation Measuring and managing the outcomes of a project after delivery to ensure intended benefits are achieved. (Plan)

Benefits register A list of expected benefits with measures, baselines, targets, owners and how and when each will be measured. (Plan)

Blocker age How long an impediment has remained unresolved; a rising age signals a need for a different intervention. (Adapt)

Bottom-up estimating Estimating each work package or activity in detail and adding them up. (Resource)

Budget at completion (BAC) The total approved budget for the work, the sum of all planned value. (Resource)

Burndown chart A chart of work remaining against time, used to see whether committed work will finish. (Deliver)

Burnup chart A chart showing completed work and total scope over time, making scope change visible. (Deliver)

Bus factor An informal measure of how many people could be unavailable before part of the work stops; a key-person risk. (Engage)

Business case A documented justification for an investment that compares options, benefits, costs and risks and recommends a way forward. (Frame)

C

Capstone A culminating project in which learners apply the whole ZPMP Professional Loop to a real or realistic project and present the resulting artefacts and reflections. (Adapt)

Change champion A respected person in the affected group who promotes and supports the change locally. (Adapt)

Change control board (CCB) A group with authority to review, approve, defer or reject change requests. (Govern)

Change impact assessment An analysis of who is affected by a change, how much, and what support they need. (Adapt)

Change log A record of change requests, their impacts, decisions and status. (Govern)

Change request A formal proposal to modify a baseline, deliverable, plan or contract. (Govern)

Change saturation The point at which people are asked to absorb more change than they can handle, so adoption drops. (Adapt)

Closure criteria Agreed conditions that must be met before a phase or project is formally closed. (Deliver)

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Coaching Helping someone improve by asking questions that lead them to their own insight and action, rather than giving instructions. (Lead)

Collaborating (problem-solving) A conflict mode that works with both sides to find a solution meeting everyone's interests; slow but durable. (Lead)

Committed cost Money already promised through contracts or purchase orders, even if not yet invoiced or paid. (Resource)

Common vision A short, shared picture of the future the project will create, held by the team and key stakeholders. (Frame)

Communication channels The number of possible two-way links between people, calculated as n × (n − 1) ÷ 2. (Engage)

Communication management plan The part of the project plan that defines who needs what information, when, how and from whom, and how feedback is gathered. (Engage)

Community of practice A group of people across teams who regularly share knowledge about a common discipline. (Engage)

Competing (forcing) A conflict mode that imposes one view; fitting for emergencies or safety and compliance matters. (Lead)

Compliance Meeting the laws, regulations, standards, contracts and policies that apply to a project and its result. (Govern)

Compliance by design Building obligations into requirements, design and acceptance criteria from the start rather than checking at the end. (Govern)

Compliance obligation A specific requirement the project must meet, such as obtaining consent or a safety permit. (Govern)

Compliance register A record of obligations, their sources, owners, evidence and status. (Govern)

Compromising A conflict mode in which each side gives up something to reach a quick, acceptable agreement. (Lead)

Configuration management Controlling versions of deliverables and their documentation so that everyone works from the same approved version. (Govern)

Conflict of interest A situation in which personal interests could improperly influence, or appear to influence, professional judgement. (Govern)

Context cue A detail in a scenario, such as 'sprint' or 'baseline', that signals the approach and shapes the best answer. (Adapt)

Contingency reserve Budget or time within the baseline set aside for identified risks. (Resource)

Continuous improvement An ongoing effort to improve products, services or processes through many small, measured changes. (Adapt)

Contract closure Completing all contract obligations, settling claims and payments and formally ending the contractual relationship. (Deliver)

Cost baseline The approved, time-phased budget, including contingency but excluding management reserve, used to measure cost performance. (Resource)

Cost of delay The value lost for each period that a piece of work is not delivered. (Plan)

Cost of quality The total cost of conformance (prevention and appraisal) and non-conformance (internal and external failure). (Deliver)

Cost performance index (CPI) EV ÷ AC — the cost efficiency of work performed; below 1.0 means over budget. (Resource)

Cost-reimbursable contract A contract that pays the seller's allowable costs plus a fee, leaving most cost risk with the buyer. (Deliver)

Crash cost slope The extra cost per unit of time saved when an activity is shortened: (crash cost − normal cost) ÷ (normal duration − crash duration). (Resource)

Crashing Shortening the schedule by adding resources to critical activities, usually at extra cost. (Resource)

Critical information requirement Information the project must produce or track, such as regulatory or sustainability data, identified early so it can be collected reliably. (Plan)

Critical path The longest sequence of dependent activities through a schedule network; it determines the shortest possible project duration. (Resource)

Cumulative flow diagram A stacked area chart showing the number of items in each workflow stage over time. (Deliver)

Customer satisfaction (CSAT) The share of customers who rate an experience as satisfying on a simple scale. (Engage)

Cycle time The elapsed time from when work starts on an item until it is finished. (Deliver)

D

DACI A decision-rights model naming the Driver who runs the decision, one Approver who decides, Contributors who advise and people who are Informed. (Lead)

Data minimisation Sharing only the data strictly needed for a task, with personal or confidential details removed or masked. (Adapt)

Decision log A record of key decisions, their reasons and the alternatives considered. (Engage)

Decision rights Agreed rules about who may make which decisions and when a decision must be escalated. (Lead)

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Definition of done A shared quality standard that every increment of work must meet before it is considered complete. (Plan)

Definition of ready An optional team agreement on the conditions a backlog item should meet before it is taken into a sprint. (Plan)

Delegation level An agreed degree of decision authority, from the leader deciding alone to the team deciding without reporting back. (Lead)

Demonstration Showing working results, samples or prototypes to customers so they can give feedback on the real thing. (Engage)

Development approach The way the product is created within the life cycle: predictive, adaptive (agile) or hybrid. (Frame)

Dis-benefit A measurable negative consequence of a change, perceived as adverse by one or more stakeholders. (Plan)

Distractor A plausible but weaker option in a multiple-choice question. (Adapt)

Domain A broad area of project management practice; the July 2026 outline uses People, Process and Business Environment. (Frame)

E

Earned value (EV) The budgeted value of the work actually completed to date. (Resource)

Elicitation Drawing out stakeholders' real needs through techniques such as interviews, observation, workshops and prototypes. (Plan)

Emotional intelligence The ability to recognise, understand and manage one’s own emotions and to recognise and influence others’ emotions. (Lead)

Empathy Understanding another person's feelings and perspective and taking them into account. (Lead)

Empowerment Giving people the authority, information and support to make decisions about their own work. (Lead)

Enabler An illustrative example of the work involved in a task; not an exhaustive list. (Frame)

Engagement assessment matrix A table that shows each stakeholder's current and desired engagement level, from unaware to leading. (Engage)

Environmental scanning Regularly surveying political, economic, social, technological, legal and environmental changes that may affect a project. (Adapt)

Epic A large body of work that is too big for one iteration and is split into smaller stories. (Plan)

Escalation Passing an issue to a higher authority, with facts and options, when it is beyond the current level's power to resolve. (Adapt)

Escalation path The agreed route by which an issue or variance moves to a higher level of authority for decision. (Govern)

ESG Environmental, social and governance considerations used to assess the wider impact and responsibility of projects and organisations. (Adapt)

Estimate at completion (EAC) The expected total cost of the project when all work is complete, based on performance so far. (Resource)

Exam content outline A published description of the domains and tasks an exam tests, with the share of questions each domain receives. (Frame)

Execution strategy The recommended way the work will be delivered: approach per component, sourcing, team structure, release strategy and coordination. (Plan)

Expectation A stakeholder's belief about what a project will deliver, how or when; often unstated. (Engage)

Expectation alignment Bringing different stakeholders' expectations together into one agreed set that fits the project's objectives. (Engage)

Expectation gap The difference between what a customer expects and what is delivered or understood. (Engage)

Expected monetary value (EMV) Probability multiplied by monetary impact; summed across risks or branches to compare options and size reserves. (Govern)

Explicit knowledge Knowledge that can be written down and shared through documents, code or records. (Engage)

External customer A client, citizen, patient or user outside the organisation who receives the project's result. (Engage)

Extrinsic motivation Motivation driven by outside rewards or pressures such as pay, bonuses, deadlines or praise. (Lead)

F

Failure cost The cost of defects found before delivery (internal) or after delivery (external). (Deliver)

Fallback plan A prepared plan B to use if the primary risk response fails. (Govern)

Fast-tracking Shortening the schedule by overlapping activities normally done in sequence, at a higher risk of rework. (Resource)

Feedback loop A built-in way of checking whether communication was received, understood and useful, and adjusting it. (Engage)

Firm fixed price contract A contract in which the seller delivers defined scope for an agreed price and carries most of the cost risk. (Deliver)

First-time acceptance rate The share of deliverables accepted at their first review, a sign of how well expectations are understood. (Engage)

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Fist-of-five A quick group technique in which people show zero to five fingers to signal their level of support for a proposal. (Lead)

Formal acceptance Documented confirmation by an authorised stakeholder that deliverables meet agreed acceptance criteria. (Deliver)

Free float The amount of time an activity can be delayed without delaying the early start of any successor. (Resource)

Funding limit reconciliation Comparing planned spending with available funding by period and adjusting the schedule if needed. (Resource)

G

Generative AI copilot An AI assistant that generates text, summaries, code or other content from natural-language prompts to support a person's work. (Adapt)

Go or no-go review A formal decision point that checks readiness criteria before a release or handover proceeds. (Deliver)

Gold plating Adding features or quality that were not requested, usually by the team, without approval. (Govern)

Grade A category for products with the same use but different features; low grade can be acceptable, low quality cannot. (Deliver)

Ground rules Explicit expectations about acceptable behaviour that a team agrees and enforces itself. (Lead)

GROW model A coaching structure: Goal, Reality, Options, Will (way forward). (Lead)

H

Hallucination Output from a generative model that is fluent and plausible but factually wrong or invented. (Adapt)

Hybrid approach A deliberate mix of predictive and adaptive practices, each applied where it fits best. (Frame)

Hygiene factor A condition such as pay, policy or working environment whose absence causes dissatisfaction but whose presence does not by itself motivate. (Lead)

Hypercare A defined period of intensified support immediately after go-live, ending when agreed stability criteria are met. (Deliver)

I

Impediment Anything that slows or blocks the team's progress. (Adapt)

Impediment board A visible list or board of current blockers, their owners and ages, reviewed regularly. (Adapt)

Inclusion Working practices that let every team member contribute fully and be heard, whatever their background. (Lead)

Increment A usable step toward the product goal that meets the definition of done. (Deliver)

Incremental Delivering usable pieces of the result one after another, each adding value. (Frame)

Influence The ability to shape others' decisions and actions without relying on formal authority. (Engage)

Information radiator A large, visible display of project information, physical or digital, that anyone can see at a glance. (Engage)

Integrated change control Reviewing every change for its combined effect on scope, schedule, cost, quality, risk and benefits before deciding. (Govern)

Integrated project management plan The single, consolidated plan that defines how the project will be executed, monitored, controlled and closed, combining all subsidiary plans and baselines. (Plan)

Integration Combining decisions across scope, schedule, cost, people, risk and value so they work together. (Adapt)

Interactive communication Real-time, two-way exchange such as meetings, calls and workshops. (Engage)

Internal customer A team or department inside the organisation that receives or depends on the project's result. (Engage)

Intrinsic motivation Motivation that comes from the satisfaction of the work itself — interest, growth and meaning. (Lead)

INVEST Six qualities of a good user story: Independent, Negotiable, Valuable, Estimable, Small and Testable. (Plan)

Issue A current problem or condition that is affecting a project objective and needs resolution. (Adapt)

Issue log A record of issues with their impact, owner, actions, target dates and status. (Adapt)

Iterative Improving the same result through repeated cycles of build, feedback and refinement. (Frame)

K

Kaizen A philosophy of continuous, incremental improvement involving everyone. (Adapt)

Kano model A model that classifies features as basic, performance or delighter by how they affect satisfaction. (Plan)

Key performance indicator (KPI) A measure that tracks how well an ongoing activity or capability is performing. (Plan)

Knowledge transfer Making sure knowledge the project depends on is identified, captured and moved to the people who need it. (Engage)

Kotter's eight steps A model for leading organisational change from creating urgency to anchoring the change in the culture. (Adapt)

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L

Lag A required waiting time between a predecessor and its successor. (Resource)

Lead An overlap that lets a successor start before its predecessor finishes. (Resource)

Lead time The elapsed time from when an item is requested or committed to until it is delivered to the customer. (Deliver)

Lessons-learned register A record of what happened, its impact and the action taken so the lesson changes future practice. (Adapt)

Life cycle The series of phases a project passes through from start to finish. (Frame)

M

Make-or-buy analysis Comparing doing work in-house with buying it, on cost, capability, risk and control. (Deliver)

Management by exception A governance style in which senior people are involved only when agreed tolerances are forecast to be exceeded. (Govern)

Management reserve Budget outside the cost baseline for unforeseen work; its use requires management approval. (Resource)

Mentoring A longer-term relationship in which an experienced person helps another build understanding and capability. (Engage)

Minimum viable product (MVP) The smallest release that lets the team test its main value assumption with real users. (Plan)

Model drift The decline in an AI model's performance over time as real-world data diverges from its training data. (Adapt)

Monte Carlo simulation A technique that runs a schedule or cost model thousands of times with random values from estimate ranges to produce probabilities. (Resource)

MoSCoW A prioritisation method that sorts requirements into Must have, Should have, Could have and Won't have this time. (Plan)

Motivator A factor such as achievement, recognition or responsibility that creates genuine satisfaction and effort. (Lead)

N

Net present value (NPV) The value today of future net cash flows, discounted at a chosen rate; a positive NPV means the investment earns more than the rate. (Frame)

Net promoter method A loyalty measure that subtracts the percentage of detractors from the percentage of promoters. (Engage)

Non-compliance Failure to meet an applicable obligation, which can lead to penalties, stoppages or harm. (Govern)

O

OKR Objectives and Key Results: a qualitative objective paired with a few measurable outcomes that show progress toward it. (Frame)

OPA update Changing an organisational template, checklist, data set or policy so that a lesson is applied on future projects. (Adapt)

Operational readiness The state in which the receiving organisation has the people, processes, tools and data to run the result. (Deliver)

Operations Ongoing, repetitive work that sustains the organisation's existing products and services. (Frame)

Opportunity An uncertain event that would benefit at least one project objective if it happened. (Govern)

Organisational change management (OCM) Structured methods for helping people move from current to future ways of working so that project outputs are adopted and benefits realised. (Adapt)

Organisational culture The shared values, beliefs and habits that shape how people in an organisation behave. (Adapt)

Organisational process assets (OPAs) An organisation's existing plans, policies, procedures, templates and knowledge that projects can use and tailor. (Govern)

Outcome The change in behaviour or performance that happens when people use the outputs. (Frame)

Output A deliverable the project produces, such as an app, a building or a trained group of staff. (Frame)

P

P80 The value, such as a date or cost, that 80% of simulated outcomes meet or beat. (Govern)

Parametric estimating Estimating by multiplying a quantity by a unit rate derived from historical data, e.g. cost per square metre. (Resource)

Pareto chart A bar chart that ranks causes by frequency or impact to show the few causes behind most problems. (Deliver)

Payback period How long it takes for the money spent to be recovered from the project's net inflows. (Frame)

Performance domain A group of related activities, such as stakeholders, planning or uncertainty, that must go well for the project to deliver value. (Frame)

PERT estimate A weighted three-point estimate calculated as (Optimistic + 4 × Most likely + Pessimistic) ÷ 6. (Resource)

PESTLE analysis A checklist for scanning the external environment: political, economic, social, technological, legal and environmental factors. (Adapt)

Pilot A full solution deployed to a limited site or group to prove it works before wider rollout. (Plan)

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Plan-do-check-act (PDCA) A four-step improvement cycle: plan a change, try it, check the result, then standardise or adjust. (Adapt)

Planned value (PV) The budgeted cost of the work scheduled to be done by a given date. (Resource)

Planning poker A team technique in which estimators reveal relative sizes at the same time and discuss differences before re-voting. (Resource)

PMO Project (or portfolio/programme) management office: a function that standardises governance, supports or directs projects and improves delivery capability. (Govern)

Portfolio The full set of projects, programs and operational investments an organisation manages to achieve its strategy, selected and balanced as a whole. (Frame)

Power distance The degree to which people accept and expect unequal distribution of power and deference to seniority. (Lead)

Power–interest grid A two-by-two grid that plots stakeholders by their power over the project and their interest in it, suggesting how closely to engage each. (Engage)

Pre-mortem A workshop in which the team imagines the project has failed and works backwards to identify the reasons, surfacing risks early. (Govern)

Predictive approach An approach that defines scope, schedule and cost early and then executes the plan in sequence, controlling change through baselines. (Frame)

Prevention cost Money spent to stop defects happening, such as training, standards and design reviews. (Deliver)

Principle A broad statement of what matters that guides judgement when no process or template fits. (Frame)

Probabilistic forecast A forecast stated as a date or amount with a confidence level, such as '85% likely by 2 June'. (Deliver)

Probability–impact matrix A grid that combines probability and impact scores to rank risks. (Govern)

Product A long-lived offering, such as an app or a service, that evolves through many releases or projects and is usually owned by a stable team. (Frame)

Product backlog An ordered, evolving list of everything that might be needed in a product, owned by the product owner. (Plan)

Product Owner The accountable person in Scrum who maximises product value and owns the ordering of the product backlog. (Deliver)

Product roadmap A high-level, time-based view of how a product is expected to evolve, used for planning and alignment. (Plan)

Product vision A statement of who a product is for, what need it meets and how it is different, used to guide backlog decisions. (Frame)

Program A group of related projects and activities managed together so their combined benefits are greater than if they were managed separately. (Frame)

Progressive elaboration Refining plans and details step by step as information and learning increase, rather than fixing everything on day one. (Frame)

Project A temporary, purposeful effort that creates a unique product, service, capability or change to move an organisation toward a better future state. (Frame)

Project artifact Any document, board, log, chart or model the project creates or maintains to plan, track or record its work. (Deliver)

Project charter A document issued by the sponsor that formally authorises a project and gives the project manager authority to apply resources. (Frame)

Project governance The framework of authority, decision rights, rules and oversight within which a project is directed and controlled. (Govern)

Project manager The person assigned by the performing organisation to lead the team and integrate the work so that the project achieves its objectives and enables its intended value. (Frame)

Psychological safety A shared belief within a team that it is safe to take interpersonal risks such as speaking up, asking questions or admitting mistakes. (Lead)

Pull communication Information placed where people can access it when they need it, such as wikis and dashboards. (Engage)

Push communication Information sent to specific recipients, such as emails, reports or notifications. (Engage)

Q

Qualifier A word in a question such as FIRST, NEXT, BEST or EXCEPT that defines what kind of answer is wanted. (Adapt)

Quality assurance Activities that check whether the processes used to create deliverables are effective. (Deliver)

Quality control Inspecting and testing deliverables to confirm they meet requirements. (Deliver)

R

RACI A responsibility assignment matrix showing who is Responsible, Accountable, Consulted and Informed for each activity or decision. (Lead)

RAG status A red, amber or green rating of status, which is useful only when its criteria are defined objectively in advance. (Deliver)

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Reconciliation Comparing figures from different sources, explaining differences and correcting the cause. (Deliver)

Reference-class forecasting Adjusting an estimate using the actual outcomes of a group of similar past projects. (Resource)

Regulatory horizon scanning Regularly checking for new or changing rules that may affect the project. (Govern)

Release plan A forecast of which backlog items are likely to be delivered in upcoming releases, based on velocity ranges. (Resource)

Requirements traceability matrix A table that links each requirement to its source, business objective, design and test, so nothing is lost or added unnoticed. (Plan)

Residual risk The risk that remains after responses have been implemented. (Govern)

Resource breakdown structure A hierarchy of the resources a project needs, grouped by category and type. (Resource)

Resource calendar A record of when each resource is available, including shifts, holidays and part-time allocations. (Resource)

Resource levelling Adjusting start and finish dates to fit limited resources; may extend the project end date. (Resource)

Resource smoothing Adjusting activities within their float to even out resource demand without changing the end date. (Resource)

Retrospective A regular team meeting to inspect how the team worked and agree concrete improvements for the next cycle. (Adapt)

Reverse mentoring Mentoring in which a less senior person coaches a senior colleague, often on new tools or perspectives. (Engage)

Reverse shadowing A transfer method in which the receiver performs the task while the expert observes and corrects. (Engage)

Risk An uncertain event or condition that, if it occurs, affects at least one project objective — negatively (threat) or positively (opportunity). (Govern)

Risk appetite The amount of uncertainty an organisation is willing to accept in pursuit of its objectives. (Govern)

Risk breakdown structure (RBS) A hierarchical grouping of risk sources used as a prompt to identify risks systematically. (Govern)

Risk management plan The part of the project plan that defines how risks will be identified, analysed, responded to, monitored and reported. (Govern)

Risk owner The one named person who monitors a risk and makes sure its response is carried out. (Govern)

Risk score Probability score multiplied by impact score, used to rank risks on a heat map. (Govern)

Risk threshold A measurable level of risk exposure above which a risk must be escalated or treated. (Govern)

Rolling-wave planning Planning near-term work in detail and later work at a higher level, adding detail as it approaches. (Plan)

Root cause analysis Techniques, such as asking 'why?' repeatedly, used to find the underlying cause of a problem. (Adapt)

Rough order of magnitude (ROM) An early, wide-range estimate used for initial decisions, before detail is known. (Resource)

Runbook A step-by-step guide for operating, supporting and recovering a system or process. (Engage)

S

S-curve The typical shape of cumulative planned spend over time: slow start, steep middle, flat end. (Resource)

Salience model A stakeholder model that classifies stakeholders by power, legitimacy and urgency. (Engage)

SBI feedback A feedback structure describing the Situation, the observable Behaviour and its Impact. (Lead)

Schedule baseline The approved version of the schedule used to measure progress, changed only through formal change control. (Resource)

Schedule performance index (SPI) EV ÷ PV; below 1.0 means less work has been done than planned. (Resource)

Scope baseline The approved scope statement, WBS and WBS dictionary, changed only through formal change control. (Plan)

Scope creep Uncontrolled expansion of scope without adjusting time, cost or resources and without proper approval. (Govern)

Scope statement A description of the project's product and project scope, deliverables, acceptance criteria, exclusions, constraints and assumptions. (Plan)

Scrum Master The accountable person in Scrum for team effectiveness, coaching the team and organisation and removing impediments. (Deliver)

Secondary risk A new risk that arises as a direct result of implementing a risk response. (Govern)

Servant leadership A leadership approach that focuses on serving the team — removing obstacles, growing people and enabling them to deliver value. (Lead)

Service level agreement (SLA) An agreement defining the measurable level of service a supplier must provide. (Deliver)

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Situational leadership Matching the mix of direction and support to a person's competence and commitment on a specific task. (Lead)

Situational-judgement question An exam item presenting a realistic scenario and asking for the best action among plausible alternatives. (Adapt)

Sprint goal A short statement of what a sprint aims to achieve, giving the team focus and flexibility in how it gets there. (Lead)

Stage gate A decision point between phases where governance reviews progress and decides whether to continue, change or stop. (Govern)

Stakeholder Any person, group or organisation that can affect, be affected by, or believe they are affected by the project. (Engage)

Stakeholder engagement plan The planned actions, messages and methods for moving each stakeholder from current to desired engagement. (Engage)

Stakeholder register A living record of stakeholders, their interests, influence, engagement levels and planned approach. (Engage)

Statement of work (SOW) A description of what a supplier must deliver: scope, deliverables, standards, acceptance criteria and timeline. (Deliver)

Story map A layout of the user journey from left to right, with stories stacked under each step by priority to plan releases. (Plan)

Story point A unitless, team-relative measure of the size of a backlog item, combining effort, complexity and uncertainty. (Resource)

Subsidiary plan A component of the project plan covering one area, such as risk, quality or procurement. (Plan)

Success criteria Agreed, measurable conditions that show whether the project has achieved what stakeholders expected. (Engage)

Systems thinking Seeing the project as a set of interacting parts so that the knock-on effects of a decision are anticipated. (Frame)

T

Tacit knowledge Know-how held in people's experience and judgement that is hard to write down. (Engage)

Tailoring Deliberately adapting the approach, processes and artefacts to fit the project's context. (Frame)

Task A specific responsibility of the project manager within a domain. (Frame)

Team charter A short document, written by the team, that records its purpose, values, ground rules, decision rights and ways of working. (Lead)

Thomas–Kilmann modes Five conflict-handling approaches — competing, collaborating, compromising, avoiding and accommodating — mapped on assertiveness and cooperativeness. (Lead)

Throughput The number of work items finished per unit of time. (Deliver)

Time and materials contract A contract that pays agreed rates for time and materials used, with most cost risk on the buyer. (Deliver)

Time-phased budget A budget spread over the periods in which costs are expected, used to plan funding and measure performance. (Resource)

To-complete performance index (TCPI) The cost efficiency that must be achieved on remaining work to meet a target (BAC or EAC). (Resource)

Tolerance The permitted deviation from a plan, such as ±5% cost, within which the project manager can act without escalating. (Govern)

Total float The amount of time an activity can be delayed without delaying the project finish date. (Resource)

Trade-off An agreed choice to give up some of one thing, such as scope, to protect another, such as the date. (Engage)

Transformational leadership Leading by inspiring people with a compelling purpose and encouraging them to grow beyond current expectations. (Lead)

Transition Moving responsibility for a deliverable from the project to operations, support or the next phase. (Deliver)

Trigger An observable warning sign that a risk is about to occur or has begun to occur. (Govern)

Tuckman model A model of team development through forming, storming, norming, performing and adjourning. (Lead)

U

Unstated assumption Something a stakeholder takes for granted without saying, such as 'it will work offline'. (Engage)

User story A short description of a need from the user's point of view, used as a placeholder for conversation and refined with acceptance criteria. (Plan)

V

Value The worth that stakeholders gain from a project, which may be financial, social, environmental or strategic. (Frame)

Value delivery system The connected set of portfolios, programmes, projects, products and operations through which an organisation turns strategy into value. (Frame)

Value-based delivery Ordering and releasing work so that stakeholders receive the most value as early as possible, and checking that value throughout. (Frame)

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Value-based prioritisation Ordering work by the value it creates for stakeholders, adjusted as feedback arrives. (Plan)

Velocity The amount of work, usually in story points, a team completes to the definition of done in one iteration. (Resource)

Virtual team A team whose members work mainly from different locations and collaborate through digital tools. (Lead)

W

Watermelon reporting Status that appears green externally while the underlying reality is amber or red. (Deliver)

WBS dictionary A document that describes each WBS element: its work, owner, acceptance criteria, milestones and estimates. (Plan)

Weighted shortest job first (WSJF) A prioritisation method that divides cost of delay by job size, so that high-value, small items go first. (Plan)

WIP limit A cap on the number of work items allowed in a workflow stage at one time. (Deliver)

Work breakdown structure (WBS) A hierarchical, deliverable-oriented decomposition of the total scope of a project. (Plan)

Work package The lowest level of a WBS, which can be estimated, scheduled, assigned and controlled. (Plan)

Workaround A temporary solution that lets work continue while the root cause is fixed. (Adapt)

Working agreement A team-created set of norms describing how members will collaborate, communicate and make decisions. (Lead)

Z

Zone of possible agreement (ZOPA) The range between the two parties' walk-away points in which a deal is possible. (Deliver)

ZPMP Professional Loop ZyeemZia's eight-stage framework for this course: Frame, Lead, Engage, Plan, Resource, Deliver, Govern and Adapt. (Frame)

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CAPSTONE

Capstone: Run a project through the ZPMP Professional Loop

Pick a real project from your work or community, or a realistic one (for example: launching a mobile banking feature in Dhaka, opening a new clinic, or moving a university to a new learning platform). Take it through all eight stages of the ZPMP Professional Loop, covering the people, process and business environment domains, and produce a compact project pack that a sponsor could approve, a team could execute and a governance board could oversee.

Deliverables

☐ Business case, charter and shared vision with measurable success criteria (Frame)

☐ Team charter with ground rules, RACI and leadership approach (Lead)

☐ Stakeholder register, expectation alignment map, knowledge transfer plan and communication plan (Engage)

☐ Integrated plan summary with the chosen approach, plus a prioritised backlog or WBS with acceptance criteria (Plan)

☐ Schedule or release forecast, resource plan and budget with contingency and management reserve (Resource)

☐ Procurement and quality approach and a simulated mid-point status dashboard using earned value or flow metrics (Deliver)

☐ Governance and escalation thresholds, compliance register, change log and top-10 risk register (Govern)

☐ Impediment log, change-adoption plan, external change radar and lessons learned with process-asset updates (Adapt)

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Value & vision	○ Objectives vague	○ Clear objectives and business case	○ Measurable outcomes tied to benefits, owners and a shared vision
Leadership & team	○ Roles unclear	○ Team charter and RACI	○ Empowering leadership, ground rules and conflict approach that fit the team
Stakeholders & communication	○ Few stakeholders listed	○ Register and communication plan	○ Expectations aligned and monitored, knowledge transfer planned
Integrated planning & scope	○ Separate, inconsistent plans	○ Approach chosen and scope defined	○ Integrated, tailored plan that delivers value incrementally
Schedule, resources & finance	○ Single-point dates and costs	○ Estimated schedule, resources and budget	○ Ranges, reserves and a forecasting method
Delivery, quality & status	○ No quality or supplier approach	○ Quality and procurement planned	○ Honest metrics, supplier management and transition readiness
Governance, compliance & risk	○ Not addressed	○ Risk register and change process	○ Thresholds, compliance evidence and quantified top risks
Adaptation & improvement	○ No learning or change plan	○ Lessons and adoption plan	○ External changes tracked, impediments removed and lessons built into process assets

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CAPSTONE

Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 40 questions at random from a separate, secret bank that tests the same skills, with a 60-minute limit and a 80% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Frame — Foundations: Value, Approaches & the PMP® Outline

1. A sponsor asks you to start work immediately on a new customer portal. Funding is verbally agreed, but no document authorises the project or names you as project manager. What should you do FIRST?

   (A) Begin requirements workshops with key users to save time

   (B) Select a delivery vendor so that work can start quickly

   (C) Develop a detailed project management plan for sponsor approval

   (D) Work with the sponsor to create and sign a project charter

2. Midway through a project, market research shows the expected customer demand that justified the investment has fallen sharply. Deliverables are on track. What should the project manager do NEXT?

   (A) Continue as planned, because scope, time and cost are all still on track

   (B) Accelerate delivery so the product reaches the market before demand falls further

   (C) Raise it with the sponsor and review the business case: continue, change or stop

   (D) Reduce the remaining testing effort to lower costs and protect the expected return

3. A company must deliver a new tax-reporting module with requirements fixed by regulation, and a customer mobile app whose features are still being explored. Which approach should the project manager recommend?

   (A) Fully predictive delivery for both components, with one integrated baseline and milestone plan

   (B) Fully agile delivery for both components, using short sprints and no fixed milestones

   (C) Postpone the app until the regulatory module is complete, then reassess the app features

   (D) Hybrid: predictive for the regulatory module, adaptive for the app, integrated milestones

4. A project delivered a new HR system on time and on budget. Six months later, only 20% of managers use it and the expected time savings have not appeared. What does this situation BEST illustrate?

   (A) Outputs were delivered, but the intended outcomes and benefits were not realised

   (B) The project was a complete success because it met its time and cost constraints

   (C) The schedule baseline was poorly estimated, so the system was not ready for users

   (D) The procurement strategy chose the wrong system for the managers' working needs

5. A newly appointed PM in a large bank is told to use the organisation's full 40-document project methodology for a six-week, low-risk internal dashboard. What should the PM do?

   (A) Propose a tailored, lighter set of practices and agree it with the governance body

   (B) Follow every step and document exactly, as the methodology requires of all projects

   (C) Set the methodology aside and run the work informally, as the risk is low

   (D) Deliver first and ask the team to complete all required documents after go-live

Lead — People I: Leading the Project Team

6. Two senior engineers have argued for a week about which database to use, and the team is now split into camps. The decision is important and the deadline is three weeks away. What should the project manager do FIRST?

   (A) Meet both to understand their criteria, then facilitate an evidence-based joint decision

   (B) Choose the database yourself, based on your own analysis, to end the argument

   (C) Ask them to compromise by using both databases, one for each part of the system

   (D) Escalate the disagreement to the sponsor and ask for a binding decision by the end of this week

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7. A new team member is very enthusiastic but has never used the team's testing framework. According to situational leadership, which style should the PM use for this task?

   (A) Avoiding — step back until they ask for help
   (B) Delegating — let them work it out independently
   (C) Supporting — focus on encouragement with little instruction
   (D) Directing — clear instructions and close follow-up

8. In a retrospective, team members say they hesitate to raise problems because a previous manager publicly criticised people for mistakes. What is the BEST action for the project manager?

   (A) Reassure the team that the previous manager has left, so the problem is now solved
   (B) Share your own mistakes, thank people for raising problems and use blameless reviews
   (C) Make raising problems mandatory and track who contributes in each retrospective
   (D) Arrange a team-building outing so that people get to know each other better

9. Three months into a hybrid ERP rollout, warehouse supervisors describe the project's purpose as 'replacing old software', while the sponsor describes it as 'halving order-to-dispatch time'. Sprint reviews keep ending in disputes about what matters. What should the project manager do FIRST?

   (A) Ask the sponsor to send the written vision statement to every supervisor again, with a short note on why it matters
   (B) Add the sponsor's goal to the backlog as a new high-priority user story
   (C) Bring both groups together to re-establish the shared vision and agree its success measure
   (D) Continue delivery, since supervisors do not need to understand the business goal

10. A distributed team's charter says disagreements about design are raised openly in the team channel. A developer has twice complained privately to the sponsor about a colleague's design choices instead. What should the project manager do?

    (A) Ask the sponsor to refuse all future complaints from team members
    (B) Remove the ground rule, because it is clearly not working for the team
    (C) Raise the breach in front of the whole team at the next stand-up so everyone sees that the rule applies to all
    (D) Talk privately with the developer, refer to the agreed rule and agree how she will raise disagreements

11. An experienced Scrum team asks to choose its own code-branching strategy. Company standards allow any of three approved strategies, and the project manager personally prefers one of them. What should the project manager do?

    (A) Let the team choose among the approved strategies and support its decision
    (B) Choose her preferred strategy herself, because as project manager she remains accountable for delivery
    (C) Ask the sponsor to make the decision to avoid any disagreement
    (D) Let the team adopt any strategy it likes, approved or not

12. A team spans Dhaka, Kuala Lumpur and Frankfurt with only one hour of shared working time a day. Many decisions wait a full day for the next call. What is the BEST improvement?

    (A) Move every team member onto Dhaka working hours
    (B) Agree async decision rules, such as written proposals with an objection deadline
    (C) Hold two daily calls at different times so that each region can attend at least one of them
    (D) Let the most senior member decide alone between calls

13. A new tester from a culture where juniors rarely contradict seniors never disagrees in meetings, but his written reports show serious defects that others missed. What should the project manager do?

    (A) Tell him that speaking up in meetings is required for promotion
    (B) Move him to documentation work where speaking up matters less
    (C) Invite his input through private and written channels and recognise his findings openly
    (D) Ask his line manager to enrol him in an assertiveness course so that he learns to challenge seniors in meetings

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Engage — People II: Stakeholders, Expectations, Knowledge & Communication

14. Two months into an ERP project, you discover that the internal audit department was never consulted and has compliance requirements for the finance module. What should you do FIRST?

(A) Submit a change request covering all possible audit requirements for the finance module

(B) Add internal audit to the stakeholder register and meet them to learn their needs

(C) Escalate to the sponsor that the project is now at risk of a compliance failure

(D) Continue as planned and address any audit concerns in a phase after go-live

15. A senior regulator has high power over your project but shows little interest in it. Using the power–interest grid, what engagement approach is BEST?

(A) Monitor them with minimal effort, as their low interest means low influence

(B) Manage them closely with weekly detailed meetings and full access to all project data

(C) Keep them satisfied with concise, relevant updates and consult them when affected

(D) Keep them informed by adding them to every team email and distribution list

16. A project team grows from 8 to 12 members. By how many communication channels does the team's communication complexity increase?

(A) 4

(B) 28

(C) 66

(D) 38

17. Before planning a clinic network's patient app, the finance director expects lower call-centre costs, doctors expect fewer missed appointments and a patients' group expects Bangla-language access. Not all of these fit in the first release. What should the project manager do FIRST?

(A) Plan the first release around the finance director's expectations, because she controls the budget and must approve it

(B) Facilitate a session in which the groups surface their expectations and agree priorities and success criteria

(C) Include every expectation in the first release to avoid conflict

(D) Ask the vendor which features are easiest to build first

18. A warehouse system passed every acceptance test, but three weeks after go-live the operations team, an internal customer, says it 'slows everything down' and satisfaction scores are falling. What should the project manager do?

(A) Point to the signed acceptance and close the discussion

(B) Ask the vendor to add servers and tune the database, since slowness usually means the system lacks capacity

(C) Restart hypercare for three months as a precaution

(D) Work with operations to find the gap between their expectations and the delivered workflow, then agree actions

19. A contractor who configured all the payment interfaces leaves in four weeks. Her contract requires 'handover documentation'. What BEST ensures the knowledge stays with the project?

(A) Have named staff perform the key tasks with her observing, using updated runbooks, before she leaves

(B) Ask her to deliver the handover documentation in her final week

(C) Record all of her remaining meetings for future reference

(D) Extend her contract by a further year so that her knowledge remains available if problems arise

20. A donor funding a rural water project requires quarterly reports in its own format, with indicators the team's dashboard does not track. What should the project manager do?

(A) Send the team's dashboard, because it is more informative

(B) Ask the donor to accept the team's measures instead

(C) Add the donor's reporting requirements to the communication plan and collect its indicators from the start

(D) Compile the donor's indicators by hand from project records at the end of each year, when full data is available

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Plan — Process I: Integrated Planning, Scope & Value

21. During project initiation, a stakeholder asks what measurable result will show the project succeeded. Where should this be documented FIRST?

(A) In the lessons-learned register, so future projects can reuse the measure

(B) In the project charter, as success criteria linked to the business case

(C) In the team's working agreement, so the team commits to the result

(D) In the first sprint retrospective notes, once the team has started work

22. A requirements workshop is dominated by two senior managers while the frontline users who will use the system stay silent. What should the facilitator do?

(A) Let the senior managers decide, since they hold authority and know the strategy

(B) End the workshop and send everyone a survey to collect their views in private

(C) Use silent brainstorming and breakouts, then interview or observe the users

(D) Ask the senior managers to leave so the frontline users can speak more freely

23. The product owner brings a story estimated at three sprints of work: 'As a customer I want the full loan application journey.' What should the team do?

(A) Split it into thin vertical slices that each deliver testable user value

(B) Accept it as one story and track its progress across the three sprints

(C) Split it by technical layer: database, API and user interface, one per sprint

(D) Ask the product owner for a complete written specification before estimating

24. Feature P has a cost of delay of 24 and a job size of 12. Feature Q has a cost of delay of 15 and a job size of 5. Using WSJF, what should the product owner do?

(A) Sequence P first, because its cost of delay (24) is higher than Q's (15)

(B) Sequence Q first, because its WSJF score (3.0) is higher than P's (2.0)

(C) Sequence P first, because its larger size makes it the more important job

(D) Treat them as equal priority, because their scores differ by too little

25. Your WBS for a new clinic has level-2 elements named 'Plan', 'Design', 'Build', 'Test' and 'Deploy', with tasks underneath. What is the BEST feedback?

(A) Restructure it around deliverables and move activities into the schedule

(B) Accept it, because a phase-based structure is a valid way to organise a WBS

(C) Add more levels until every element represents less than one hour of work

(D) Replace the WBS with a Gantt chart, which shows the same work over time

26. The approved procurement plan shows network switches arriving in week 18, but the approved schedule starts integration testing in week 14. What should the project manager do?

(A) Start integration testing in week 14 without the switches

(B) Tell procurement to deliver the switches by week 14 whatever it costs, since the schedule is already approved

(C) Leave both plans unchanged and report the conflict at go-live

(D) Resolve the conflict across the plans and update the integrated plan through change control

27. A bank wants new teller software live in all 200 branches over one weekend. The software is new to staff and the support team is small. Which execution strategy should the project manager recommend?

(A) A single weekend switch-over for all branches at once

(B) A pilot in a few branches, then phased waves with lessons applied between waves

(C) Letting each branch manager choose when to switch, because managers know their own staff's readiness best

(D) Waiting until the software has no known defects before any rollout

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28. Field inspectors say the most painful part of their day is re-typing paper notes. The full inspection app is planned for release in ten months. What should the product owner consider FIRST?

(A) Whether a small early release that digitises notes could deliver this value sooner

(B) Adding more features to the ten-month release to make it more attractive

(C) Hiring more developers so that every planned feature can be finished and released several months sooner

(D) Keeping the current plan, because the roadmap has been approved

29. A project promises to cut loan processing time from 10 days to 4. Halfway through, the PM learns that nobody is measuring current processing times. What should the project manager do?

(A) Wait until go-live and start measuring then

(B) Use the vendor's estimate of the improvement in progress reports until real data becomes available after go-live

(C) Work with the benefit owner to set the baseline and a way to measure processing time now

(D) Remove the benefit from the business case to avoid embarrassment

Resource — Process II: Schedule, Resources & Finance

30. An activity has an optimistic estimate of 6 days, most likely 9 days and pessimistic 18 days. The sponsor asks for the expected duration. Using PERT, what should you report?

(A) 9 days, as the most likely value is the best single estimate

(B) 11 days, the simple average of the three estimates

(C) 18 days, to protect the sponsor against any overrun

(D) About 10 days, with a standard deviation of about 2 days

31. A non-critical activity has 5 days of total float and 0 days of free float. A supplier says it will be 3 days late. What is the MOST accurate impact?

(A) The project end date will move by 3 days, as will the successor's start

(B) There is no impact on any other activity, because float absorbs the delay

(C) The end date is unaffected, but the successor's early start moves by 3 days

(D) The activity immediately joins the critical path and drives the end date

32. A Scrum team's velocity over the last four sprints was 30, 34, 28 and 32. The remaining release backlog is 186 points. A stakeholder asks when the release will be ready. What is the BEST answer?

(A) Commit to exactly 6 sprints, using the average velocity of the last four sprints

(B) Forecast a range of about 6–7 sprints from observed velocity, updated each sprint

(C) Ask the team to raise velocity to 40 points so the release finishes sooner

(D) Decline to forecast, because agile teams do not commit to delivery dates

33. A stakeholder suggests funding a new feature from the project's contingency reserve because 'the risks it covered have not happened yet'. What should the project manager do?

(A) Explain that contingency covers identified risks; process it as a change request

(B) Move the money into management reserve until a decision is needed

(C) Approve the use, since the money is unused and the feature adds value

(D) Fund the feature from management reserve instead, without informing the sponsor first

34. Two activities scheduled in parallel both need the only certified electrician, causing an overload. The activities are on the critical path. Which technique applies, and what is a likely consequence?

(A) Resource smoothing, which never changes the project end date

(B) Resource levelling, which may extend the project end date

(C) Crashing, which removes the dependency on the electrician

(D) Fast-tracking, which reduces demand on the shared resource

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35. At month 5 of a project, PV = 200,000, EV = 180,000 and AC = 225,000. Which statement BEST describes the project's status?

(A) Behind schedule and over budget (SPI 0.90, CPI 0.80)
(B) Ahead of schedule and over budget (SPI 1.11, CPI 0.80)
(C) Behind schedule and under budget (SPI 0.90, CPI 1.25)
(D) On schedule and over budget (SPI 1.00, CPI 0.80)

36. A project has BAC = 800,000, EV = 300,000 and AC = 400,000. The sponsor asks for a forecast assuming current cost performance continues. What is the estimate at completion?

(A) About 900,000
(B) About 1,000,000
(C) About 1,066,667
(D) About 600,000

37. A project has BAC = 500,000, EV = 200,000 and AC = 250,000. The sponsor insists on finishing within the original budget. What cost performance must the team achieve on the remaining work?

(A) A TCPI of 0.83
(B) A CPI of at least 0.80
(C) A TCPI of 1.20
(D) A TCPI of 1.00

Deliver — Process III: Procurement, Quality, Status & Closure

38. During a sprint, a senior manager asks a developer to start an urgent new feature. What should the developer do?

(A) Start the feature immediately, because the request comes from a senior manager
(B) Add it to the sprint backlog without telling anyone and absorb it with extra effort
(C) Ask the team to cancel the sprint and replan it so the urgent feature can be included
(D) Refer it to the product owner, who decides priority while the sprint goal is protected

39. A Kanban team's Test column is at its WIP limit and developers say they are idle because they cannot pull new work. What should the team do?

(A) Raise the WIP limit in Test so that more work can flow through
(B) Start new development work and park it until Test has capacity
(C) Help finish items in Test by swarming on the bottleneck
(D) Remove WIP limits entirely, since they are causing idle time

40. A team's average WIP is 30 items and average throughput is 10 items per week. Management wants average cycle time to fall to about 1.5 weeks without adding people. What is the BEST approach?

(A) Ask the team to work overtime until cycle time falls
(B) Reduce WIP to about 15 items by limiting new starts
(C) Add more columns to the board to make work visible
(D) Measure lead time instead, as it better reflects delays

41. A buyer needs a supplier for a research-like data platform whose scope is highly uncertain, but wants the supplier motivated to control costs. Which contract type is MOST suitable?

(A) Fixed price with economic price adjustment
(B) Firm fixed price
(C) Cost plus incentive fee
(D) Purchase order for a fixed quantity

42. A garment factory's defect data shows that three causes account for most rejected pieces. Which tool should the quality lead use to communicate where to focus improvement efforts?

(A) A control chart
(B) A scatter diagram
(C) A flowchart
(D) A Pareto chart

43. A programme dashboard has shown every project green for six months. Last week two projects turned red overnight, each reporting a slip of more than two months. What should the PMO do to prevent this in future?

(A) Require PMs to justify any red status in writing to the board before reporting
(B) Introduce objective, forecast-based RAG criteria and encourage early bad news
(C) Remove the red category so projects escalate through amber before failing
(D) Report status quarterly rather than monthly to reduce reporting pressure

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44. The project manager is negotiating a hardware contract. The supplier insists on a 15% price increase. The PM has a confirmed offer from another qualified supplier at a 5% increase. How should this alternative be used?

(A) Keep it hidden, because revealing alternatives weakens a negotiating position

(B) As the PM's BATNA, setting a walk-away point and strengthening the negotiation

(C) As a reason to end the negotiation now and sign with the other supplier

(D) As leverage to threaten reporting the supplier to the industry regulator

45. A software system has been deployed and users are working with it. The vendor is asking for final payment, but no formal acceptance has been recorded. What should the project manager do FIRST?

(A) Hold a formal acceptance review against the agreed criteria and obtain sign-off

(B) Withhold payment until every outstanding enhancement request is completed

(C) Archive the project documents and pass the vendor's invoice to finance for payment

(D) Approve final payment, since users are already working with the system

Govern — Business Environment I: Governance, Compliance, Change & Risk

46. A team member has added an attractive analytics screen that no stakeholder requested, delaying a planned feature. What does this BEST describe, and what should the PM do?

(A) Gold plating; agree with the team that work must trace to approved items

(B) Progressive elaboration; record it as normal refinement of the requirements

(C) Scope creep; raise a change request against the requesting stakeholder

(D) Healthy innovation; recognise it publicly to encourage creative initiative

47. A project's forecast shows it will finish 2% over budget. The governance board has set a cost tolerance of ±5%. What should the project manager do?

(A) Request an urgent exception meeting with the board to approve the overrun

(B) Re-baseline the budget immediately to reflect the new cost forecast

(C) Stop non-essential work until the board formally approves the overrun

(D) Manage the variance within the project and report it in the normal cycle

48. An organisation wants a PMO that sets mandatory reporting standards and reviews project compliance, but leaves day-to-day management with project managers in business units. Which PMO type fits?

(A) Supportive

(B) Directive

(C) Portfolio-only

(D) Controlling

49. A risk owner reports that the warning sign defined for a high-priority risk has just appeared, and an approved contingency plan exists within the PM's authority. What should the project manager do NEXT?

(A) Schedule a risk workshop to reassess all risks before taking action

(B) Implement the contingency plan and update the risk register and issue log

(C) Ask the sponsor for permission before executing the contingency plan

(D) Wait to see whether the risk actually materialises before using reserves

50. Two vendors are being considered. Vendor A costs 50,000 more but has a 10% chance of a 200,000 delay penalty. Vendor B has a 40% chance of the same penalty. Based on expected monetary value, which is the better choice?

(A) Vendor B, because it is 50,000 cheaper upfront

(B) Vendor A, because it has the lowest probability of delay regardless of cost

(C) They are equal once the penalty risk is taken into account

(D) Vendor A, because its expected cost is 70,000 versus 80,000 for Vendor B

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51. Before a board meeting, the sponsor asks the PM to report the project as amber instead of red, promising that the problems will be fixed next month. What should the PM do?

(A) Report green to avoid alarming the board before the fixes are in place

(B) Report amber as asked, since the sponsor is accountable for the project

(C) Report the true status and offer a recovery plan with the decisions needed

(D) Avoid attending the board meeting and let the sponsor present the status

52. An inspection of a garment-factory extension finds that the fire exits do not meet the national building code, although they match the approved drawings. The buyer's compliance audit is in six weeks. What should the project manager do FIRST?

(A) Assess the non-compliance and its consequences with the safety engineer, then agree corrective actions and owners

(B) Ask the buyer to postpone its audit until the exits are rebuilt

(C) Record the finding as a lesson learned for future projects

(D) Instruct the contractor to rebuild the exits immediately at its own cost and report the outcome to the auditors

53. A change adding a new payment method was approved two weeks ago. Testers are still using the old requirements and the training material has not changed. What was MOST likely missed?

(A) Analysing the change's impact on scope, schedule, cost and risk before the board approved it

(B) Obtaining the sponsor's approval for the change

(C) Adding the change to the risk register

(D) Updating project documents and communicating the approved change to everyone affected

Adapt — Business Environment II: Impediments, Improvement, Change & the Outside World

54. Six weeks after a new procurement system went live, usage is low. Surveys show users understand why the change matters and want it, but feel unsure how to complete key tasks. Which ADKAR element should the next intervention target?

(A) Awareness — more communication about why the change is needed

(B) Reinforcement — recognition for heavy users

(C) Desire — incentives to use the new system

(D) Knowledge and Ability — hands-on training and floor support

55. A senior supervisor publicly criticises a new digital process, and adoption on her team is the lowest in the organisation. What should the project manager do FIRST?

(A) Report her behaviour to HR as a breach of the change programme’s standards

(B) Exclude her team from the rollout until adoption elsewhere is fully complete

(C) Meet her privately to understand her concerns and involve her in solving them

(D) Double the training sessions for her team to build skills and confidence

56. A team member wants to upload a spreadsheet of customer names, phone numbers and complaints to a free online AI tool to get a summary of complaint themes. The organisation has an approved enterprise AI assistant. What should the project manager advise?

(A) Use the approved enterprise tool and mask personal data before analysis

(B) Avoid AI altogether and have the team read every single complaint manually

(C) Use the free tool, but upload only half of the data to limit exposure

(D) Use the free tool, because generating summaries does not store any data

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57. A project manager uses an AI assistant to draft a status summary. The draft states that the project is '12% under budget', but the PM is not sure where that figure came from. What should the PM do BEFORE sending it?

(A) Send it as drafted, since AI tools are usually accurate with simple numbers

(B) Verify the figure against the cost reports and correct the draft if needed

(C) Remove all the numbers from the report to avoid any risk of inaccuracy

(D) Ask the AI assistant to confirm the figure and cite where it came from

58. Delays in municipal permits are blocking a 5G site rollout, and only the city office can grant them. The project team has no influence there. What is the BEST action?

(A) Pause the whole rollout until every permit arrives

(B) Start construction on the blocked sites without permits and seek retrospective approval to protect the schedule

(C) Engage the sponsor and regulatory-affairs team to work with the city office, and re-sequence other sites

(D) Remove the affected sites from the project's scope

59. Midway through a hospital project, a currency devaluation raises the cost of imported medical equipment by 18%. What should the project manager do FIRST?

(A) Assess the impact on budget, scope and the business case, then present options to governance

(B) Cancel the equipment order and look for local suppliers

(C) Absorb the increase quietly from the contingency reserve so that the sponsor and board are not alarmed

(D) Wait for the exchange rate to recover before acting

60. A PMO's lessons-learned library holds 400 entries, but project managers say they never use it. Which change would MOST improve continuous improvement?

(A) Require every project manager to read all 400 entries at kick-off and sign to confirm they have done so

(B) Turn the most frequent lessons into updates to templates, checklists and gate criteria

(C) Add more entries from current projects to make the library complete

(D) Delete the library and start again with a new tool

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EXAM PREPARATION

Practice exam answers & explanations

1. D The charter formally authorises the project and the PM's use of resources. Workshops, detailed planning and vendor selection all assume authority and budget that do not yet exist.

2. C The business case is the yardstick for continued investment; when its assumptions change, decision-makers must review it. Delivering on plan does not create value if the justification has gone. Speeding up or cutting testing ignores the real issue.

3. D Each component needs the approach that matches its uncertainty, while integrated milestones manage dependencies. Fully predictive freezes uncertain features; fully agile ignores the fixed regulatory deadline; postponing delays value unnecessarily.

4. A Success in modern project management is judged by outcomes, benefits and value. Meeting time and cost constraints is necessary but not sufficient. Nothing suggests estimating or procurement problems.

5. A Tailoring fits the approach to the context while keeping governance informed and in agreement. Applying everything wastes effort; ignoring governance creates compliance risk; writing documents after the fact adds no value.

6. A The issue is important and there is time, so collaborating on interests and objective criteria gives the best, most committed outcome. Forcing a choice loses buy-in, a forced compromise may produce a poor technical result, and escalation is premature.

7. D High enthusiasm with low competence calls for directing. Delegating and supporting assume skills they do not yet have, and waiting leaves them to struggle.

8. B Psychological safety is rebuilt through consistent leader behaviour and blameless practices. Simply stating that things have changed, tracking compliance or a one-off social event does not create lasting safety.

9. C The groups hold different pictures of success, which is a vision problem. Re-establishing the shared vision together, with a measure both accept, removes the root cause. Re-sending a document or adding a story does not create shared understanding, and ignoring it lets the disputes continue.

10. D Ground-rule breaches are handled promptly and privately first, with reference to what the team agreed. Public criticism harms safety, dropping the rule undermines the charter, and blocking the sponsor misses the behaviour itself.

11. A Empowering a capable team means delegating decisions within agreed boundaries, here the approved standards. Overruling the team, pushing the decision upward or ignoring the standards are all weaker.

12. B Async-first working agreements let distributed teams decide without waiting for overlap time, while live time is kept for real discussion. The other options are unfair, add meetings or concentrate decisions in one person.

13. C Supporting varied perspectives means offering ways to contribute that fit the person, and showing the team that his contributions count. Pressure or sidelining wastes his insight.

14. B First understand the stakeholder and their actual requirements; only then can impact be assessed and a change request raised if needed. Escalating before understanding is premature, and ignoring compliance creates serious risk.

15. C High power and low interest calls for keeping the stakeholder satisfied without overloading them. Weekly detailed meetings or mass emails overload; minimal monitoring underestimates their power.

16. D 8 people: $8 \times 7 \div 2 = 28$ channels. 12 people: $12 \times 11 \div 2 = 66$ channels. The increase is $66 - 28 = 38$. The other options are the head-count change or one of the totals.

17. B Aligning expectations means bringing the groups together to agree one set of priorities and success criteria that fit the objectives. Siding with one group, overloading the release or letting ease of build decide all leave the conflict unresolved.

18. D Managing expectations means monitoring satisfaction and responding to gaps. The cause may be workflow rather than performance, so it must be investigated with the customer before choosing a fix.

19. A Knowledge transfer is proven when receivers can perform the work. Documents and recordings help, but tacit knowledge moves through practice; extending the contract only postpones the risk.

20. C Understanding reporting requirements and planning for them is part of communication management. Collecting the indicators from the start makes reports accurate and timely.

21. B Success criteria belong in the charter at initiation so everyone agrees on what success means from the start. The other documents serve different purposes and come later.

22. C Structured facilitation balances participation, and follow-up elicitation captures users' real needs. Deferring to rank misses user needs; abandoning the session wastes it; excluding managers damages relationships and loses expertise.

23. A Vertical slices keep stories small, valuable and testable within one sprint. Layer splits deliver no user value on their own; multi-sprint stories hide progress; heavy specification delays learning.

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24. B WSJF = cost of delay ÷ job size: P = 24 ÷ 12 = 2.0, Q = 15 ÷ 5 = 3.0. The higher score goes first. Looking at cost of delay or size alone ignores the ratio WSJF relies on.

25. A A deliverable-oriented WBS supports the 100% rule, ownership and acceptance; activities belong in the schedule. Over-decomposition adds overhead, and a Gantt chart shows timing rather than scope decomposition.

26. D An integrated plan must be consistent. The PM assesses consolidated plans for dependencies and gaps, resolves the conflict and updates the baseline formally.

27. B With new software and limited support, a pilot followed by phased waves reduces risk and lets the bank learn before scaling. A big-bang switch concentrates the risk; the other options lose control or delay value indefinitely.

28. A Value-based delivery looks for chances to deliver the most valuable part incrementally. An early release can create benefits months sooner while the rest continues.

29. C A benefits measurement system needs a baseline captured before the change. Working with the benefit owner now makes later improvement measurable.

30. D PERT = (6 + 4×9 + 18) ÷ 6 = 60 ÷ 6 = 10 days; σ = (18 − 6) ÷ 6 = 2 days. The most likely value ignores the risk of overrun, 11 days is the simple triangular mean, and reporting the pessimistic value overstates the estimate.

31. C A 3-day delay within 5 days of total float does not delay the project, but zero free float means its successor starts later, consuming shared float. The activity becomes critical only if the delay uses all its float.

32. B 186 ÷ 34 ≈ 5.5 and 186 ÷ 28 ≈ 6.6, so about 6–7 sprints. A range reflects uncertainty and is refined as data arrives. A single number hides uncertainty; velocity targets inflate estimates; agile teams do forecast.

33. A Contingency protects against identified risks; new scope requires change control and an authorised funding decision. Spending contingency leaves risks unfunded, and using management reserve requires management approval.

34. B When a resource limit is hard and the activities are critical, levelling delays one of them and can push out the end date. Smoothing only moves work within float, fast-tracking increases overlap, and crashing adds resources rather than resolving a single-specialist limit.

35. A SPI = EV ÷ PV = 180,000 ÷ 200,000 = 0.90 (behind). CPI = EV ÷ AC = 180,000 ÷ 225,000 = 0.80 (over budget). The other options invert one or both ratios.

36. C CPI = 300,000 ÷ 400,000 = 0.75, so EAC = BAC ÷ CPI = 800,000 ÷ 0.75 ≈ 1,066,667. 900,000 is the atypical formula AC + (BAC − EV); the other values do not follow from any standard formula here.

37. C TCPI = (BAC − EV) ÷ (BAC − AC) = 300,000 ÷ 250,000 = 1.20 — every unit of money must produce 20% more work than planned. 0.80 is the current CPI, and 0.83 and 1.00 do not reflect the remaining work and funds.

38. D The product owner owns the order of work, and the developers manage the sprint plan to protect the sprint goal. Seniority does not bypass this; only the product owner can cancel a sprint; silent additions destroy transparency.

39. C Swarming on the bottleneck improves flow — 'stop starting, start finishing'. Raising or removing limits hides the constraint, and starting new work increases WIP and lengthens cycle time.

40. B By Little's Law, cycle time = WIP ÷ throughput: 30 ÷ 10 = 3 weeks; at the same throughput, 15 ÷ 10 = 1.5 weeks. Overtime is unsustainable, extra columns do not change flow, and changing the metric does not change performance.

41. C Cost-reimbursable contracts suit high uncertainty, and an incentive fee shares overruns and savings, motivating cost control. Fixed-price types need clear scope; a simple purchase order fits standard goods.

42. D A Pareto chart ranks causes by frequency so the team can focus on the vital few. A scatter diagram tests relationships between variables, a flowchart shows process steps, and a control chart shows process stability over time.

43. B This is watermelon reporting. Objective thresholds tied to forecasts, plus a culture that welcomes bad news early, keep status honest. Adding justification burdens to red, removing red or reporting less often all make hidden problems more likely.

44. B A strong BATNA sets the walk-away point and strengthens the negotiating position. Hiding it gives no benefit, ending talks immediately may forgo a better deal, and threatening regulators is inappropriate.

45. A Formal acceptance against agreed criteria should precede contract closure and payment. Paying without acceptance invites disputes, demanding new enhancements is outside the contract, and archiving comes after acceptance.

46. A Adding unrequested extras is gold plating, which adds cost and risk. It is not stakeholder-driven scope creep, nor normal elaboration; rewarding it would encourage more unapproved work.

47. D Within tolerance, the PM manages by exception and reports normally. An urgent board meeting, stopping work or re-baselining are reactions reserved for forecasts outside tolerance or approved changes.

48. D A controlling PMO requires compliance and assures quality with moderate control. A supportive PMO only advises; a directive PMO manages projects directly; ‘portfolio-only’ is not one of the standard types.

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49. B Triggers exist so that pre-approved responses can be executed promptly. A workshop or seeking fresh permission wastes the planning already done, and waiting defeats the purpose of having a trigger.

50. D A: 50,000 + (0.10 × 200,000) = 70,000. B: 0 + (0.40 × 200,000) = 80,000. A is better by 10,000 in expected cost. Upfront price or probability alone ignores the full picture, and the values are not equal.

51. C Honesty requires accurate reporting; adding a recovery plan keeps the conversation constructive. Changing the status at anyone's request misleads the board, and avoidance leaves the board uninformed.

52. A Non-compliance must be analysed for consequences and corrected with clear owners. Instructing the contractor before understanding the cause and contract position, delaying the audit or merely recording a lesson are weaker first steps.

53. D After approval, change control includes communicating the decision and updating all affected documents. The approval itself happened; the follow-through did not.

54. D Awareness and desire are already present; the first barrier is knowing how and being able to perform. More why-messaging, incentives or recognition will not fix a skills gap.

55. C Resistance often contains useful information, and influential sceptics can become strong champions. Disciplinary action, exclusion or more training skip understanding the root cause.

56. A Approved tools plus data minimisation protect customers and the organisation. Free consumer tools may retain data, avoiding AI altogether wastes a legitimate capability, and uploading half the data still exposes personal information.

57. B AI-generated figures must be checked against authoritative data; the PM remains accountable. Assuming accuracy, removing all numbers or asking the same tool to confirm itself does not verify anything.

58. C Some impediments can only be resolved by stakeholders outside the team. Collaborating with those who can influence the outcome, while re-planning to limit the damage, is better than waiting, breaking rules or cutting scope.

59. A External changes must be assessed for their impact and brought to the right decision-makers with options. Acting before assessment, hiding the variance or waiting are weaker.

60. B Lessons change practice when they are built into the organisation's process assets. Reading or adding more entries does not change behaviour on its own.

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NOTES

My notes

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